Projectile Tracking Chronograph (PTC)

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1 Executive Summary

The Projectile Tracking Chronograph (PTC) is a senior design project developed by Group 5 at the University of Central Florida for the 2026 Spring/Summer semesters. The interdisciplinary team consists of students from Optical Engineering, Electrical Engineering, and Computer Engineering. The PTC is an integrated ballistic measurement system that combines precise velocity measurement with optical trajectory estimation in a single, affordable platform.

Traditional commercial chronographs provide only muzzle velocity and offer limited or no information about the projectile's flight path. The PTC addresses this limitation by using a laser-based optical gate to accurately measure velocity while simultaneously capturing synchronized high-speed images of the projectile in flight. These images are processed in real time to extract spatial position data, enabling the system to reconstruct and visualize the projectile's trajectory, launch angle, and predicted impact point. The system will operate reliably in both indoor and outdoor environments by employing modulated laser detection and synchronized strobe illumination to minimize the effects of ambient light.

The PTC is designed to be user-friendly and suitable for hobbyists and educational use. It features an intuitive graphical interface that displays both numerical results and a graphical trajectory visualization after each shot. By integrating optical sensing, embedded processing, and real-time image analysis, the PTC provides a complete picture of projectile performance that is better than conventional chronographs at a significantly lower cost.

This document presents the complete design and development process of the PTC. It begins with the project background and motivation, followed by a detailed review of the current technologies. Subsequent chapters cover the research and component selection for all major subsystems, engineering standards and design constraints, hardware and software implementation, system fabrication, testing and evaluation, and administrative content.

The PTC represents a practical and innovative advancement in ballistic instrumentation, demonstrating how optical, electrical, and computer engineering can be combined to create an accessible measurement tool for modern shooting sports and research applications.

2 Project Description

2.1 Background and Motivation

The ability to accurately measure projectile velocity has long been fundamental to the study of ballistics and the advancement of shooting sports. Early methods relied on manual observation and empirical estimation, which were often inconsistent and dependent on environmental conditions. The development of mechanical ballistic chronographs in

the late nineteenth century introduced a scientific approach by measuring velocity from distance over time, laying the groundwork for modern ballistic instrumentation.

In contemporary applications, ballistic chronographs have become essential tools for competitive shooters, hunters, and militaries as they provide quantitative data about projectile performance. Even small variations in muzzle velocity can significantly impact a projectile's trajectory and bullet drop, specifically at long ranges. Competitive shooters depend on accurate velocity measurements to refine ballistics and ensure the target is consistently hit. Other measurements of extreme spread and standard deviation are of interest, where lower variability in these metrics directly correlates with tighter shot grouping and improved accuracy.

Modern commercial chronographs further enhance usability by performing real-time calculations of muzzle energy, allowing users to simply gather projectile data rather than manual processing. However, many advanced chronograph systems rely on technology that can be expensive for hobbyists. Additionally, the commercially available systems focus primarily on velocity measurement and provide limited insight about the projectile's trajectory without the use of specialized or high-cost radar technology.

The PTC project is motivated by the need for an accessible and transparent ballistic measurement system that goes beyond the basic velocity calculation. By leveraging optical sensing, precise timing, and embedded computation, PTC aims to provide accurate projectile velocity measurements while enabling trajectory estimation through angular and spatial analysis. Beyond its technical objectives, PTC serves as an interdisciplinary senior design experience; this effort brings together optical engineering expertise focused on laser optics and beam deflection, electrical engineering contributions centered on analog sensing and signal processing, and computer engineering responsibilities of embedded firmware, timing analysis, and user interface implementation. Through collaboration, the team aims to develop a ballistic measurement system while fostering teamwork and problem-solving skills essential for professional engineering practice.

2.2 Current Technologies and Relevant Projects

2.2.1 The First Patented Ballistic Chronograph

Early projectile velocity measurement can be traced back to the late nineteenth century through purely mechanical systems such as the Le Boulenge ballistic chronograph, which was patented in 1874. This device relied on electromagnetic release mechanisms to drop two metal rods at different times as a projectile passed through two spaced wire screens. Knowing the acceleration due to gravity and the fixed spacing between the screens allowed the projectile's velocity and flight time to be calculated. Although entirely mechanical, this approach established the fundamental concept of velocity measurement through precise timing, a principle that remains central to modern electronic and optical chronographs. [1]

2.2.2 Optical Chronograph

These devices operate by detecting the passage of a projectile across two optical sensors spaced at an identified distance. As the projectile passes over each sensor, the light received through the sensor is interrupted, generating electrical signals that are time-stamped to compute the projectile's velocity. [2]

While optical chronographs are relatively affordable and straightforward to operate, they are sensitive to environmental conditions. Measurement errors are introduced when there are changes in sunlight intensity, shadows, or improper projectile positioning. These commercial chronographs only display the velocity data. PTC builds upon the fundamental principle of measurement used by optical chronographs, but seeks to mitigate environmental sensitivity by designing laser-based detection and precise electronic timing, enabling improved consistency and expanded trajectory analysis beyond simple velocity measurement.

2.2.3 Magnetic Chronographs

Magnetic chronographs measure projectile velocity by detecting disturbances in a magnetic field caused by a projectile passing near a sensor array. These systems are typically mounted directly to the firearm barrel and do not rely on external lighting conditions. [2]

Magnetic chronographs introduce certain limitations despite their reliability. The attachment of the device to the barrel can slightly alter the harmonics of the barrel, affecting where the projectile impacts. Magnetic chronographs are not suited for spatial position movement since their purpose is to measure velocity close to the muzzle. In contrast, PTC is designed as a non-contact measurement system, allowing projectile velocity and angular trajectory to be determined without physical interaction with the firing mechanism, preserving ballistic integrity.

2.2.4 Doppler Radar Chronographs

Doppler radar chronographs represent the most advanced of the commercially available projectile measurement systems. These devices transmit radar signals that reflect off the projectile during flight, allowing the system to continuously track velocity based on Doppler frequency shifts. [2]

While radar-based chronographs offer excellent accuracy and ease of use, they are costly and rely on complex signal processing hardware, limiting user access to customization of raw data. PTC aims to explore an alternative approach by leveraging optical scanning techniques and real-time signal processing to achieve comparable spatial resolution at a lower cost. By using laser interruption timing and angular position mapping, PTC seeks to bridge the gap between basic optical chronographs and high–end radar systems.

2.3 Goals and Objectives

The PTC project is designed to extend traditional ballistic chronograph functionality by combining optical sensing, embedded computation, and user-friendly data presentation into a single system. The goals define the overall vision of the system and its intended capabilities, while the objectives describe measurable outcomes that guide implementation. The stretch goals are identified to encourage innovation beyond the minimum functional scope of the project.

2.3.1 Goals

Basic Goals:

- Develop an optical-based system capable of detecting a projectile as it passes through a defined measurement region.
- Accurately measure projectile velocity using precise timing and optical sensing techniques.
- Ensure the system operates reliably in indoor environments without dependence on ambient lighting conditions.
- Provide calculated results, including projectile velocity, to the user within 10 seconds of projectile detection.
- Present projectile-related information in an intuitive manner suitable for novice users.

Advanced Goals:

- Extend the system's measurement capabilities beyond velocity to include trajectory-related information, such as the direction of projectile travel.
- Display processed data on an integrated screen, enabling standalone operation without the need for external devices.
- Reconstruct and visualize a 3-dimensional representation of the projectile's trajectory.

Stretch Goals:

- Determine whether a projectile's trajectory would intersect a target with a one-meter diameter at a distance of ten meters.
- Integrate environmental sensing (such as temperature and humidity) to support improved ballistic analysis in outdoor environments.
- Incorporate atmospheric effects such as wind, temperature, humidity, and pressure into trajectory and offset calculations based on user-input target distance.
- Support a wide range of projectile velocities, from as low as 5 m/s to as high as 100 m/s.
- Determine and report the compass direction of projectile travel using onboard orientation and navigation sensors.

2.3.2 Objectives

Basic Objectives:

- Implement an optical detection system that does not require precise alignment by the end user.
- Capture projectile event data using optical sensors, cameras, or laser-based detection methods.
- Present results to the user via an integrated display with minimal user interaction required.
- Utilize a microcontroller such as a Raspberry Pi-based platform to coordinate sensing, timing, computation, and user interface tasks.

Advanced Objectives:

- Develop a calibration procedure that is intuitive, such as using known reference objects with the camera's field of view.
- Synchronize optical detection events with strobe lighting and camera/image acquisition to capture multiple images of the projectile in flight.
- Implement image processing and mathematical analysis to extract spatial information from captured images.

Stretch Objectives:

- Integrate a magnetometer to determine the compass orientation of the device and infer the compass direction of the projectile motion.
- Use an accelerometer to determine the vertical orientation of the system and calculate the projection elevation angle.
- Combine orientation, environmental, and optical data to improve trajectory estimation accuracy.

2.3.3 Projectile Tracking Chronograph: Features/Functionalities

The PTC is designed to provide a comprehensive set of features that extend beyond the traditional ballistic chronograph by incorporating various subsystems such as optical sensing, embedded computation, and spatial analysis. PTC implements an optical light detection system (light gate) that identifies a projectile as it enters the sensing region. By interrupting a modulated beam from a light source, the system generates high-speed timing events that are captured using hardware interrupts on the microcontroller. These events can be used to enable accurate time-of-flight measurements across a known distance, allowing for precise velocity calculations. The use of a modulated carrier frequency along with photodetector(s) that have a sufficient bandwidth will allow the system to operate in indoor and ambient lighting conditions. Additionally, PTC uses an optical chamber with synchronized strobe light illumination and camera imaging to capture multiple instances of the projectile as it is moving through the chamber. Once the initial velocity is obtained from the light gate, the microcontroller activates the strobe flashes and camera exposures. This in turn produces a single image containing multiple shadows of the projectile, each corresponding to a known interval of time. Image processing algorithms can then be executed on a Raspberry Pi to extract the spatial position data from the shadows of these images, allowing for the direction of the

projectile and its angular trajectory to be approximated.

This system also supports embedded ballistic and trajectory computation, combining the previous timing data and spatial measurements to calculate the projectile velocity, angular deviation, and estimated trajectory path through the measurement region. In our advanced and stretch goals, we would like to further enhance these calculations by incorporating other environmental data from other types of sensors. Integrating information from temperature, humidity, and pressure sensors can be used to further refine the ballistic calculations to increase the overall accuracy of the system by accounting for atmospheric effects. Furthermore, combining other sensors such as an accelerometer and magnetometer would allow the system to determine its own orientation. This capability would support other advanced features, like determining the compass heading of the projectile's travel path.

To enhance the usability of PTC, a user interface consisting of simple input controls and a display has been incorporated into the system. Processed ballistic results are to be presented to the user within seconds of the detection of the projectile. This interface is designed to be user-friendly while still providing meaningful data to the user. Collectively, these features make the PTC a powerful tool for projectile measurement and analysis.

2.4 Required Specifications

2.4.1 System Specifications

General System Specifications:

TABLE 2.1: Specification of System Components

Parameter	Description	Specification
Speed Measurement Error	The error of the speed detected of the projectile	$< 20\%$
System Response Time	The time from a projectile traveling through the system until information is displayed to the user	$< 5\,\text{s}$
Detectable Projectile Size	The diameter of a spherical projectile that can be detected and characterized	$15\,\text{mm}$ to $50\,\text{mm}$
Detectable speeds	The speed of a projectile that can be detected and characterized	$10\,\text{m/s}$ to $50\,\text{m/s}$
Angular Resolution Error	The error of the angle detected of the path of the projectile	$< \frac{1}{5}$ radian

Power Requirements	The power the system will draw on average over the course of 10 seconds	$< 300\,\text{W}$
Sensing Area	The area of the system that will be able to sense projectiles through it	$> 10\,\text{cm} \times 10\,\text{cm}$
System Size	Total size of the system	$< 3\,\text{m} \times 3\,\text{m} \times 3\,\text{m}$
System Weight	Total weight of the system	$< 50\,\text{pounds}$
Ambient Light Tolerance	The amount of ambient light the system can tolerate	$< 20\,\text{lumens/ft}^2$
Cost	The total cost to make the system	$< \$1000$

2.4.2 Component Specifications

Specific Component Specifications:

TABLE 2.2: Specification of Project Components

Component(s)	**Description**	**Specification**
Camera(s)	The sensor area, the quantum efficiency of detecting the photons in the wavelength of the strobe, and the resolution of the sensor	$\geq 6\,\text{mm}$ sensor width; 600–700 nm spectral sensitivity; 50–70% quantum efficiency @ 650 nm; 1600×1200–1920×1200 pixel target resolution; 200–500 fps target frame rate; $\geq 60\,\text{dB}$ dynamic range
Photodetector(s)	The responsivity of the photodiode in the wavelength of the emitting laser or LED wavelength, the noise floor of the photodiode, and the bandwidth of the photodiode	$> 100\,\text{kHz}$ bandwidth; 0.3–0.6 A/W @ 650 nm; 5–10 mm width; $\geq 20\,\text{dB}$ SNR during beam interruption
Gate light source(s)	The light source for the initial light gate for detecting the projectile entering into the system	635–670 nm; $\leq 1\,\mu\text{s}$ rise/fall time; 10–100 kHz carrier frequency; $\geq 80\%$ modulation depth; $\geq 100\,\text{mm}$ line length

Environmental Sensor(s)	Accuracy of the humidity, temperature, and pressure sensors	$< \pm 10\,\text{hPa}$ at $1000\,\text{hPa}$; $< \pm 5\%$ humidity at 30–80%; $< \pm 5°\text{C}$ at 10°C–50°C
Power Supply	Provides a regulated voltage and current to power the electronic devices	$\geq 50\,\text{A}$ continuous current; 12–48 V output; $\leq 2\%$ ripple; $\leq 5\,\text{ms}$ transient response recovery
Strobe Light Mechanism	Capabilities of the subsystem that creates the strobe flashes	$< 200\,\mu\text{s}$ flash time; $< 100\,\mu\text{s}$ turn-on time; $< 100\,\mu\text{s}$ turn-off time; $> 1000\,\text{lm}$ output
Physical System Structure	Physical layout of the system	$> 10\,\text{cm}$ sensing width/height; $< 2\,\text{m}$ sensing length; $< 3\,\text{m}$ height; $< 2\,\text{m}$ width; $< 3\,\text{m}$ length
User Interface	User interface specifications	$> 2\,\text{cm}$ screen size; $< 1\,\text{m}$ screen size
Microcontrollers and Computers	Microcontroller and computer specifications	$> 1\,\text{MHz}$ MCU clock speed; image processing capability for computer

2.4.3 House of Quality

In this section is a House of Quality showing the relationships (main body) between various customer and engineering requirements, as well as the correlations between those engineering specifications (roof). The relationships in the main body between customer and engineer have different weights depending on how closely one affects the other, with the strongest relationship having a weight of 7 and the weakest having a weight of 1. Each requirement also has a polarity, denoted by an up or down arrow, that displays the desirability said requirement. For example, "Affordability" would have a positive polarity because a cheaper product is a more desirable trait. In the roof of the House, the correlations between engineering tradeoffs can be seen. A positive correlation means that improving one engineering requirement also benefits another, while a negative correlation means that improving one engineering requirement comes at the detriment of another.

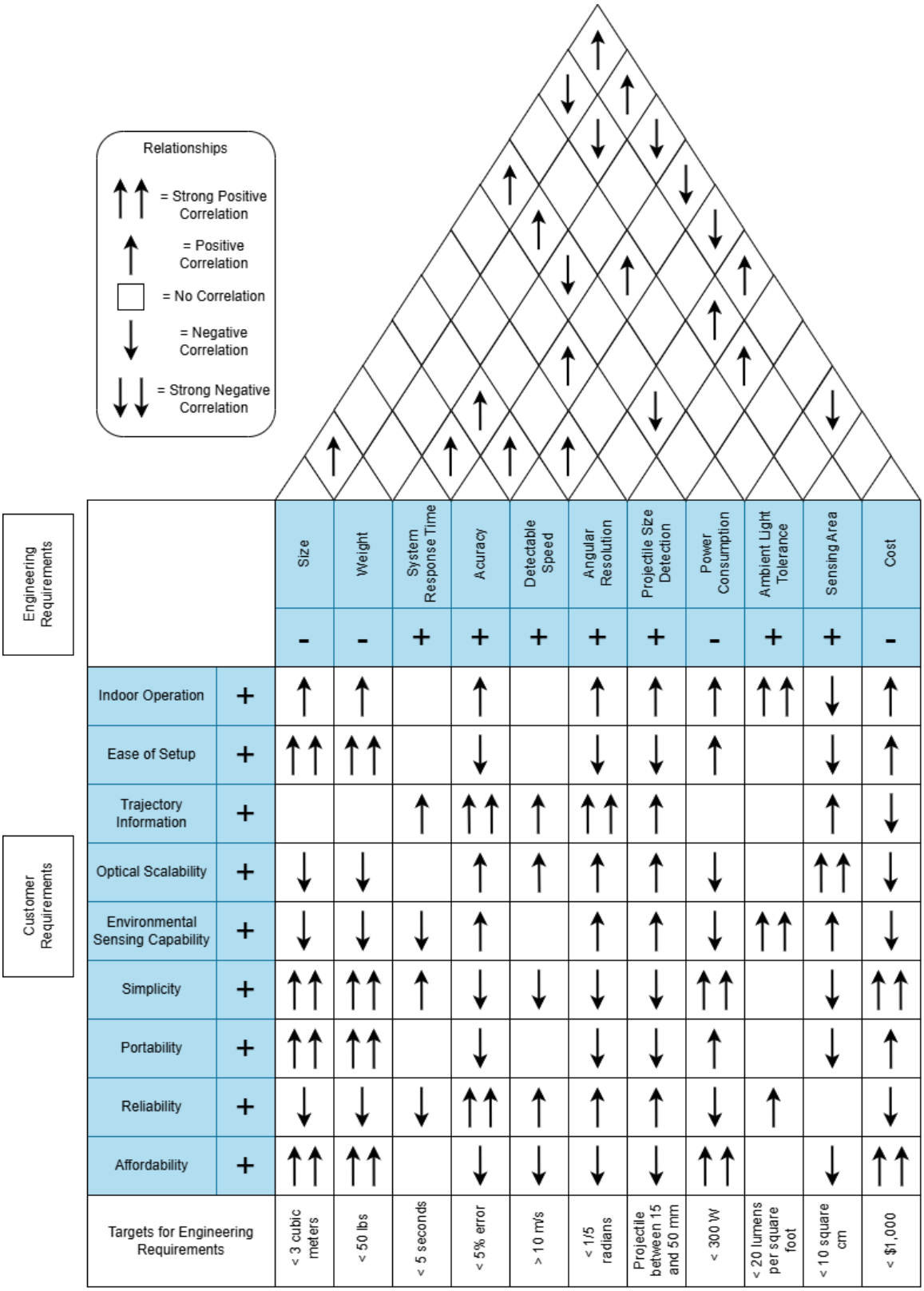


Fig. 2.1: House of Quality diagram that shows the tradeoffs between the engineering and customer requirements of the project. The customer requirements (left-most column) depict the features that are more important to the user, while the engineering requirements (top row) show the features that are considered for various engineering aspects of the project.

2.5 Optical Hardware

The system will consist of a light gate (or sensor gates) and an area for strobe photography of the projectile (optical chamber). The initial light gate will get a projectile speed that will determine the timing of the strobe flashes. The camera(s) will take one photo that contains multiple shadows of the projectile at the points in time where the strobe is triggered.

The cameras, lenses, and strobe LEDs depend on each other. The LEDs must be bright enough to drown out the ambient light leaking into the box in order to provide a good signal to the cameras. The lens system must be designed to capture enough light to provide a sufficient signal to show the projectile on the cameras. The LEDs will need to flash for a short enough time to avoid too much blurring of the projectile in the chamber; however, they should flash for a long enough time to provide sufficient light to activate the camera sensor enough to get an acceptable picture. A rough estimate of a camera that operates at 30 fps can comfortably film at typical bedroom levels of illumination, which is about 10 lumens/square foot. Given an approximate strobe area of 2 square feet, and about 20% illumination efficiency, that would be 100 lumens for the 1/30 s or 33 ms that a camera frame lasts. With a 100 microsecond strobe time, that's 330 times the brightness needed for an equivalent amount of light to hit the camera sensor during the 100 microsecond strobe, which gives at least 3000 lumens required from the strobe for a simple camera.

The initial light gate will be used to detect the projectile entering the system and its approximate speed. The photodiode must be of sufficient bandwidth to detect a fast-moving projectile (a paintball is about 17mm and travels at 90 m/s, which gives 180 microseconds of signal time) would require at least 5 kHz of bandwidth. A projectile that is 1/4 inch long and traveling at 1000 feet/s would require a 50 kHz bandwidth sensor to detect the projectile. In order to filter ambient light, a carrier wave of sufficiently higher bandwidth would be required, so 100 kHz would be the minimum for detecting a bullet.

In the following, the diagram elaborates on the design of the sensor gates and optical chamber responsible for detecting a projectile and taking a picture of the projectile as it moves through the enclosure. By interrupting the line of sight of the light source to its corresponding photosensor, a projectile can be detected for initial calculations to be made. As the projectile moves through the optical chamber, multiple cameras can be used to make further calculations that estimate position and trajectory.

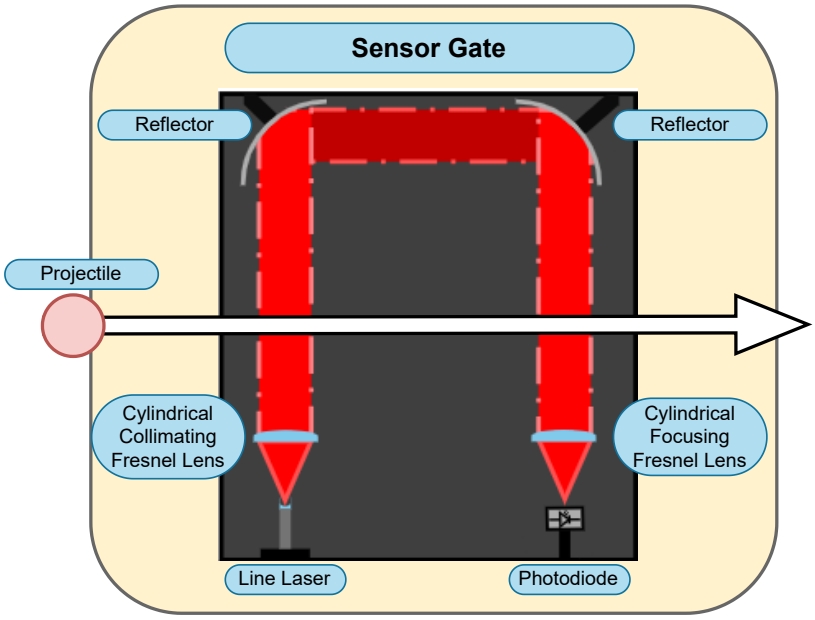


Fig. 2.2: Sensor Gate Design

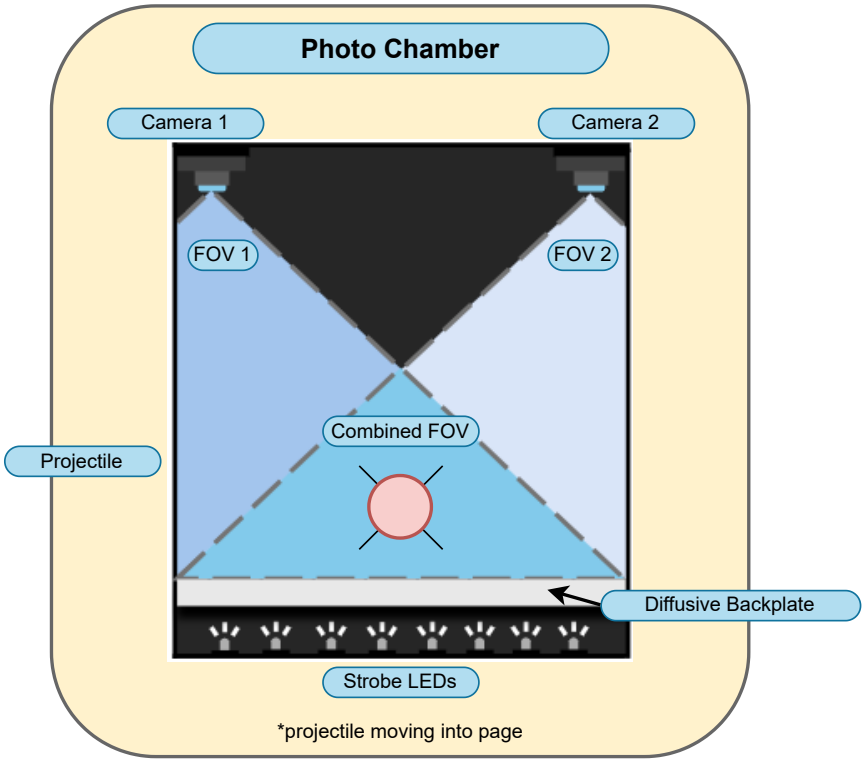


Fig. 2.3: Figures 1 and 2 elaborating on the sensor gates and optical chamber of the PTC system, illustrating the interaction between external projectile input and the optical system in place.

2.6 Electronics and Software

The hardware block diagram illustrates the necessary hardware and connections to facilitate our goals and objectives, as well as their current status at the time of this writing. Each block in the diagram is color-coded to indicate which team member will be primarily responsible for its research, design, and operation. The RP2350 MCU will be primarily responsible for handling sensor information and hardware-level interactions, whereas the Raspberry PI computer will be used for image processing of the projectile and to control the user interface. An optical sensor and chamber will be used to detect a projectile and take pictures, and these signals will be used by the RP2350 and the Raspberry PI to make calculations.

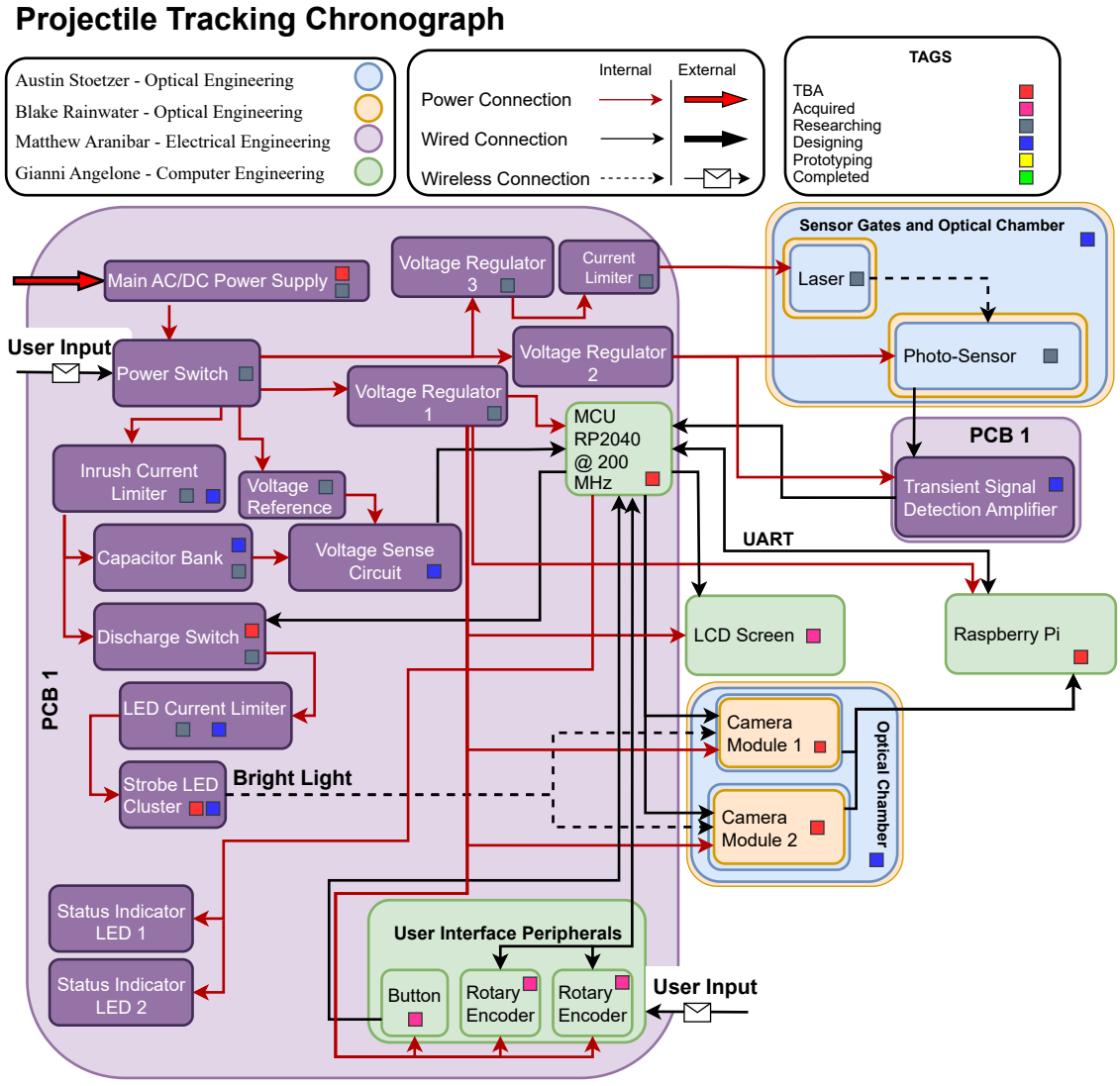


Fig. 2.4: Hardware Diagram of the PTC system that illustrates component connections, status, and division of workload.

The software flowchart begins with the system power-on and initialization of the Rasp-

berry Pi subsystems. After loading the configuration settings and performing the necessary calibrations, the system enters a charging state to prepare the capacitors for the strobe LED cluster. Once the capacitors are fully charged, the software transitions into an armed ”idle” mode that continuously monitors for a projectile detection signal from the sensor gate. Once a projectile is detected, the system will trigger synchronized strobe pulses and camera captures, followed by image processing to extract positional data from the projectile. The ballistic and trajectory calculations are then performed, with the results being displayed through the user interface.

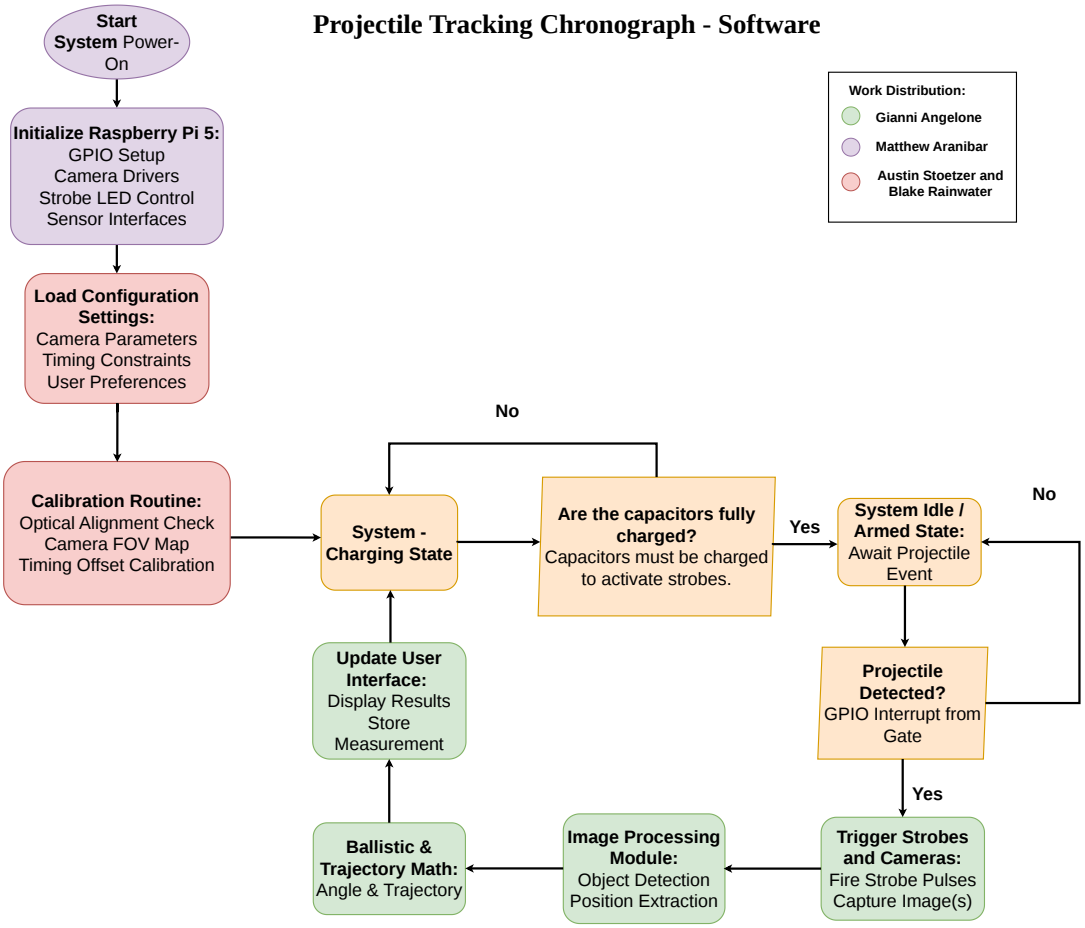


Fig. 2.5: The software flow of the Projectile Tracking Chronograph, detailing system initialization, projectile detection, data acquisition, processing, and reset to an idle state.

The data flowchart of the PTC system, details the flow of data from initial projectile detection to final system output. The diagram outlines how optical interruptions at the gates trigger hardware interrupts that prompts the MCU to synchronize the cameras and LED strobes, enabling the capture of multiple images of the projectile as it moves through the system. These images are then processed using computer vision techniques to extract positional data, allowing for the velocity and trajectory to be calculated in the ballistic and trajectory engine. The resulting data from these subsystems is then communicated to the user interface, where the results are displayed.

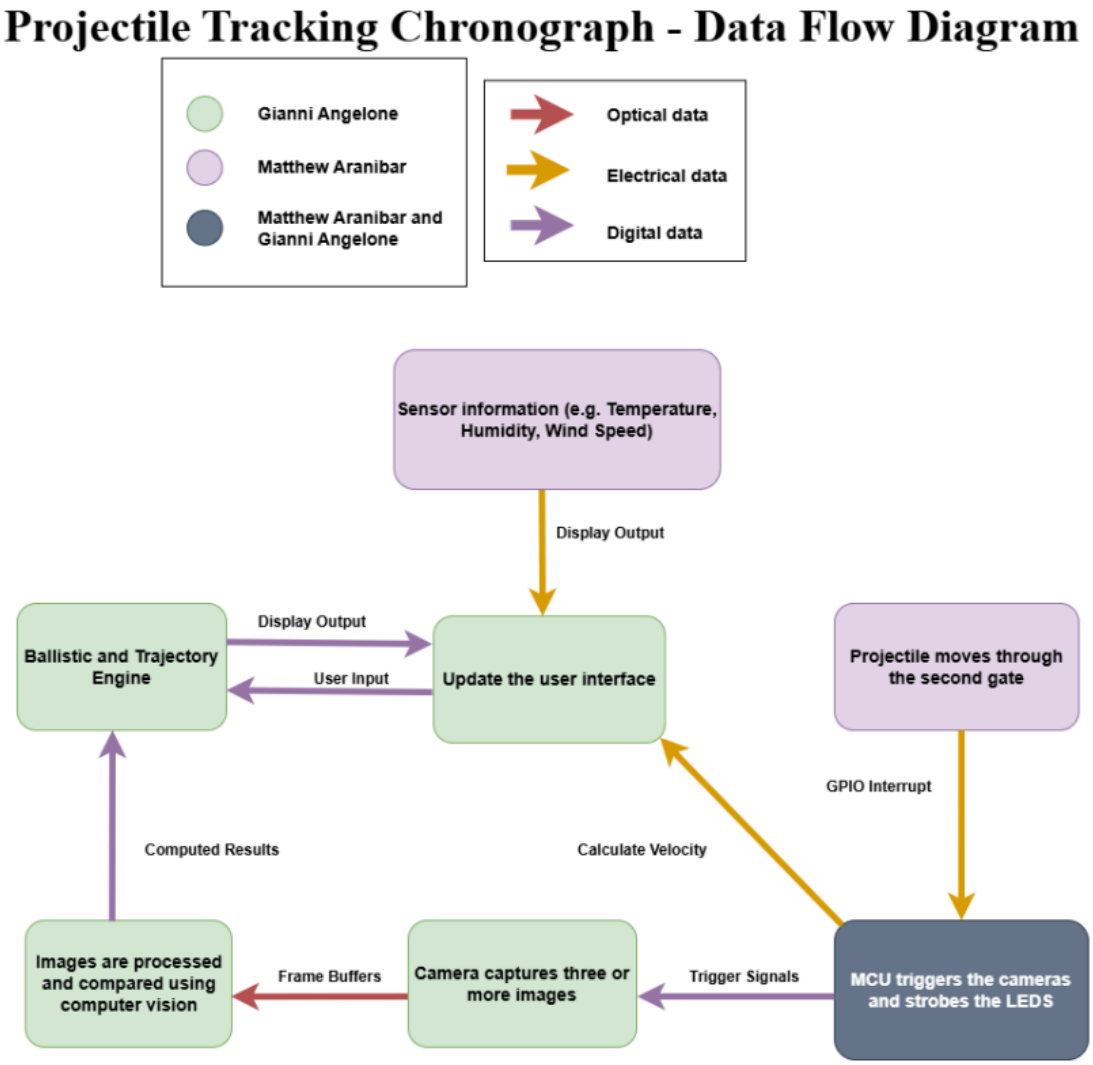


Fig. 2.6: Flowchart of the PTC system that illustrates the end-to-end data flow from optical event detection and hardware interfacing through synchronized strobe and cameras, image processing, and ballistic computation, outputting to the user via the user interface.

The illustrations below present the current prototype configurations developed for the PTC. These figures provide visual context for the mechanical layout of the various system components described in the previous sections. These sketches are intended to provide a clear framework for system functionality during development and testing.

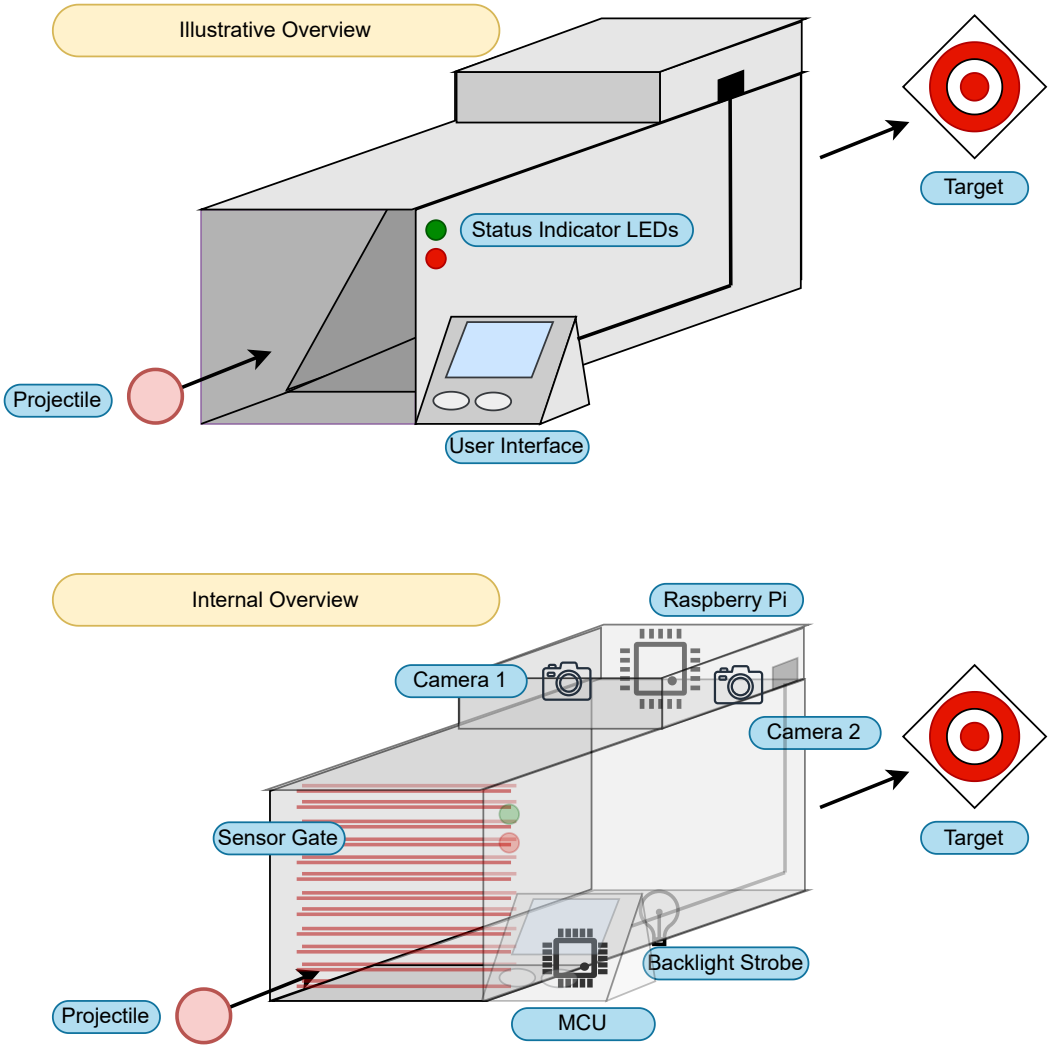


Fig. 2.7: Prototype illustrations detailing the various subsystems from different viewing points.

3 Research and Investigation

3.1 Illuminative Technology - Light Gate

3.1.1 Light Sources

3.1.1.1 LEDs

LEDs are semiconductor devices that are capable of emitting light through a process called electroluminescence. This happens when charge carriers (electrons and holes) recombine

within a semiconductor junction and produce photons. In an LED, this type of emission is mostly driven by spontaneous emission, which means that the emitted photons have a random direction, phase, and polarization upon emission. These types of light emitter lack an optical cavity as well, leading to an incoherent light source that has a broad spectral distribution. [3], [4]

Due to their broad emission angle, LEDs are useful for many applications that require wide or diffuse illumination. Examples of this include general lighting, displays, and indicators. LEDs generally operate with low to moderate efficiency and can achieve long operating lifetimes as well. These light sources typically have lower optical power density and broader spectral output compared to laser diodes, making them less suited for high-precision optical measurements. [3], [4]

3.1.1.2 Laser Diodes

Laser diodes, like LEDs, are semiconductor light emitters that make use of carrier recombination to produce photons. Unlike LEDs, however, laser diodes utilize stimulated emission within an optical resonant cavity to produce monochromatic light that is coherent and highly directional. In stimulated emission, an incoming photon causes the release of an identical photon in phase and direction. Since the optical cavities operate as a positive feedback loop, these photon interactions get amplified, leading to a narrow beam of light that is concentrated at a single wavelength. [3]

Due to these properties, laser diodes can generate significantly higher optical power, maintain a narrow spectral linewidth, and support fast modulation at timescales much shorter than LEDs. These characteristics make laser diodes ideal for applications such as communications, sensing, and high-speed measurement systems. Drawbacks of these components are that they typically tend to be much more complex, have higher costs, and require better thermal management techniques. [3]

3.1.1.3 Technology Comparison - Light Sources

For the PTC project, we want modulated illumination across the sensing region, enabling reliable projectile detection when the gate is disturbed by a passing object. This light source must maintain sufficient intensity and uniformity to ensure consistent triggering while also being resistant to variations in ambient lighting. Due to these conditions, the light source chosen for this subsystem was the laser diode.

The coherent and greatly directional output of laser diodes enables the design to illuminate specified detection zones with minimal spreading and high intensity. Additionally, the advantages of laser diodes will improve timing accuracy and reduce the susceptibility of the sensor gate to ambient lighting variations. In contrast, LEDs produce broad light that would require large optics or complex lens systems to achieve a similar power density, increasing overall system complexity without delivering equivalent precision to the laser diode.

TABLE 3.1: Illuminative Technology Comparison

Feature	**LED**	**Laser Diode**
Emission Type	Spontaneous	Stimulated
Coherence	Incoherent	Coherent
Spectral Width	Broad	Narrow
Beam Directionality	Broad	Very Directional
Optical Power Density	Lower	High
Modulation Capability	Lower	Higher
Thermal Sensitivity	Low	Higher
Cost	Low	Higher

3.1.2 Wavelengths (Light Gate)

3.1.2.1 Ultraviolet

Ultraviolet (UV) light occupies the portion of the electromagnetic spectrum with wavelengths between 100 and 400 nm. Due to its relatively short wavelength, UV light carriers higher photon energy compared to visible and infrared radiation. This higher energy enables strong interactions with biological compounds and photosensitive materials, making it very useful in applications for fluorescence analysis, photolithography, and sterilization. UV light can be divided into three main sub-categories: UV-A (315-400 nm), UV-B (280-315 nm), and UV-C (200-280 nm). [5], [6]

Although UV light is very useful, there are several challenges associated with using this light for optical sensing systems. For instance, many common optical materials and protective coatings absorb UV radiation, reducing the amount of light that can be sensed on photodetectors. Additionally, common silicon photodiodes have low sensitivity in the UV region because much of the incident light is absorbed by the protective window materials or the surface of the detector itself before photoelectric conversion can take place. These limitations can significantly increase the difficulty of trying to use ultraviolet detection for optical sensing without the use of specialized optical components. [7], [8]

3.1.2.2 Infrared

Infrared (IR) light consists of wavelengths longer than visible light, ranging from 700 nm all the way to 1 mm. In optical sensing systems, the near-IR region (700-1000 nm) is most commonly used because it can be easily generated by semiconductor emitters and efficiently detected by silicon photodiodes. Infrared emitters are used for a wide range of applications, such as remote control, night-vision imaging, and proximity sensors. [9]

One advantage of IR illumination is its ability to penetrate materials and scattering media more effectively than shorter wavelengths. Silicon photodiodes are able to maintain high responsivity throughout much of the near-IR region, often reaching peak sensitivity between 800 nm and 900 nm. This makes infrared illumination particularly viable for optical sensing systems that use photodiode detection. However, one drawback of IR

light is that it is invisible to the human eye, making beam alignment and system testing particularly difficult. [8]

3.1.2.3 Visible (Red)

Visible light spans wavelengths from approximately 400 to 700 nm, corresponding to colors that are perceptible to the human eye. The red portion of the spectrum (about 620-700 nm) is particularly useful for optical sensing systems due to its compatibility with silicon photodetectors. Red light sources are frequently used in optical sensors, alignment tools, and various measurement systems because they provide a good balance between practical usability and detector sensitivity. [5]

Silicon photodetectors exhibit strong responsivity across the visible spectrum, especially the red region, allowing efficient conversion of photons into electrical signals. Furthermore, visible red light offers practical advantages during system development and operation, since the light can be easily observed by users. This visibility simplifies beam alignment and calibration, making troubleshooting processes much simpler compared to infrared systems. For these reasons, red wavelengths are commonly selected for optical sensing applications that require reliable detection and simpler system setups. However, one limitation of visible light is its sensitivity to ambient light. Since many common indoor lighting sources also emit strongly within the visible spectrum, background light can introduce noise and reduce the signal-to-noise ratio of optical detectors. [8]

3.1.2.4 Technology Comparison - Wavelengths (Light Gate)

Several factors are to be considered when choosing the wavelength for our light source, such as detector sensitivity, ambient light susceptibility, alignment convenience, and overall system complexity. UV light offers high photon energy and strong material interactions, but often requires specialized optical components due to factors such as reduced detector sensitivity and material absorption. IR light provides strong compatibility with many photodetectors and is widely used in sensing systems, but its invisibility to the human eye can make system alignment more difficult. [7], [9]

Visible red light provides many of the advantages of infrared illumination and adds more as well. Silicon photodiodes maintain high responsivity in this region, and the emitted light is visible, allowing simple system alignment and calibration. Due to these factors, the visible red wavelength was chosen as the primary wavelength for the sensor gate subsystem.

TABLE 3.2: Wavelength Technology Comparison

Feature	**UV**	**Infrared**	**Visible (Red)**
Typical Wavelength Range	100–400 nm	700 nm – 1 mm	620–700 nm
Photon Energy	High	Low	Moderate
Detector Compatibility	Lower for standard Si detectors	High for Si detectors	High for Si detectors

Visibility to Human Eye	No	No	Yes
Common Applications	Sterilization, fluorescence, photolithography	Remote sensing, night vision, proximity sensors	Optical sensing, alignment systems
Alignment Convenience	Difficult	Difficult	Easy
System Complexity	Higher	Moderate	Lower

3.1.3 Types of Lasers

3.1.3.1 Point Laser

A point laser is a light source that emits a highly focused, collimated beam concentrated into a small spot. This type of laser typically produces a Gaussian beam profile, where the intensity is highest at the center of beam and decreases radially outward. Due to its high optical power density and narrow beam divergence, a point laser is useful in applications requiring precise alignment, localized illumination, or high-accuracy targeting. Examples of this can be found in systems such as interferometers, laser cutting tools, and laser-targeting technologies. [10]

The high coherence nature of a point laser allows for extremely accurate spatial resolution, making it particularly useful in systems where only a single point in space being disturbed is desired. However, it is this same property that limits the point laser in applications that require coverage over a large sensing region. For applications like this, additional optical components will be required to expand the effective detection area, increasing overall system complexity and costs. [10]

3.1.3.2 Line Laser

A line laser is a laser source that has been modified to project the laser beam in a line rather than a point. This is typically achieved through the use of optical components, such as cylindrical lenses or other diffractive elements. These optics spread the beam in one dimension while maintaining collimation in the perpendicular direction. Due to this modification, the emitted light forms a continuous line across a surface, providing a much larger area of illumination than that of a point laser. [11]

These types of laser are commonly used in applications such as alignment systems, dimensional measurement systems, and machine vision technologies. Line lasers are particularly beneficial in systems where you want to illuminate an entire cross-sectional region simultaneously. However, since the optical power is distributed over a larger area, the intensity at any given point across the line is lower than that of a point laser, which can increase the need for more robust sensing in some systems. [11]

3.1.3.3 Technology Comparison - Types of Lasers

When selecting between point and line lasers, the primary consideration for the PTC project is the tradeoff between spatial coverage and optical intensity. The sensing system benefits from illumination across a defined detection region rather than a single point. A line laser enables the creation of an optical ”gate” that can be interrupted by a passing projectile. While the reduced intensity of a line laser compared to a point laser is a consideration, the benefits provided by the increased coverage of a line laser outweigh the drawbacks.

TABLE 3.3: Laser Type Comparison

Feature	**Point Laser**	**Line Laser**
Beam Shape	Focused point	Expanded line
Optical Power Density	High	Lower (distributed)
Coverage Area	Small	Large (1D spread)
Alignment Sensitivity	High	Lower
Need for Scanning	Often required for coverage	Not required
System Complexity	Higher (for large coverage)	Lower (built-in coverage)

3.1.4 Component Selection - Laser Diodes

3.1.4.1 Adafruit 1057 Laser Diode

The Adafruit 1057 laser diode module is a compact red line laser designed for low-power optical applications. This device operates at a wavelength of approximately 650 nm and provides an optical output power of up to 5 mW. It is designed to operate over an input voltage range of 2.8-5.2 V, with a maximum current draw of 25 mA. This is a Class IIIR laser that is made to have direct compatibility with low-voltage embedded systems without the need for complex driver circuitry. This module incorporates an integrated cylindrical lens that converts the emitted beam into a line pattern with a wide fan angle of approximately 120°, allowing for broad area coverage. The physical dimensions of this device are 10 mm in diameter and 32.8 mm in length, with wire length of 100 mm. This makes the module very easy to mount and integrate into compact systems. While this module provides low power consumption and overall simplicity, it’s performance is primarily suited for general-purpose sensing tasks. The datasheet for this particular device is limited in optical specifications, indicating that it is not optimized for high-precision measurement applications. [12]

3.1.4.2 Quarton Inc. VLM-650-23 LPO

The VLM-650-23 LPO is a miniaturized red line laser diode designed for precision alignment and sensing applications. This device operates in a wavelength range of about 640-655 nm and produces a line-shaped beam using a cylindrical lens system. It is designed to operate at an input voltage of 3-6 V with a maximum operating current of

50 mA. The module produces a total optical power of less than 5 mW, while the accessible laser output is limited to less than 1 mW, classifying this diode as a Class 1 laser device. The optical system consists of a collimating glass lens and a cylindrical quartz line lens, producing a line beam with a fan angle of $50° \pm 5°$. The resulting line has a width of less than 1.5 mm at a distance of 2.5 meters and a line accuracy of approximately 4/1000, which is a deviation of less than 1.6 mm at 400 mm. Mechanically, this device has a cylindrical diameter of 5 mm and a length of 22.8 mm, with an approximate mass of 6 g. It operates in a temperature range of -20°C to $+65$°C and includes various safety features such as surge protection and reverse polarity features. This device has suitable characteristics for sensing applications, but can be a little restricted due to its smaller fan angle. [13]

3.1.4.3 Quarton Inc. VLM-650-27 LPT-10

The VLM-650-27 LPT-10 is an industrial red line laser module designed for applications requiring higher optical output and a large coverage area. This device operates in a wavelength range of 645-655 nm and requires an input voltage of 2.6-6 V. The VLM-650-27 LPT-10 is able to support two output regimens, with the first being a low-power configuration with an optical output of less than 2 mW and the second being a higher-power configuration producing 12-15 mW. These two modes have operating currents of below 35 mA and below 100 mA respectively. For this project, the low-power configuration was considered as this is a safer laser being categorized as a Class 1 laser device. The optical system consists of an ashperical plastic collimating lens and a quartz cylindrical lens, producing a line-shaped beam with a fan angle of $90°$. The projected line has a width of 3 ± 0.5 mm at 5 m with an accuracy of approximately ± 1 mm at 5 m. Physically, this device has a cylindrical brass housing with a diameter of 12.5 mm and a length of 30 mm and operates in a temperature range of $+10°$ to $+40$°C. This device is intended for general industrial alignment applications, with a recommended working distance of more than 2 meters on average. This device has a fairly moderate fan angle, making it better suited for more compact systems. [14]

3.1.4.4 Quarton Inc. VLM-635-55 LPO

The VLM-635-55 LPO is an industrial-grade red line laser module that is optimized for precision alignment and high visibility. This device operates in a wavelength range of 630-645 nm and requires an input voltage of 3-6 V, with a maximum operating current of less than 80 mA. This module produces a total optical output power of 4-5 mW, while the accessible laser output is limited to less than 0.39 mW, making this a Class 1M laser device. The optical system uses both a glass collimating lens and a glass line lens to produce a uniform, line-shaped beam. Similar to the VLM-650-27 LPT-10, the VLM-635-55 LPO generates a line that has a width of 3 ± 0.5 mm at 5 m with an accuracy of approximately ± 1 mm at 5 m. This laser has a fan angle of $60°$, which is less than that of the VLM-650-27 PLT-10. This laser module features an aluminum housing with a diameter of 10 mm and a length of 38 mm, with a mass of about 6 g. It operates in a wide range of temperatures, from -20°C to $+65$°C and can be stored in temperatures up to $+85$°C. This laser was designed for precision alignment applications as well, but with a bigger working distance of up to 20 meters. Like the VLM-650-23 LPO, the fan

angle is a limiting factor due to its smaller beam spread over short distances. [15]

3.1.4.5 Component Comparison - Laser Diodes

The Adafruit 1057 module emphasizes simplicity, low power consumption, and a very wide fan angle, but offers limited optical precision characterization compared to the industrial-grade alternatives. The VLM-650-23 LPO provides a balanced combination of compactness, low power operation, and a well-defined line being generated, making it well-suited for applications requiring greater optical control. The VLM-650-27 LPT-10, on the other hand, delivers higher optical power and wider coverage, improving performance in more compact systems and in environments with greater amounts of ambient light. One drawback of these improvements though is the increased cost and and power demand required by this laser. The VLM-635-55 LPO offers enhanced visibility due to its shorter operating wavelength and higher brightness, but also has the same disadvantages as the VLM-650-27.

Cost further differentiates these lasers diodes. Listed on the DigiKey website, the Adafruit 1057 is the cheapest, being prices at $8.95, whereas the Quarton Inc. VLM-650-23 LPO, VLM-650-27 PLT-10, and VLM-635-55 LPO are significantly more expensive at $79.40, $75.16, and $83.22, respectively. Although the Quarton Inc. modules provide improved optical performance, the increased cost is not justifiable for a system focused on reliable projectile detection rather than high-precision measurements. For the PTC project, the primary objective of the sensor gate is to achieve consistent detection across a defined sensing region while maintaining a compact, low-power design while also being cost-effective. The Adafruit 1057 meets our requirements by providing a very wide fan angle ($120°$) for sufficient coverage in a compact system, while also having adequate optical output power.

TABLE 3.4: Laser Diode Component Comparison

Characteristic	**Adafruit 1057**	**VLM-650-23 LPO**	**VLM-650-27 LPT-10**	**VLM-635-55 LPO**
Wavelength (nm)	∼650	640–655	645–655	630–645
Optical Output Power (mW)	≤ 5	< 5 (total), < 1 (accessible)	< 2 (low power), 12–15 (high power)	4–5 (total), < 0.39 (accessible)
Laser Class	Class IIIR	Class I	Class I (low power mode)	Class 1M
Operating Voltage (V)	2.8–5.2	3–6	2.6–6	3–6
Operating Current (mA)	≤ 25	≤ 50	< 35 (low) < 100 (high)	< 80
Beam Shape	Line	Line	Line	Line
Fan Angle (°)	~ 120	50 ± 5	~ 90	60
Line Width	N/A	< 1.5 mm @ 2.5 m	3 ± 0.5 mm @ 5 m	3 ± 0.5 mm @ 5 m

Line Accuracy	N/A	$\sim 4/1000$ (< 1.6 mm @ 400 mm)	± 1 mm @ 5 m	± 1 mm @ 5 m
Dimensions (mm)	$\phi 10 \times 32.8$	$\phi 5 \times 22.8$	$\phi 12.5 \times 30$	$\phi 10 \times 38$
Mass (g)	N/A	~ 6	N/A	~ 6
Operating Temperature (°C)	N/A	-20 to $+65$	$+10$ to $+40$	-20 to $+65$
ESD Protection	N/A	Yes	30 kV	30 kV
Recommended Distance	N/A	N/A	> 2 m	1–20 m
Cost ($)	8.95	79.40	75.16	83.22

3.2 Internal Sensor Technology

3.2.1 Photodetectors (Internal)

3.2.1.1 Photodiodes

Photodiodes are semiconductor devices that convert incident light into an electrical current through the generation of electron-hole pairs within a p-n (or p-i-n) junction. Photodiodes are commonly operated in a reverse-bias mode, meaning that the diode provides a high resistive path to the flow of current. When photons are absorbed in the depletion region, charge carriers are generated and flow across the junction under a reverse-bias, thus producing a photocurrent proportional to the incident optical power. Operation in this mode allows for improved response speed and reduced junction capacitance, making them suitable for high-speed detection tasks.

Silicon photodiodes have strong responsivity across the visble red and near-IR spectrum, with peak responsivities typically ocurring between 600-900 nm. At wavelengths near our desired wavelength of 650 nm, responsivity values typically range from 0.4 to 0.6 A/W. These devices are very desirable to precision timing applications due to their very fast response times, often in the nano- to microsecond range. They are able to respond this quickly due to their low capacitance and lack of internal gain mechanisms, however, this also results in a naturally small photocurrent being generated from the device. This necessitates the inclusion of transimpedance amplifiers (TIA) or other gain amplification circuits in order to generate a usable signal. [7]

3.2.1.2 Phototransistors

Phototransistors are another light-sensitive semiconductor device like the photodiode. However, these photodetectors operate more similarly to a bipolar junction transistor, with the incident light acting as the base current. When photons are absorbed in the base-collector junction, a small photocurrent is generated and then amplified by the current gain of the transistor. This internal gain results in significantly higher output current

compared to photodiodes, often reaching micro- to milliampere ranges under similar lighting conditions. [16]

The increased sensitivity of phototransistors allows for the detection of much lower light levels without the need of additional amplification circuitry. One drawback of this, however, is that the internal amplification often leads to slower response times due to the charge storage effects within the device. Additionally, phototransistors tend to exhibit reduced linearity, greater sensitivity to thermal fluctuations, and have increased noise compared to photodiodes. As a result, phototransistors are commonly used in applications where sensitivity in low-light situations are more important than speed and precision. [16]

3.2.1.3 Technology Comparison - Photodetectors

When comparing photodiodes and phototransistors, several key tradeoffs are evident, including response speed, sensitivity, and circuit complexity. Photodiodes provide a much higher response speed and low noise, with the drawback being the necessity for additonal amplification circuitry. The phototransistor offers much larger output signals compared to photodiodes, at the cost of increased noise and lower response speeds. For the PTC project, the primary requirement for the photodetector subsystem is to detect rapid interruptions of the optical signal as a projectile passes through the sensing region. Therefore, we believe that photodiodes are the more suitable choice for this system as they provide the necessary speed and precision required to meet our goals for projectile detection.

TABLE 3.5: Photodetector Technology Comparison

Characteristic	**Photodiode**	**Phototransistor**
Operating Principle	Photon-generated carriers in reverse-biased junction	Photon-generated base current amplified by transistor gain
Internal Gain	None	High ($\beta \approx 50$–300)
Output Current Range	nA to μA	μA to mA
Responsivity @ 650 nm	~ 0.4–0.6 A/W	Gain-dependent
Response Time	ns to μs	μs to ms
Bandwidth	High (MHz range)	Low (kHz range)
Linearity	High	Nonlinear
Noise	Low	Higher
Sensitivity	Moderate	High
Saturation Behavior	Minimal	Susceptible at high illumination
Temperature Sensitivity	Lower	Higher
Circuit Complexity	Requires amplifier	Simple
Power Consumption	Low	Moderate

3.2.2 Component Selection - Photodiodes

3.2.2.1 PDB-C142

The PDB-C142 is a silicon PIN photodiode designed for general visible and near-IR light detection applications. This device operates over a wavelength range of approximately 400-1100 nm, with a responsivity around of $\approx 0.53 \frac{A}{W}$ at 650 nm. This photodiode exhibits a relatively fast response time of around 50 ns under a 10 V reverse bias with a 1 $k\Omega$ load, making suitable for moderate-to-high speed sensing applications. This device supports a maximum reverse voltage of up to 100 V and exhibits a low dark current that is on the order of nanoamperes (5-30 nA), with a junction capacitance in the range of 18-25 pF. Additionally, this device exhibits a shunt resistance of approximately 100-500 MΩ, indicating low leakage characteristics. Under standard illumination conditions, this photodiode produces a short-circuit current of approximately 100-150 μA and has a noise equivalent power (NEP) of $2 \times 10^{-14} \frac{W}{\sqrt{Hz}}$ at the 650 nm wavelength. [17]

This photodiode is packaged in a 5 mm (T1 $\frac{3}{4}$) through-hole package with a clear lens. The active area of this device is approximately 4.12 mm^2 with a viewing angle of about 40°. Larger active areas allow for improved light collection efficiency, which allows for the alignment tolerances of the application to be less strict, but the small viewing angle can increase difficulty in alignment. The device operates over a temperature range of -40°C to $+80$°C, making well suited for applications both indoors and outdoors. While this photodiode offers a good balance between speed and sensitivity, but its relatively higher capacitance, compared to smaller surface-mount devices, limits its performance in high frequency applications. [17]

3.2.2.2 PDB-C171SM

The PDB-C171SM is silicon PIN diode designed for surface-mount optical sensing applications in compact systems. This device operates over a spectral range of approximately 350-1100 nm, with a stronger sensitivity in the red and near-IR regions. Specifically, this photodiode has a peak responsivity of 0.68 $\frac{A}{W}$ at 940 nm, with responsivities of 0.4 $\frac{A}{W}$ and 0.3 $\frac{A}{W}$ at 650 nm and 1050 nm respectively. This diode exhibits a typical rise time of 100 ns with a junction capacitance of 25 pF at a 3 V bias, supporting moderate-to-high speed detection. The PDB-C171SM produces a short-circuit current of 50 μA and NEP of $4 \times 10^{-14} \frac{W}{\sqrt{Hz}}$ at the 950 nm wavelength. The dark current was characterized to be 2-30 nA with a maximum reverse voltage of 60 V. [18]

This photodiode features a ceramic surface-mount package with an active area of about 7.5 mm^2 and a wide viewing angle of approximately 130°. Similar to the PDB-C142, this device operates over a temperature range of -40°C to $+80$°C. Compared to through-hole alternatives, this photodiode has a strong optical sensitivity with reduced parasitic effects. Overall, this device has a good balance between a large active area, good sensitivity, and reliable system integration, making it a good fit for compact sensing systems that require reliable detection like the PTC. [18]

3.2.2.3 PDB-C152SM

Like the PDB-C171SM, the PDB-C152SM is another silicon PIN photodiode designed for compact surface-mount applications needing reliable optical detection. This device operates over a wavelength range of about 400-1100 nm, with a peak responsivity of $\approx 0.52 \frac{A}{W}$ at 900 nm. At the wavelength of our selected laser diode (650 nm), the responsivity of the photodiode is $\approx 0.33 \frac{A}{W}$, which is not that great compared to other photodiodes at this wavelength. This photodiode has a typical response time of 50 ns with a junction capacitance of of 15-20 pF at a 10 V bias, which supports moderate speed operation. The short-circuit current of this device is in the range of 8-10 μA and the NEP is $1.5 \times 10^{-13} \frac{W}{\sqrt{Hz}}$ at the peak responsivity. The PDB-C152SM has a typical dark current of 2 nA, with a maximum of 10 nA and a breakdown voltage in the range of 50-100 V. [19]

The device is packaged in a compact surface-mount form factor (1206) with an active area of approximately 1.66 mm^2 and a viewing angle of $120°$. This photodiode has a very small active area, greatly reducing the light collecting capabilities compared to larger photodiodes. This device, like the others before, can be operated in the temperature range of approximately $-40°$C to $+80°$C. While this device offers fast response times and compactness, the reduced active area of this device makes it more suitable for applications that require wide-area sensing as opposed to a focused point. [19]

3.2.2.4 SFH 2713

The SFH 2713 is a high-speed silicon PIN photodiode designed for precise optical detection specifically in the visible spectral range. Specifically, the SFH 2713 operates at a wavelength range of 400-660 nm, with a peak sensitivity at 635 nm. This device exhibits very fast response characteristics, with a rise time of 22 ns and fall time of 31 ns (at a measured wavelength of 530 nm). The junction capacitance is approximately 50 pF and supports a maximum reverse voltage of 16 V. At a forward voltage of 5 V, the generated photocurrent of this device is 0.95 μA. The dark current is very low (around 100 pA), making the SNR of this device very high in low-light environments. [20]

This photodiode is packaged in a 2-SMD compact surface-mount configuration with a moderate sensing area of 3.27 mm^2. The SFH 2713 has a viewing angle of approximately $126°$, which allows for a higher light collection yield despite the smaller active area compared to the other photodiodes listed above. This device operates at the temperature range of about $-40°$C to $+85°$C, allowing this device to be operated in slightly higher temperature conditions. While the SFH 2173 offers the fastest speed and reduced noise characteristics, the reduced active area makes it slightly harder to align. Ultimately, this photodiode is still usable for this project due to its high speed sensing in the visible wavelength range. [20]

3.2.2.5 Component Comparison - Photodiodes

The PDB-C142 photodiode has a moderate active area with a through-hole package, making it more suited for applications where the alignment tolerance can be more lax.

However, the larger junction capacitance limits its performance in higher-speed detection circuits. Both the PDB-C171SM and PDB-C152SM provide improved integration capabilities through their surface mount designs, enabling them to be easier to use in compact system layouts. Between the two, the PDB-C171SM offers a favorable combination of sensitivity and large sensing area, making it the more versatile choice in general detection applications. The SFH 2713 stands out in terms of speed and overall performance, with very low junction capacitance and fast response characteristics. However, one major issue with this photodiode is the very small viewing angle, making alignment especially difficult. The cost of the photodiodes is another factor that has great consideration. On the DigiKey website, the costs of one for these photodiodes are as follows:

- **PDB-C142**: $3.32
- **PDB-C171SM**: $4.49
- **PDB-C152SM**: $1.96
- **SFH 2713**: $1.86

For the PTC project, the primary objective of the sensor system is to achieve reliable and repeatable detection. Overall, we have decided that the PDB-171SM provides the best compromise between size and performance. Its surface-mount packaging allows for decent PCB integration, while still maintaining sufficient sensitivity. We decided that one of the most important factors for consideration was sensing area, as it greatly increases the light collection capabilities of our system. Additionally, the moderate junction capacitance and low dark current are able to support stable and repeatable detection measurements without complex amplification circuits. Even though this was the most expensive photodiode, the PDB-171SM was selected as the optimal photodiode for the PTC project.

TABLE 3.6: Photodiode Component Comparison

Characteristic	**PDB-C142**	**PDB-C171SM**	**PDB-C152SM**	**SFH 2713**
Spectral Range (nm)	400–1100	350–1100	400–1100	400–660
Peak Wavelength (nm)	$\sim$ 650–660	940	900	635
Responsivity (A/W)	$\sim$ 0.53 @ 650 nm	0.68 @ 940 nm; 0.4 @ 650 nm	$\sim$ 0.52 @ 900 nm; 0.33 @ 650 nm	N/A
Active Area (mm^2)	4.12	$\sim$ 7.5	1.66	3.27
Response Time (ns)	$\sim$ 50	$\sim$ 100	$\sim$ 50	22 (rise), 31 (fall)
Junction Capacitance (pF)	18–25 @ 10 V	25 @ 3 V	15–20 @ 10 V	$\sim$ 50
Dark Current (nA)	5–30	2–30	2–10	0.1-5

Max Reverse Voltage (V)	100	60	50–100	16
Breakdown Voltage (V)	15–25	N/A	50–100	N/A
Shunt Resistance (MΩ)	100–500	N/A	N/A	N/A
Short-Circuit Current (μA)	100–150	~ 50	8–10	0.95
NEP ($W/\sqrt{Hz}$)	2×10^{-14}	4×10^{-14}	1.5×10^{-13}	N/A
Viewing Angle (°)	~ 40	~ 130	~ 120	~ 126
Operating Temperature (°C)	-40 to $+80$	-40 to $+80$	-40 to $+80$	-40 to $+85$
Cost ($)	3.32	4.49	1.96	1.86
Light Collection Efficiency	High	High	Low	Moderate

3.3 Reflector Technology

3.3.1 Types of Reflectors

3.3.1.1 Flat Mirrors

Flat mirrors are reflective surfaces where, for an incoming beam, the angle of incidence equals the angle of reflection. These mirrors produce virtual images that appear to be located behind the reflective surface at a distance of equal length to the object in front of them. Flat mirrors maintain the size and orientation of the reflected images, making them useful for applications requiring very accurate undistorted reflections. However, these types of reflectors provide little to no focusing capability, which limits their utility for applications requiring beam collimation or focusing. [21]

3.3.1.2 Diffuse Reflectors

Diffuse reflectors scatter incident light in many directions. This scattering occurs due to the microscopic irregularities on the reflector's surface, causing the light rays to reflect at various angles. These types of reflectors are useful in applications requiring uniform illumination such as display backlighting or photography lighting. These types of reflectors are unable to concentrate optical energy onto a focal point due to their scattering nature, making them unsuitable for applications where beam collimation or focusing is important. [21]

3.3.1.3 Retroreflectors

Retroreflectors are optical devices that are designed to reflect incident light from an oncoming beam directly back toward the source, regardless of the angle of incidence. Devices such as corner cubes or arrays of glass beads that make use of the laws of

internal reflection and refraction are used to achieve this effect. These types of reflectors are commonly found in technologies where you want to spot the reflector, such as bicycle reflectors and road signs. While these reflectors are very effective at returning light to its origin, they do not focus light onto a separate focal point, limiting its use in application requiring light collimation or focusing. [21]

3.3.1.4 Parabolic Reflectors

Parabolic reflectors are surfaces that use a parabolic shape to ensure that all of the wave traces in a reflected beam light runs parallel to each other. This means that the parabolic reflector is able to transform an incoming plane wave traveling parallel to the optical axis into a spherical wave converging at the focal point. Conversely, this also means that a point source placed at the focus produces a collimated parallel beam upon reflection. This makes this type of reflector very useful in technologies such as the receiving dishes of radio telescopes or car headlights. [22], [23]

These types of reflectors can be configured in both on-axis and off-axis configurations. On-axis mirrors have the focal point located along the axis of symmetry, while off-axis parabolic mirrors are sections from a larger paraboloid. These off-axis configurations are able to position the focal point away from the incoming beam path, eliminating obstruction of the incident light by detector hardware placed at the focus. These types of reflectors are very useful for applications where the collimation and de-collimation of light is important. [21]

3.3.1.5 Ellipsoidal Reflectors

Ellipsoidal reflectors are surfaces shaped as an ellipsoid of revolution. These reflectors are characterized by their unique property of having two focal points instead of just one. Light originating from one focal point is reflected by the reflector, and this reflected light then converges at the second focal point. This property makes this reflector useful in application requiring the transfer of light from one point to another, such as medical illumination systems. However, these types of reflectors cannot collimate light into a parallel beam, requiring external optical components to achieve this effect. [21]

3.3.1.6 Technology Comparison - Types of Reflectors

When comparing these different reflector types for the PTC system, the primary requirements are the ability to reflect a sufficient amount of light and concentrate it efficiently onto a photodetector. Flat mirrors, diffuse reflectors, and retroreflectors lack the focusing capability that is required for the design of the system. Ellipsoidal reflectors are able to transfer the light between two points, but ultimately fails at collimating or focusing parallel rays from extended sources. Parabolic reflectors uniquely combine the ability to both collimate light from a point source into a parallel beam while also being able to focus that same light back down to a single focal point. This bidirectional focusing property makes parabolic reflectors the optimal choice for the PTC system.

TABLE 3.7: Reflector Technology Comparison

Characteristic	**Flat Mirror**	**Diffuse Reflector**	**Retroreflector**	**Parabolic Reflector**	**Ellipsoidal Reflector**
Focusing Capability	None	None	None	Single focal point	Two focal points
Collimation Capability	None	None	None	Yes	No
Spherical Aberration	N/A	N/A	N/A	None	None
Light Concentration	None	None	None	High	Moderate
Reflection Pattern	Specular (single angle)	Diffuse (scattered)	Back toward source	Converging to focus	Point-to-point transfer
Alignment Sensitivity	Low	Very Low	Low	Moderate	High
Suitability for Light Collection	Poor	Poor	Poor	Excellent	Limited

3.3.2 Reflective Materials

3.3.2.1 Glass

Glass mirrors consist of a glass substrate with a metallic coating applied to one surface. This substrate provides great flatness and stability for the reflective coating. Front-surface glass mirrors place the metallic coating on the side incident to the oncoming beam, eliminating double reflections that occur in rear-surface mirrors where the light passes through the glass before and after reflection. These mirrors can have varying reflectivities based on the applied coatings, but aluminum-coated glass mirrors can have reflectivities in the range of 85-90% in the visible spectrum. One drawback of these mirrors though is that for the PTC system, a mirror that fits the exact sizing requirements for our sensing region can be significantly more expensive due to the large sensing area. [21]

3.3.2.2 Aluminum

Aluminum is a reflective metal that is commonly used in optical applications due to its high reflectivity in the visible spectrum. Polished aluminum surfaces can achieve reflectivity values of approximately 88-92%, with stronger performances falling in the red and near-IR regions. Aluminum maintains a relatively consistent reflectivity across a broad wavelength range of wavelengths though, from UV to IR. Other advantages of aluminum include low cost and good corrosion resistance. This material also has a good thermal conductivity, which helps dissipate heat in high-power applications. These factors make aluminum a preferred choice for many different applications. [21], [24]

3.3.2.3 Silver

Silver has the highest reflectivity of any metal across most of the visible spectrum, achieving reflectivities of 95% at wavelengths above 450 nm and nearly 99% in the IR region. This makes silver an extremely useful coating material for applications where maximum optical efficiency is required, such as precision optical instruments or telescope mirrors. One drawback of silver, however, is that silver tarnishes over time due to reactions with sulfur compounds in the atmosphere, significantly degrading reflectivity. This can make the use of silver in applications that are outdoors cost ineffective due to this tarnishing effect. Protective overcoatings can be used to extend the life of silver mirrors, but these additional coatings add cost and complexity to the system while potentially reducing peak reflectivity as well. Ultimately, the higher cost of silver compared to aluminum, combined with its maintenance requirements, make this material unsuitable for a project where maximum optical efficiency is not the primary focus. [21], [24]

3.3.2.4 Stainless Steel

Stainless steel is a metal that has exceptional durability and corrosion resistance, making it a good material for reflectors operating in harsh environments. However, this material has a significantly lower reflectivity in the visible spectrum compared to silver or aluminum, with values in the range of 60-70%. The primary advantages of this material are its mechanical robustness, resistance to scratches, and ability to withstand rough environments with the need to additional protective coatings. These properties make stainless steel suitable for industrial applications where durability outweighs optical performance. For applications like the PTC, however, the significantly lower reflectivity offered by this material represents a great disadvantage in terms of light collection efficiency. [21]

3.3.2.5 Technology Comparison - Reflective Materials

For our project, the reflective material must balance the reflectivity at the wavelength of our laser diode (650 nm), durability, cost, and ease of integration into the system. Silver offers the highest reflectivity, but requires additional protective coatings to prevent tarnishing and can be costly. Stainless steel offers great durability at the cost of significantly lower reflectivity, reducing the overall efficiency of this material to collect the light. Glass mirrors provide good optical performance, but can be hard to integrate due to sizing constraints. Aluminum provides a good balance between all of these factors, being fairly durable with good reflectivity in the visible range. Aluminum can also be easily integrated into custom parabolic geometries due to the various types of aluminum products, making it the ideal choice as the reflective material to be used in the PTC system.

TABLE 3.8: Reflective Material Comparison

Characteristic	**Glass (Al-coated)**	**Aluminum**	**Silver**	**Stainless Steel**
Reflectivity @ 650 nm	85–90%	85–92%	>95%	60–70%

Spectral Range	Depends on coating	Broad	Visible to IR	Visible to IR
Corrosion Resistance	Good	Good	Poor	Excellent
Durability	Fragile	Good	Requires protection	Excellent
Surface Quality Achievable	High	High	Very High	Moderate
Cost	Moderate	Low	High	Moderate
Thermal Conductivity	Low	High	High	Low
Maintenance Requirements	Low	Low	High	Low

3.3.3 Component Selection - Reflectors

3.3.3.1 Romeda Aluminum Foil Tape

The Romeda Foil Tape is a reflective material designed for insulation, sealing, and light reflecting applications. Depending on the surface finish, aluminum surfaces can exhibit reflectance values around 88-92%. This flexible tape consists of a thin aluminum foil layer with an acrylic adhesive backing. This backing allows for straightforward installation without the need for additional mounting hardware. The physical dimensions of this tape is typically 2-3 inches with a thickness on the order of 0.05-0.1 mm. One drawback of this product is that the reflective surface is relatively soft and can develop wrinkles during application, which can introduce unwanted diffuse scattering. Additionally, the long-term durability of the adhesive may degrade under environmental exposure, making the lifetime of this reflector somewhat short. While this material provides a low-cost and easily deployable solution, its optical performance is less consistent due to application difficulties. [25]

3.3.3.2 Anodized Aluminum Sheet Clear (5005 Alloy)

The 5005 alloy anodized aluminum sheet is a rigid material designed for structural and decorative applications. This material consists of primarily aluminum with a clear anodized oxide layer, which enhances corrosion resistance and overall durability. While polished aluminum surfaces have high reflectance values, the anodized finish slightly reduces the overall reflectivity of the material. This sheet has a thickness of about 1.6 mm, which provides good structural stability, which ensures consistent reflection geometry and minimized distortion. A drawback of this though is that this material requires significant machining and mounting mechanisms, increasing system complexity and cost compared to adhesive-based solutions that can be easily form fitted into a parabolic reflector. [26]

3.3.3.3 Rust-Oleum Gloss Aluminum Spray Paint

The Rust-Oleum spray paint is a coating solution designed to provide a metallic, reflective finish on a variety of surfaces. This paint contains aluminum pigments that create a reflective appearance, which results in a reflectivity that is significantly lower than both the anodized metal sheet and aluminum tape alternatives. It can be easily applied to a wide-range of materials such as wood, plastics, or metals, allowing very easy system integration. One drawback to this though is that the reflective quality is highly dependent on the application technique, as spray uniformity and surface preparation can greatly affect the reflective property of the paint. While this option provides a low-cost and easy-to-implement solution, the optical performance is the least consistent among the options previously considered. [27]

3.3.3.4 Component Comparison - Reflectors

For our system, the primary objective of the reflector is to maximize the return signal of the laser to the photodiode while maintaining consistent performance. This means that when selecting a reflector, several key factors such as reflectivity, ease of integration, and cost are to be considered. The aluminum foil, while being easy to integrate and low-cost ($6.98), has only moderate consistency in maintaining a good reflective surface due to the potential wrinkling that may occur during the application of this material. This could be counteracted with post-application polishing procedures, since polished aluminum has the highest reflectance of any of the current reflector options. The aluminum sheet, although significantly more complex to integrate, provides a stable and uniform reflective surface that ensures a more consistent optical behavior with a moderate cost ($13.82). While the spray paint is a low-cost ($6.98) solution that offers the greatest flexibility in application, but offers greater diffuse reflection characteristics, reducing its effectiveness for optical sensing applications. Considering all of these factors, the aluminum foil tape was chosen as the optimal reflector for this design due to its high potential as a reflector and its ease of integration capabilities.

TABLE 3.9: Reflector Component Comparison

Characteristic	**Aluminum Foil Tape**	**Anodized Aluminum Sheet**	**Spray Paint (Aluminum)**
Reflectivity (%)	High	Moderate	Low
Surface Type	Thin foil	Rigid anodized metal	Coated surface
Thickness	0.05-0.1 mm	1.6 mm	$\sim \mu m$
Surface Quality	Variable	Uniform	Dependent on application
Reflection Type	Mostly specular	Specular	More diffuse
Ease of Integration	Very high	Low	Very high
Mechanical Stability	Low	High	Moderate
Durability	Moderate (adhesive degradation)	High (corrosion resistant)	Moderate (wear/fading)

Consistency	Moderate	High	Low
Cost	$6.98	$13.82	$6.98

3.4 Camera Technology

3.4.1 Camera Outcome

The cameras are primarily responsible for capturing an image of the projectile in flight within the strobe imaging chamber. Due to the potential speed of the projectile, these images need to be taken quickly, which is possible with a larger amount of incoming light.

3.4.2 Color vs Monochrome

Monochrome cameras are simpler, and are capable of higher resolutions than color cameras. Monochrome cameras do not provide color information, and therefore do not provide as beautiful of images compared to color cameras. For the goals of this project, a monochrome camera is advantageous for a number of reasons.

First, the camera lens design is significantly simpler due to the elimination of chromatic dispersion, so elements such as doublets and triplets are unnecessary to correct for this. Chromatic dispersion is the difference in effective refractive index of a material for different wavelengths (or colors) of light. This difference in refractive index causes a change in the focal length of simple lenses for different colors of light, so a lens system set up perfectly for yellow light images will blur the blue and red light, causing chromatic aberration. Chromatic aberration can be compensated for by using doublets or triplets, which are lenses with different materials, which have different chromatic dispersions, and are arranged in such a way as to have nearly identical focal lengths for multiple colors of light across the visible spectrum.

Second, monochrome imaging would allow for a bandpass color filter which would greatly reduce the effect of ambient light on the system. With a narrow bandwidth strobe illumination, a bandpass color filter would not significantly reduce the light into the cameras from the strobe system, but would reduce light from outside, such as sunlight reflecting into the strobe imaging chamber during outdoor usage. Sunlight reflecting into the chamber would cause the imaging of the projectile to include times other than when the strobe is illuminated, which would cause a blurry streak that could interfere with the processing of the image.

Third, monochrome imaging allows for higher resolution. Color cameras use a grid color filter in front of a monochrome 2d sensor array in order to get color information. A process of demosaicing is used to recover the color information for the full resolution of the camera sensor; however, information or resolution is lost in the process. Monochrome

imaging does not require a color filter grid for retrieving color information, which allows a higher resolution image to be produced. The higher resolution of a monochrome camera would allow more precise projectile tracking, giving higher accuracy on the position, angle, and speed of the projectile from the strobe photography stage of the system. [28]

3.4.2.1 Technology Comparison - Camera Color

While color imaging would allow us to produce more striking strobe photographic images to include with our presentation, the advantages of monochromatic imaging outweigh those of color imaging.

TABLE 3.10: Camera Color Technology Comparison

Feature	**Color**	**Monochrome**
Lens Design Complexity	High	Low
Ambient Light Sensitivity	High	Low
Resolution	Lower	Higher
Aesthetics	Better	Worse

3.4.3 NoIR vs Standard

Camera sensors are sensitive to IR wavelengths that are invisible to humans, and therefore many cameras have a filter that blocks those IR wavelengths in order to better replicate human vision. Sometimes it is valuable to be able to detect those IR wavelengths, such as when debugging a TV remote it can be advantageous to point your cell phone camera at the remote to see if it is transmitting. Additionally, since we are using IR LEDs for lighting, ambient white light will have less of an affect on our camera's resolution.

3.4.3.1 Technology Comparison - Camera Filter

The selection of NoIR vs standard is not of great importance, except for the showcase high-speed strobe photographic images. Since we have decided to use IR LEDs as the strobing mechanism, the NoIR choice is the one we are proceeding with for the PTC.

TABLE 3.11: Camera NoIR vs Standard

Feature	**NoIR**	**Standard**
IR Sensitive	Yes	No
Color Fidelity	Worse	Better

3.4.4 Global Shutter vs Rolling Shutter

Global Shutter cameras have a shutter that changes from transparent to opaque or from opaque to transparent all at the same time, and samples all of the pixels at the same time. Rolling Shutter cameras use a technique that either samples the lines of pixels or blocks the light of the lines of pixels one line at a time, which causes a slight time delay between the top of the image being taken and the bottom of the image. Rolling Shutter

is a less expensive technique, it allows for higher resolutions at a lower price, but there are significant issues for high speed photography.

3.4.4.1 Technology Comparison - Shutter Type

For early system integration, the low cost and high resolution offered by the rolling shutter type make this a desirable choice for the PTC. However, we would like to produce high quality images by using the high-speed photography capabilities provided by the global shutter type. Therefore, we have chosen the global shutter type for this project.

TABLE 3.12: Camera Shutter Technology Comparison

Feature	**Global Shutter**	**Rolling Shutter**
High-Speed Photography	Better	Worse
Cost	Higher	Lower
Resolution	Lower	Higher

3.4.5 Component Selection - Cameras

3.4.5.1 Raspberry Pi Camera Module 2

A low cost option, the camera module 2 is the camera we will use for integration testing while the proper cameras are designed and created. There may be problems with the rolling shutter in the high speed photography, which can be seen for example in the bend of moving propellers when imaging with a cell phone camera. The Raspberry Pi Camera Module v2 is based on the Sony IMX219 sensor, which is an 8 megapixel sensor. The maximum image resolution is 3280×2464 pixels, but the video it can take is 1640×1232 at 41 frames per second. Pixel size is on the small side, at $1.12[\mu\text{m}] \times 1.12[\mu\text{m}]$. It does not have an interchangeable lens, so we would be unable to change out the lens for a custom one. There are versions with and without an IR filter which could be used for IR imaging. [29]

3.4.5.2 Raspberry Pi High Quality Camera

The high quality camera has a higher resolution than the global shutter camera, but has rolling shutter distortion that would interfere with proper capture of the high speed projectiles in the system. The Raspberry Pi High Quality Camera is based on the Sony IMX477 sensor with a resolution of 4056×3040 pixels. The module has an external trigger, which allows the capture of images to be synchronized and controlled by a simple electrical signal. It comes with either a M12 or a C/CS mount for the lens, allowing for custom interchangeable lenses. The IR filter can be removed, which will allow sensitivity to IR light. Video can be captured at a maximum resolution of 2028×1080 at 50 frames per second. The pixel size is $1.55[\mu\text{m}] \times 1.55[\mu\text{m}]$, allowing moderate low-light performance. [29]

3.4.5.3 Raspberry Pi Global Shutter Camera

The Raspberry Pi Global Shutter Camera is based on the Sony IMX 296 sensor. It has a resolution of 1456×1088 pixels. The key defining feature of this camera is the global shutter, which allows the entire frame to be exposed simultaneously, significantly enhancing photography of high-speed objects. The pixels size is $3.45[\mu\mathrm{m}] \times 3.45[\mu\mathrm{m}]$, allowing for excellent low-light performance. It has a C/CS lens mount, allowing for custom interchangeable lenses. The IR filter can be removed, allowing for IR imaging. The resolution is significantly lower at 1.58 megapixels. [29]

The global shutter camera is specialized for high-speed photography. While it has a lower resolution than the High Quality camera, the global shutter camera is capable of taking photographs over 500 times per second (at low resolution), or a single photograph as short as 30 μs using an external trigger. The advantages and rather low cost of the Raspberry Pi Global Shutter camera an ideal choice for the system. [30]

3.4.5.4 e-CAM21_CURB - 2MP OV2311

The e-CAM21_CURB is a global shutter monochrome camera compatible with the Raspberry Pi computer. It uses the OmniVision OV2311 image sensor. It uses a M12 lens mount, so any lenses would need to be made compatible with an M12 thread. 2MP resolution video can be streamed at 60 fps over the 2-lane MIPI connector. The image sensor has $3[\mu\mathrm{m}] \times 3[\mu\mathrm{m}]$ pixels. The field of view (FOV) is $58.6°$ by $48.77°$. [31]

3.4.5.5 ArduCam OV9281 Global Shutter NoIR Mono Camera

The Arducam OV9281 is a 1 megapixel global shutter NoIR monochrome camera. It has a 130 degree field of view (FOV), and uses a M12 lens mount. Though it has an M12 lens mount, it claims that the lens is not interchangeable. Compatible with the Raspberry Pi ecosystem. It is sensitive to IR light due to a lack of an IR filter. [32]

3.4.5.6 Component Comparison - Cameras

For the PTC, the Raspberry Pi Global Shutter Camera provides the best overall balance between performance, system compatibility, and cost. Its global shutter mechanism eliminates motion distortion by exposing the entire frame at once. Additionally, its relatively large pixel size of $3.45\ \mu m$ improves its performance in low-light environments, which is beneficial for detecting fast-moving objects with short exposure times. While the resolution is lower than some of the alternatives listed, it is sufficient for the spatial requirements of the PTC system.

TABLE 3.13: Camera Selection Comparison

Name	Price	Resolution	Shutter	External Trigger	Color
Raspberry Pi Global Shutter Camera	$50	1456×1088	Global	Yes	Color

Raspberry Pi Hi Quality Camera	$55	4056 × 3040	Rolling	Yes	Color
Raspberry Pi Camera Module 2	$15	3280 × 2464	Rolling	No	Color
e-CAM21_CURB	$90	1600 × 1300	Global	Yes	Monochrome
ArduCam OV9281 Global Shutter NoIR Mono Camera	$42	1280×800	Global	No	Monochrome

3.5 Illuminative Technology - Strobe Imaging

Strobe Imaging uses short bright flashes of light in order to image things faster than the framerate of the camera. The two most common technologies for strobe illumination is LED and gas discharge. LEDs, or light emitting diodes, are a common technology for lighting, and often have a rise time in the realm of 10 to 100 nanoseconds. This makes LEDs more than fast enough for strobe illumination. Gas discharge strobe illumination can be much brighter, but require high voltage and other considerations. A third alternative technology to investigate could be a Q-switched laser, however that is not common in practice and would therefore be too close to a research task for our senior design project.

3.5.1 LED vs Gas Discharge

LEDs as a light source for strobe photography is a cheap and inexpensive light source. The major challenge for LEDs is to provide a large current in a short period of time. Discharge flash lamps are a good way to create short flashes of light, however they are broadband, expensive, and require high voltages. Different gasses can be used for different wavelengths of light, e.g. Xenon often produces a white light. There are details about the high voltage electric spark discharge triggering that make for a complex use case. Gas discharge lamps are often used to pump lasers. [33]

3.5.1.1 Technology Comparison - Strobe Illumination Source

Due to the low cost and simplicity of using LEDs, as well as the wider range of wavelengths available, we have selected to use LED lights for the PTC. Gas discharge lamps are too expensive and broadband to be used for a reliable illumination source.

TABLE 3.14: Strobe Illumination Technology

Feature	**LED**	**Gas Discharge**
Wavelength selectivity	High	Low
Cost	High	Low
Voltage Requirements	High ($> 1\,\text{kV}$)	Low (2–20 V)

Intensity	High	Lower

3.5.2 Wavelengths (Strobe Imaging)

There are advantages and disadvantages to using different wavelengths, or colors, of light.

3.5.2.1 UV

UV leds are typically more expensive due to the exotic high-bandgap materials required. They require a higher forward voltage than other LEDs due to the higher energy of UV photons. The UV light will breakdown materials, causing lower operating lifespans for UV LEDs. However, UV LEDs have the shortest wavelength, which allows for higher resolution imaging than other wavelengths, but the sizes and distances we are working with do not get anywhere near the diffraction limits imposed by the wavelength of light. Cameras are less sensitive to UV light due to a few effects, including the higher energy per photon creating a lower response per unit power, and the absorption of UV light from the optics.

3.5.2.2 Blue

Blue LEDs are inexpensive due to their common use in white LEDs. White LEDs are typically made with a blue LED coated with a phosphor that converts the blue light into a wide range of wavelengths. Cameras are moderately sensitive to Blue light. While blue LEDs require gallium nitride (GaN), a slightly exotic material, with high temperature requirements for fabrication. Due to the high volume of manufacturing for use in white LED lights, the economies of scale have significantly decreased the costs for high power blue LEDs.

3.5.2.3 Red

Red LEDs are ubiquitous, but are often low powered. There are very few high-power red LEDs available, and those that are are quite expensive. Red light is often used for night time and camping due to it not ruining the night vision and low-light vision of humans. Cameras are often very sensitive to red light.

3.5.2.4 IR

IR LEDs in the 850 nm - 960 nm range are some of the cheapest LEDs due to the fact the bandgap of Silicon is in the energy range of IR photons. One of the advantages of IR is the invisible nature, so that it wouldn't interfere with vision nor would it alert enemies in military applications. Cameras are moderately sensitive to IR light in the 700 nm - 800 nm range, however they have a low sensitivity to the light in the 850 nm and up range, with about 0.4 responsivity to 850 nm light. [34]

3.5.2.5 Technology Comparison - Wavelengths (Strobe Imaging)

Each of these wavelengths have their own strengths and weaknesses, but for the PTC there are several factors that are more important than others. For our system, we want the LEDs to be cheap because we plan on ordering a lot of them, and we want the wavelength to have moderate amount of sensitivity for the cameras. Therefore, we have chosen the IR LEDs for the PTC project.

TABLE 3.15: Strobe Illumination Wavelength

Feature	**UV**	**Blue**	**Red**	**IR**
Cost	Moderate	Moderate	High	Low
Longevity	Low	Moderate	High	High
Diffraction-limited Resolution	High	Moderate	Low	Low
High Power Common	Yes	Yes	No	Yes
Visible	No	Yes	Yes	No
IMX296 Responsivity	Low (< 0.5)	Moderate (0.5–0.8)	High (0.9-1)	Moderate (0.5–0.9)

3.5.3 Component Selection - Strobe LEDs

3.5.3.1 Component Comparison - Strobe LEDs

The Solidlite IR LEDs of AG3535C, AC3535B, AC3535L, AC3535A are the IR LEDs that showed up as the lowest price per watt of generated light on Digikey. They are all 1.5 amp rated LEDs, and the manufacturer characterized the LEDs at 1 amp. They have radiant flux at 1000 mA between 1250 and 1550 mW. The rise and fall times are 10 ns and 15 ns respectively, which is more than enough for our purposes. Forward voltage for each is between 2.7 V and 3.6 V, with manufacturing or usage variation being the biggest driver of the differences (each gives a range of voltages with a $\Delta V \approx 0.6[\mathrm{V}]$). Spectral bandwidth is $\Delta\lambda = 40\,[\mathrm{nm}]$, with the peak wavelength at 850-855 [nm]. The biggest difference is the viewing angles, ranging between 60° to 150°. For a more uniform illumination, the AG3535C with it's 120° viewing angle would be good. For more concentrated light and requiring fewer LEDs, the AC3535A with a 90° viewing angle would be better. One of the advantages of these LEDs compared to some of the others is that the IV curve flattens out around 3.5 V, which allows for a simpler driving circuit due to a more linear current relationship to voltage. The selected LED is the AC3535A since the FOV is compatible with that of the cameras.

TABLE 3.16: Comparison of IR LEDs

Parameter	**AG3535C-YRR-1000mA-H1**	**AC3535B-YRR1000-mA-H1**	**AC3535L-YRR-1000MA**	**AC3535A-YRR-1000MA**

Manufacturer	Solidlite	Solidlite	Solidlite	Solidlite
Price ($)	1.1	1	1	1
Current - DC Forward (If) (Max)	1.5 A	1.5 A	1.5 A	1.5 A
Radiant Intensity (Min @ If)	430 mW/sr @ 1 A	900 mW/sr @ 1.5 A	175mW/sr @ 1 A	550 mW/sr @ 1 A
Viewing Angle	120°	60°	150°	90°
mW/$	1228.08	757.61	814.97	1012.17
mW/$ (wall-plug)	2454.55	4650	4650	4950
Total Radiant Flux @1000mA [mW]	1550	1350	1250	1350

3.6 Computational Technology - MCU

3.6.1 Microcontrollers

Sensing fast moving projectiles and measuring their physical characteristics is inherently a challenge that absolutely requires high speed processing. Therefore, it is essential that the processor we select is capable of receiving inputs and running calculations much faster than the projectile is moving. Our stretch goals list the PTC as possibly being capable of tracking projectiles moving as fast as 100 m/s. Given a projectile that is 1 cm long moving at 100 m/s, the time to detect the projectile moving past a fixed point is 0.1 milliseconds. To be able to detect a projectile in less than 0.1 milliseconds, that corresponds to a sampling rate of at least 10 kHz, which means that if the microcontroller can detect a projectile within 1 clock cycle, the minimum clock speed would be 10 kHz. The processor would likely need to have multiple clock cycles within one detection cycle, so a clock speed of 100 kHz or above is more reasonable. In order to detect the projectile and to minimize error in doing so, our processor must be running as fast as possible while remaining reasonable in price, accessibility, and complexity. Even if we decide to use an external component for the detection of projectiles, these criteria still hold true as we would want to minimize the amount of time spent running calculations, potentially throwing off the accuracy of triggered flashes and image captures. Luckily, there are a wide variety of microprocessors available to both industry and consumers that operate at significantly higher speeds than 10 kHz while remaining extremely cheap, particularly in the hobbyist space with brands such as Arduino, Raspberry Pi, Espressif, and Adafruit. We prefer microcontroller architectures in the hobbyist space because they are guaranteed to have development boards that will be easy to develop and debug our programs on.

3.6.1.1 Arduino

Arduino is a brand of development boards with a wide range of products available to consumers, at widely varying price points. The most expensive Arduino product at 77.90 euros (about $89.69) is the Arduino GIGA R1, with Wifi capability and a dual core

architecture, running at 480 and 240 MHz with the STM32H747XI MCU. This is quite expensive, and does not include the cost of a separate MCU by itself. The only option in stock for the STM32H747XI MCU costs $19.56. Among the cheapest of Arduino products is the Arduino Nano Every at 14.50 euros (about $16.69), with an ATMega4809 MCU. The ATMega4809 has a maximum frequency of 20 MHz. In the middle of the pack lies the Arduino UNO family, with the UNO R4 Minima sporting a maximum clock frequency of 48 MHz at a price of 22.00 euros ($25.33). At the time of writing, the cheapest option in stock for the ATMEGA4809 itself costs $1.74 on DigiKey, and the cheapest option in stock of the RA4M1 MCU costs $5.36. [35], [36], [37]

3.6.1.2 Raspberry Pi Pico

Raspberry Pi mostly specializes in Single Board Computers, however, they do have a series of development boards with their Pico line, the Pico 1 at $4.00 and the Pico 2 at $5.00. The Pico family consists of 2 MCU architectures, the Pico 1's RP2040, and the Pico 2's RP2350 family. The RP2040 can run up to 133 MHz, and the RP2350 series can run up to 150 MHz. The RP2040 and RP2350 chips themselves can each be found on Digikey at prices of $0.70 and $1.00, respectively. However, the RP2040 is not in stock, but the RP235X MCUs are. [38], [39]

3.6.1.3 Espressif

Espressif, like Arduino, hosts a wide variety of products, many of which include features like Wifi and Bluetooth. But with a far wider variety of products than Arduino, Espressif's website hosts a 'Product Selector' that can help narrow down choices based on a customer's needs. Since Espressif's product list still includes products that are Not Recommended for New Designs (NRND) or are only available as sample products, it is important to filter for only products under a Mass Produced status, meaning they are available and not too complex or specialized for new designs. Doing this reveals‘ that nearly all Espressif microcontrollers available have processing speeds exceeding 100 MHz and that about 90 have processor clock speeds of 240 MHz, but it does not narrow down options all that much with 162 results. Additionally, most products are System on Chip (SoC) or modules that include Wi-Fi and Bluetooth capability that we will not need for the PTC. Coupling this large variety of over 160 different products with features that are unnecessary for this project, and the higher price point as a result, makes choosing an ESP32 microcontroller difficult. Looking at DigiKey and searching up Espressif development board products, most viable options (options that are actually development boards and not particular modules, as DigiKey's search tool is unable to make the distinction) are upwards of $8.00. These cheaper $8.00 options actually are general purpose development boards. The ESP32-C6-DEVKITM is a general purpose development board that costs $8.00 and is built around the ESP32-C6-MINI-1 MCU, which itself costs $4.40, and has a clock speed of 160 MHz. Another option is the ESP32-DEVKITM-1, also $8.00, that uses the ESP32-MINI-1 MCU ($4.07) with a clock frequency of 240 MHz. [40]

3.6.1.4 Technology Comparison - MCU

The Arduino GIGA R1 will not be chosen. Not only does it include features we do not currently have a use for, it and it's MCU are far and away the most expensive options, which would not be friendly to our budget if we order more for redundancy or because of malfunction. The Raspberry Pi Pico 1 and 2 meet our needs quite well, being among the fastest options for chips and development boards, designed for simplicity, and also being extremely affordable. The Pico series retain a high clock frequency and still remain the cheapest options when ordering more individual IC's, as the cost of the Espressif MCU's is significantly higher. However, it would be worth noting that the ESP-32-MINI-1 is a great secondary option if the Raspberry Pi Picos do not work out. The Pico architecture is split into two families, the RP2040 used by the Raspberry Pi Pico 1, and the RP2350 series used by the Raspberry Pi Pico 2.

TABLE 3.17: Microcontroller Technology Comparison

Name	**Max Clock Speed (MHz)**	**Development Board Cost (USD)**	**Standalone MCU Cost (USD)**	**Total Cost (USD)**	**In Stock (Mouser, DigiKey)**
Arduino GIGA R1	480	$89.69	$19.56	$109.25	Yes
Arduino UNO R4	48	$25.33	$5.36	$30.69	Yes
Arduino Nano Every	20	$16.69	$1.74	$18.43	Yes
Raspberry Pi Pico 1	133	$4.00	$0.70	$4.70	No
Raspberry Pi Pico 2	150	$5.00	$1.00	$6.00	Yes
ESP32-C6-MINI-1	160	$8.00	$4.40	$12.40	Yes
ESP32-MINI-1	240	$8.00	$4.07	$12.07	Yes

3.6.2 Component Selection - MCU

3.6.2.1 RP2040

The RP2040 is a microprocessor that was developed by the Raspberry Pi team for use in their Raspberry Pi Pico development board, and is available to industry and the general public as a standalone IC. According to the datasheet, it runs at 133 MHz, but can be safely overclocked to 200 MHz with some setting changes specified by Raspberry Pi. This is a point in the RP2040's favor, as this overclocking ability is specifically pointed out by the designer and will not negatively affect the processor's architecture. [38]

3.6.2.2 RP2350 / RP2354

The RP2350 family of processors shares many similarities with the RP2040, but offers a variety of enhancements as well. Not only does the RP2350 have a dual core architecture, it operates by default at a clock speed of 150 MHz. This would allow us a higher base clock speed without the extra effort of overclocking, as well as provide us extra processing power to potentially facilitate concurrent calculations at a high speed. The RP2350 however, does not have internal flash storage for writing programs. The RP2354 variant of the RP2350 series is exactly the same as the RP2350, but has 2 MB of internal flash storage, reducing the number of external components we would need to incorporate in our PCB in order to get a functioning chip. [41]

3.6.2.3 RP235X A / RP235X B

Of the RP2350 and RP2354, there are also A and B variants of each type. The A type is the base, and has 30 GPIO pins, 4 of which are capable of receiving analog inputs. The B variant expands upon this with a larger form factor and more GPIO pins; 48 GPIO pins total with 8 of them being capable of analog inputs. While extra GPIO pins may be useful, they may not be necessary for our project. [41]

3.6.2.4 Component Comparison - MCU

The RP2040 remains the cheapest option all around, with the added benefit of being able to have its internals overclocked to 200 MHz. However, the RP2354 variants can decrease the complexity of implementing these chips by allowing us to use an internal Flash memory to program them, rather than buying and attaching an external device as is necessary for the RP2040 and the RP2350 variants. We don't anticipate needing all 30 GPIO pins, and so the B variants of the RP2350 series seem unnecessarily large. Additionally, while not explicitly sanctioned in the datasheet like for the RP2040, it is still possible to safely overclock the RP2350 series chips as shown on some experimental forums with repeatable experiments citeoverclocking-pico-2. And so, the RP2354A, with its higher base frequency and internal flash memory is our choice.

TABLE 3.18: Microcontroller Component Comparison

Name	**RP2040**	**RP2350A**	**RP2350B**	**RP2354A**	**RP2354B**
Rated Clock Speed	133 MHz	150 MHz	150 MHz	150 MHz	150 MHz
Cost Per Chip	$0.70	$0.90	$1.00	$1.00	$1.10
Dev Board Cost	$4.00	$5.00	N/A	N/A	N/A
GPIO Pins	30	30	48	30	48
Internal Flash Memory	No	No	No	Yes	Yes

3.6.2.5 Development Board

Instead of trying to program and constantly debug a chip soldered to a board, it makes sense to also acquire a development board so that software can be developed alongside hardware designs, and both can be tested together.

While there are many of the same options as above in earlier sections for a development board, familiarity and software compatibility are of the utmost importance. And so, the discussion of which development board to choose is a short one. It makes sense to develop functionality with hardware similar to what we will actually be using, so the Raspberry Pi Pico 2 development board was chosen to accompany our choice of RP2354A MCU, a chip in the same series as those used in Raspberry Pi Pico 2's.

3.7 Computational Technology - Image Processor

3.7.1 Single Board Computers

One objective of the PTC project is to capture multiple images of the projectile in flight and extract spatial information from these images. Due to the complexity of digital image processing, a normal microcontroller does not have sufficient processing power to handle the procedural analysis of photographs, as commonly listed hardware requirements for OpenCV are far beyond that which a microcontroller is capable of citehardware-requirements-for-opencv. To solve this issue yet still comply with project constraints, a more powerful computer will be used to handle image processing. The image processor will only be used for image processing, and all other signals and data must be handled by our RP2354A microcontroller. Much like our options for the MCU, there are a few small single board computers that will be capable of image processing, such as the Raspberry Pi, Orange Pi, and Odroid.

3.7.1.1 Raspberry Pi

Raspberry Pi is widely known for their Single Board Computers, and their latest products are variants of the Raspberry Pi Model 5, and the Zero 2 W. Raspberry Pi computers often come with their own Operating System called Raspberry Pi OS, and hardware specifications focused on image and video processing; including GPUs that support the latest hardware graphics API's, high resolution display ports and high speed USB ports, MIPI camera/display ports, as well as Wifi and Bluetooth support citeraspberry-pi-5-brief. Raspberry Pi also provides a step by step manual to install the Raspberry Pi OS and how to use Python to control GPIO pins on the Raspberry Pi device itself, alongside standard documentation and software packages available on their website. All these features allow Raspberry Pi computers to be readily used for image processing and other projects, which will be beneficial to our use case citeraspberry-pi-documentation. Additionally, Raspberry Pi also provides peripheral devices like cameras, including various lens types, which are guaranteed to work with Raspberry Pi products. It is also worth noting that Raspberry Pi

is based in the UK, and as such their products may be easier to get a hold of in today's geopolitical climate.

3.7.1.2 Orange Pi

Orange Pi is another manufacturer of single board computers, with a naming scheme reminiscent of Raspberry Pi. Among their top products are the Orange Pi 5 Ultra, Orange Pi 4 Pro, and Orange Pi 6 Plus. Orange Pi computers are advertised with high processing power, hardware optimized for AI computation, and advanced graphics capabilities. Like Raspberry Pi computers, Orange Pi computers tend to come with Wifi and Bluetooth support, high speed USB ports, and powerful GPUs. Unlike Raspberry Pi computers, Orange Pi computers tend to have more HDMI ports and do have less MIPI camera/display ports or none at all. For example, the Orange Pi 6 Plus has no MIPI ports, but the Orange Pi 5 Ultra has one, and the Orange Pi 4 Pro has 2. Orange Pi also provides their own operating system, called Orange Pi OS, with three specializations for certain applications, such as Orange Pi OS (Droid), which is based upon android software and Orange Pi OS (Arch), based upon the popular Linux distribution. Orange Pi does include a documentation page on their website where documentation and software for all available products can be found. Like Raspberry Pi, Orange Pi does offer cameras as well, but they mostly vary in image quality rather than lens type. All these features allow Orange Pi computers to be readily used for computationally and graphics intensive projects, but they do not allow us to easily connect cameras due to the variable amount or lack of MIPI ports, and since this manufacturer is based in China, is likely subject to large tariffs and shipping delays [42].

3.7.1.3 Odroid

Odroid, based in South Korea, advertises single board computers meant for advanced computing for industrial embedded systems. As such, Odroid devices are built with efficiency, affordability, and longevity in mind, as many newer devices list improvements in these areas over their predecessors. Odroid also makes it easy to see their product specifications and compare device benchmarks. Odroid devices are run on Linux based distributions or Android software, and tend to have more focus on processing power, efficiency, and I/O rather than graphics processing, and as such have less ports for carrying video information, often having only one HDMI port and one MIPI port for cameras. Odroid offers a wiki for each available product and an OS installation guide. Odroid does offer a variety pack of lenses of different sizes, but these lenses are without IR filters and Odroid only offers one camera, potentially limiting our options for image quality and what wavelengths of light to capture. While also likely subject to tariffs and shipping delays, an Odroid device may be a good middle ground option due to its competitive pricing and reliability, but lacks some variety available in other options [43].

3.7.1.4 Technology Comparison - Image Processor

A few criteria important to this selection process are the number of MIPI ports available, the cost of the devices, and of course whether or not they are actually available and will arrive in a timely manner. Orange Pi products Amazon or Aliexpress, and Odroid

products are only reliably available through their international website, whereas Raspberry Pi products can be bought either via Amazon or Digikey. Raspberry Pi Single Board Computer products start at $15, but this is for the Zero 2 W. Orange Pi's comparable Zero 2 W starts at $56 on Amazon, and Odroid starts at $55 with the Odroid C5. None of these prices include shipping. Another criteria is ease of use, and this can be classified by how much documentation is readily available for us to find and use. Given that Raspberry Pi offers documentation, software, and user guides, this makes Raspberry Pi the best option in terms of usability. Orange Pi and Odroid do however have a larger list of devices that we could potentially use, but this comes with the added detriment of extra research necessary to figure out which of these products could best suite our needs. With all of these factors considered, it makes sense to choose a Raspberry Pi product for their extensive documentation, ready availability, and competitive pricing citeraspberry-pi, orange-pi, ordroid.

TABLE 3.19: Image Processor Technology Comparison

Name	**Raspberry Pi**	**Orange Pi**	**Odroid**
Max MIPI Ports Available	2	2	1
Starting Price	$15.00	$56.00	$55.00
Range of Products	Minimal	Wide	Medium
Availability	In Stock	Almost Out of Stock	In Stock
Vendor	DigiKey, Amazon	Amazon	Odroid
Documentation and Support	Overwhelming	Extensive	Sufficient

3.7.2 Component Selection - Image Processor

Raspberry Pi has a smaller product line, and as such, this makes it far easier to compare devices and find one most suitable for our project. Since we plan to use cameras, it is imperative that the device in question have MIPI-CSI ports so that the cameras can be connected to the device and communicate image information.

3.7.2.1 Raspberry Pi Model 4

The Raspberry Pi Model 4 ,from $34, has plenty of high speed ports for USB 3.0, USB 2.0, as well as ethernet, and also has 2 MIPI ports. These two MIPI ports, however, are not of the same type, as one is a MIPI-DSI display port and the other is a MIPI-CSI display port, meaning that the Raspberry Pi Model 4 can only host one camera module, and we would likely need an adapter to use a different port to handle two at the same time citeraspberry-pi, pi-model-4-datasheet.

3.7.2.2 Raspberry Pi Model 5

The Raspberry Pi Model 5 hosts many improvements over the Model 4, namely significantly faster processing speed and better graphics capability. For the PTC project, the

most important improvement is that the Model 5 has two MIPI ports that are capable of both MIPI-DSI and MIPI-CSI communication, meaning that the Model 5 can host two cameras at the same time, as is desired for our use case. The standard Model 5 costs $45 citeraspberry-pi, pi-model-5.

3.7.2.3 Raspberry Pi Compute Module 5

The Raspberry Pi Compute Module 5 is a variant of the Model 5 that is meant for embedded applications, and is built with personalization and scalability in mind and includes a larger amount of GPIO pins for external control and communications. Naturally, the Compute Module 5 is more expensive, starting at $67.50, and since we can't use most of its features for this project, overkill citeraspberry-pi, pi-compute-5.

3.7.2.4 Raspberry Pi Zero 2 W

The Raspberry Pi Zero 2 W is the cheapest single board computer available at $15, and an excellent consideration for saving space, as it fits an entire computer on a 65 mm by 35 mm board. With such a small form factor, extra adapters may be necessary to use it, as the ports in its specifications are all of a mini or micro type, like micro USB and mini HDMI. The Raspberry Pi Zero 2 can support a single camera, but it uses a small MIPI CSI-2 port, which would require an extra cable to convert a camera's larger MIPI CSI output to a CSI-2 input. Space isn't necessarily a concern for the PTC project, so a large board may be more suitable to reduce the need for extra adapters citeraspberry-pi, pi-02-w.

3.7.2.5 Component Comparison - Image Processor

Of the listed devices, the Raspberry Pi Model 5 and Compute Module 5 are the only devices that would allow us to easily and simultaneously connect two camera modules. While the Zero 2 W has an excellent, small form factor, saving space is not a priority for this project, and we desire two MIPI-CSI ports. As such, our choice for image processor is the Raspberry Pi Model 5.

TABLE 3.20: Image Processor Component Comparison

Name	**Zero 2 W**	**Model 4**	**Model 5**	**Compute Module 5**
MIPI-CSI Ports Available	1	1	2	2
Starting Price	$15.00	$34.00	$45.00	$67.50
Size (mm x mm)	65x35	85x56	85x56	55x40
Power Requirements	5V/2.5A DC	5V/3A DC	5V/5A DC	5V/5A DC

3.8 Interfacing Technology - Tactile

3.8.1 Types of Tactile Interfacing Technologies

3.8.1.1 Buttons

Buttons are some of the most commonly seen components of user interfaces. Buttons work by either closing or opening a circuit, and this change in voltage can be detected by the MCU to perform an action. Each button will occupy a single GPIO pin of the MCU, and software will be able to control its function, as well as filter out bouncing, a phenomenon that occurs when the physical contacts of the button bounce on the conductors, sending false signals to the MCU. De-bouncing, as the process is called, can also be achieved with other methods.

3.8.1.2 Switches

Switches are also very commonly seen components of user interfaces, and work similarly to buttons, either closing or opening a circuit, or switching between circuits, but for an indefinite amount of time rather than a short moment like a button press. A switch could occupy one or more GPIO pins of the MCU, and software could be used to detect which state the switch is in. Since a switch remains in its position until an action is performed by the user, the software may be less fluid than if the component was a button.

3.8.1.3 Potentiometers

Potentiometers use a wiper on a resistive pad to create variable resistance, adjustable by the user. This wiper is also available as a third pin on the package, meaning that potentiometers inherently also function as voltage dividers. With this variable resistance and voltage division, variable voltages can be sensed by the MCU on a single GPIO pin, and such values can be processed to finely tune or adjust some setting in software. Potentiometers are, however, limited in the range that they can spin and the voltage range and resolution they can provide, e.g. a knob that can only make one rotation. These factors are all dependent on the type of potentiometer and the voltage source in use. Additionally, since this is a resistive circuit component, power will be consumed and let off as heat.

3.8.1.4 Rotary Encoders

Rotary Encoders are a more complex component of user interfaces, but are commonly seen on devices that require a large range of input values, like volume and frequency controls on a radio. While such uses cases can be serviced with a potentiometer, Rotary Encoders are more conductive than resistive, and do not consume much power like a potentiometer would. Additionally, they do not have a limited range of turns, and can be rotated indefinitely. Consequently, they do need more GPIO pins to use, requiring two for encoders without a built in button, and three for encoders with a built in button. Rotary Encoders function by alternating two voltage signals as they spin, with an inherent phase shift between the two signals. The MCU is then able to determine which signal

alternates first, essentially conveying whether the rotary encoder was rotated clockwise or counter-clockwise.

3.8.1.5 Technology Comparison - Tactile Interfacing Technology

In this instance, it would be prudent to select at least two of these technologies. By doing so, the PTC can provide multiple avenues of providing inputs to the machine, which is far more desirable than pushing a single button multiple times to select one option, or some other similarly annoying action. While potentiometers are simple, to use them as an input would require actively supplying them power, and as such they would waste power. Switches are limited in the kind of input they can supply, and so buttons and potentiometers are our choices. Buttons provide a way to quickly provide an input to a system, and rotary encoders, with they're infinite spin, would allow a user to quickly peruse menus. Furthermore, our choice of MCU already requires a few buttons for key features like debugging.

TABLE 3.21: Tactile Interfacing Technology Comparison

Name	**Buttons**	**Potentiometers**	**Rotary Encoders**	**Switches**
Minimum GPIO Pins Required	1	1	2	1
Variable Input?	No	Yes	Yes	Yes
Range of Variable Input	N/A	Limited by Outside Factors	Unlimited	Limited, Discrete
Resistive?	No	Yes	No	No

3.8.2 Component Selection - Button Technology

3.8.2.1 Grayhill 95C06A4GWRT

The Grayhill 95C06A4GWRT is a standard tactile button like those commonly seen in use for prototype circuits on breadboards. Naturally, they also work in plenty of other use cases, and can handle 50 mA of current at 12V. This particular model is square shaped and covers an area of 6 mm squared, has a life rating of 500,000 cycles (500,000 button presses), and is normally open; requiring the button to be pressed in order to complete a circuit citegrayhill-tactile-switch-datasheet. It costs $3.25 for ten of these.

3.8.2.2 E-Switch LC2255EENPTAGBLK

The E-Switch LC2255EENPTAGBLK is a black latching button that can handle up to 30 mA at 30V DC. It has a larger profile, being 29.05 mm long and 7.20 mm wide. Furthermore, it is a latching button, meaning that when pressed, the button will remain in its position instead of returning to its original position. This causes the button to act more like a switch than a button, having two states and requiring user action in order to

switch between them citee-switch-pushbutton-datasheet. It would cost $8.53 for ten of these.

3.8.2.3 C&K D6C60 F1 LFS

The C&K D6C60 F1 LFS is a blue, square shaped key switch button in a circular housing that can handle up to 100 mA at 32V DC. It is larger than the Grayhill option but still smaller than the E-Switch option with a diameter of 11.4 mm. This button is normally open, and also customizable with the option to switch out button caps from the housing citeck-key-switch-datasheet. It would cost $10.30 for ten of these, but if ordered it may be necessary to also order the button caps, as it is unclear if the vendor sells both together.

3.8.2.4 Component Comparison - Buttons

Between these three options, the Grayhill button, the E-Switch button, and the C&K button, the Grayhill button stands out as more cost effective. Therefore, our choice of button is the Grayhill 95C06A4GWRT.

TABLE 3.22: Tactile Interfacing Component Comparison - Button

Brand	**Grayhill**	**E-Switch**	**C&K**
Size	2.45mm x 2.45mm	29.05mm x 7.20mm	11.4mm diameter
Switching Type	Momentary	Latching	Momentary
Switching State	Normally Open	Normally Open	Normally Open
Cost for 10	$3.25	$8.53	$10.30

3.8.3 Component Selection - Rotary Encoder Technology

3.8.3.1 PEC11R-4220K-N0024

The PEC11R-4220K-N0024 is a rotary encoder with a bushing, a 12.5 mm by 13.4 mm base, and a 20 mm long knurled shaft. It costs $1.89 per unit, or $9.15 for 5 units. This rotary encoder does not have a momentary push switch, and so only has three signal pins: Channel A, Channel B, and Common, which are aligned in parallel with the shaft axis [44].

3.8.3.2 PEC12R-2120F-N0012

The PEC12R-2120F-N0012 is a rotary encoder that also has a 12.5 mm by 13.4 mm base, but instead has a metal-flatted shaft that is also 20 mm long. It costs $1.51 per unit, or $6.65 for 5 units. It also does not have a momentary push switch, and does not have a bushing. Its three pins are aligned radially [45].

3.8.3.3 PEC11R-4325F-S0012

The PEC11R-4325F-S0012 is another rotary encoder of the same size as the other options, except its shaft is a metal-flatted shaft that is 25 mm long. It does come with a bushing

and a momentary push switch, and as such has two extra pins to support its functionality. It costs $2.23 per unit, or $18.80, and its pins are aligned in parallel with the shaft axis [44].

3.8.3.4 Component Comparison - Rotary Encoders

While all of these components are very similar, there are clear price differences between them based on whether or not they have a built in button. Thus, it would be more cost effective to simply get dedicated buttons, as rotary encoders with switches seem to be far more expensive. These encoders are also oriented slightly differently. With radial pins, the shaft remains perpendicular to the pins, and with axial pins they remain parallel with the shaft. If we were to place these components on a PCB, it may be harder for users to interact with rotary encoders situated on radial pins (think knobs on the bottom of a box, rather than coming out the front). If the rotary encoders were to be mounted directly on some casing, a bushing would help hold it in place. Thus, the PEC11R-4220K-N0024 offers the most versatility while still being the second cheapest option, and significantly cheaper than the most expensive option.

TABLE 3.23: Tactile Interfacing Component Comparison - Rotary Encoder

Name	**PEC11R-4220K-N0024**	**PEC12R-2120F-N0012**	**PEC11R-4325F-S0012**
Pin Orientation	Axial	Radial	Axial
Bushing	Yes	No	Yes
Momentary Push Switch	No	No	Yes
Cost per Unit	$1.89	$1.51	$2.23

3.9 Interfacing Technology - Visual

3.9.1 Types of Visual Interfacing Technologies

For the PTC project, we intend on having some form of visual output for the end-user to see and interpret, in this case, information about a projectile's measured speed and trajectory. There are many varieties of visual displays, and all could potentially accomplish the same goal, but each function differently and display visual information using different methods.

3.9.1.1 LCD

LCD stands for Liquid Crystal Display, and LCDs are quite common in small electronics and applications that only require a predetermined set of visual outputs. For example, an LCD may be used in a calculator, where the output is primarily numerical. In this case, the LCD would not be able to show anything other than numbers and some specialized characters, as what areas of the screen the LCD can actually alter or display to is physically built into its construction.

3.9.1.2 ePaper

Electronic paper displays maintain an image without power after it is written, offering excellent readability in bright light and very low power consumption. However, they have slow refresh rates, making them unsuitable for real-time updates during measurement cycles. [46]

3.9.1.3 TFT

Thin-Film Transistor displays are full-color graphical LCDs that provide high resolution, fast refresh rates, and good color reproduction. They are ideal for showing detailed graphs and user-friendly interfaces, but consume more power than basic LCDS. [47]

3.9.1.4 Technology Comparison

LCD technology was selected as the primary visual interfacing method because it offers the best balance of low power consumption, adequate refresh rate for real-time results, and low cost while still meeting the display requirements of the PTC system.

TABLE 3.24: Visual Interfacing Technology Comparison

Characteristic	**LCD**	**ePaper**	**TFT**
Power Consumption	Low	Very Low	Moderate
Refresh Rate	Good	Very Slow	Excellent
Color Capability	Limited	Monochrome	Full Color
Cost	Low	Moderate	Higher
Suitability for Real-time Data	High	Low	High
Readability in Bright Light	Good	Excellent	Good

3.9.2 Component Comparison - Visual Interfacing Technology

3.9.2.1 Waveshare 2.8-inch Capacitive Touch LCD

The Waveshare 2.8-inch Capacitive Touch LCD was selected as the display for the PTC. This module features a 320×240 resolution TFT LCD with a capacitive touch overlay, allowing both information display and simple user input through the screen. It connects directly to the Raspberry Pi via GPIO and is fully supported by existing Python libraries. The display provides clear, readable output for velocity, trajectory graphs, and status messages while keeping the overall system cost and power consumption low. [48]

3.9.2.2 Adafruit TFT Touchscreen

The Adafruit 3.5-inch TFT Touchscreen offers higher resolution and full-color graphics with capacitive touch input. It provides excellent visual quality but requires more GPIO

pins, draws significantly more power, and increases overall system cost compared to smaller displays. [47]

3.9.2.3 Generic 16 x 2 Character LCD

The Generic 16 x 2 Character LCD is a low-cost, text-only display commonly used in embedded projects. It can show two rows of 16 characters and requires only a parallel GPIO interface. While inexpensive and easy to implement, it lacks graphical capability and cannot display trajectory plots or modern user interfaces. [49]

3.9.2.4 Component Comparison - Visual Interfacing Technology

The Waveshare 2.8-inch Capacitive Touch LCD was chosen because it provides the best balance of graphical capability, touch interaction, cost, and power consumption for the PTC system. It enables a modern, intuitive user interface while maintaining the overall system simplicity and cost targets.

TABLE 3.25: Visual Interfacing Component Comparison

Characteristic	**Waveshare 2.8-inch Capacitive Touch LCD**	**Generic 16 x 2 Character LCD**	**Adafruit 3.5-inch TFT Touchscreen**
Resolution	320 × 240	16 × 2 characters	480 × 320
Touch Input	Yes (capacitive)	No	Yes (capacitive)
Graphics Capability	Yes	No	Yes
Interface	GPIO (SPI)	GPIO (parallel)	GPIO (SPI)
Cost (single unit)	$18.50	$4.50	$32.00
Power Consumption	Low	Very Low	Moderate
Suitability for PTC	High	Low	Moderate

3.10 Transimpedance Amplification Technology

Since it has been established that the PTC will shoot lasers into photodiodes and wait for the interruption of the beam in order to detect projectiles, a system must be put in place to make the output of the photodiodes legible to the MCU, as photodiodes output a current rather than a voltage. The solution is to use a transimpedance amplifier that will convert the current output of the photodiodes into a stable, almost linear voltage, and then impart massive gain on that voltage to get it high enough to be read by another device.

3.10.0.1 Specialized IC

Transimpedance amplifiers are often used in fiber optic applications and high speed photodiode applications, and so are specialized in operating with bandwidths greater

than a megahertz. Since these come as a complete package in integrated circuits, they would require no extra components. Their use case is often specialized, and so they would be more expensive and exceed the PTC's specifications, allowing us to trade increased cost for reliability in signal propagation speed. However, this cost may not be worth it as we only expect the PTC to be able to measure projectiles as fast as 100 m/s, which does not quite exceed a megahertz bandwidth, and these IC's are generally quite expensive when compared to other options.

3.10.0.2 Generic Operational Amplifier

Using general purpose op-amps, a transimpedance amplifier can be designed and assembled that meets our requirements. This will of course require extra external components, as the device in use would not be a complete transimpedance amplifier alone, but an operational amplifier. Such designs will likely be cheaper, but may be more susceptible to excessive noise and may have a lower bandwidth due to the generic nature of the components, limiting response speed. Additionally, an insufficient slew rate could cause additional delays in signal propagation, and all these factors mean that great importance will be placed on our choice of operational amplifier and additional components, such that the amplifier has a large enough bandwidth and high enough slew rate, and the external components do not impart a large amount of noise.

3.10.0.3 Comparator

While the transimpedance amplifiers focus on providing a stable and somewhat linear voltage output to the MCU, another option could be reading the current output of the diode and constantly comparing to previous measurements to determine a drop in current. This could likely be accomplish with parallel outputs fed into a comparator. One parallel output of the photodiode would be fed straight into a resistance to ground and the input of a comparator, and the other would be fed into a capacitor to ground that is also attached to the other input of the comparator. With this, it may be possible to determine a change in the photodiode's output due to the delay caused by the capacitance on one track, allowing the comparator to detect a difference in voltage in outputs and send a signal to the MCU. This approach would likely require more experimentation, and may be even more susceptible to ambient light than our other options.

3.10.0.4 Technology Comparison

A preliminary search of transimpedance amplifiers shows that specialized op-amps for transimpedance amplification vary wildly in price, from about $5 for a single chip to more than $20. General Purpose Op-Amps however, are far more affordable and are capable of serving the same function, though to a lesser degree. The Comparator based circuit may in theory work, but it is likely to require much testing and experimentation, which is undesirable for this project. Therefore, a generic op-amp will be used.

TABLE 3.26: Transimpedance Amplifier Technology Comparison

Name	Generic Op-Amp	Specialized IC	Experimental Comparator Circuit
Extra Components	Required	Not Required	Required
Performance	Variable	High	Unknown
Noise Susceptance	Medium	Minimal	High
Cost	Cheap	Expensive	Cheap

3.10.1 Component Selection - Amplifiers

3.10.1.1 TL084

The TL084 comes from a family of FET-input operational amplifiers, reducing input bias currents and noise imparted by the device to a negligible amount. It can operate with a 30 V difference between its postive and negative power pins, and has a high slew rate of 20 V per microsecond and a gain bandwidth product of 5.25 MHz. This features would allow us to use the TL084 as is without further regulating a high supply voltage to a lower value, and also should provide ample bandwidth and slew rate to detect changes in a photodiode's current output. The TL084 also is a quad package, with 4 op-amps inside, allowing us more amplifiers to use if necessary. At a price of $0.34 per unit and an extended price of $2.38 for 10, the TL084 seems like an all around great option [50].

3.10.1.2 MCP6487T-E/SN

The MCP6587T dual package operational amplifier maintains a high slew rate like the TL084, at 20 V per microsecond, and supports almost double the gain bandwidth product at 10 MHz, as well as protections against electro-magnetic interference. At $0.35 per unit, it would no doubt be a cost effective choice. However, it can only support a supply voltage of up to 5.5 V, and this would likely require us to insert another voltage regulator into the circuit to deliver power without damaging this device [51].

3.10.1.3 OPA2727AIDR

The OPA2727 is a dual package amplifer advertized as a high-speed CMOS amplifier, and is the highest performing amplifier among these options, with a gain-bandwidth product of 20 MHz and a slew rate of 30 V per microsecond. It also uses a supply voltage higher than the MCP6487, with a maximum input of 12 V. At $2.90 per unit, this would be the most expensive, but highest performing option, as this would also likely require its own voltage regulator to prevent damage to the component [52].

3.10.1.4 Component Comparison - Transimpedance Amplifier

While both the MCP6487 and the OPA2727 have higher performance specifications, they are both more expensive and require the added expense and complexity of their own voltage regulator to prevent component damage. Meanwhile, the TL084 will be

capable of handling a substantially larger supply voltage, and does not fall too far behind the other options in component specifications. Additionally, the TL084 is cheaper, and comes as a quad package, allowing us to potentially use 4 Op-Amps at once with every unit, rather than just 2 as with the MCP6487 and the OPA2727.

TABLE 3.27: Transimpedance Amplifier Component Comparison

Name	**TL084**	**MCP6487**	**OPA2727**
Op-Amps in Package	4	Not 2	2
Slew Rate (V/us)	20	20	30
Gain Bandwidth Product (MHz)	5.25	10	20
Cost per Unit	$0.34	$0.35	$2.90

3.11 External Light Sensor Technology

For calibration purposes, it may be useful to have an ambient light sensor housed in the PTC's imaging chamber. With the output of this device, it would be possible to determine how bright the imaging chamber already is before operation, due to outdoor sunlight or other ambient sources, and better adjust shading manually or possibly even adjust how bright the strobe circuits strobe. Below are a few technologies that could accomplish this.

3.11.1 Photosensors

3.11.1.1 Photoresistor

A Photoresistor is a kind of resistor whose resistance changes based on the amount of light incident on itself. Photoresistors start out at high resistances and are reduced to low resistances when in bright environments, allowing a measured output current or voltage to be directly affected by an environment's light levels. This can then be interpreted by an MCU, but would likely require a software formula to quantify brightness in some way.

3.11.1.2 Photodiode

A photodiode produces a current when light reaches it. In this instance, a measured voltage and current could be directly generated by the environment's ambient lighting. This eliminates the need to actively provide a power source whose output is altered by an optoelectronic device, like what would be necessary for components like photoresistors and phototransistors. The photodiode's generated current can then be interpreted by an MCU by using a resistor connected in series with the photodiode to ground. Again, a software formula would likely be necessary to quantify the ambient brightness.

3.11.1.3 Phototransistor

Phototransistors are often specialized BJTs, and a phototransistor, instead of generating a current or changing resistance, changes how much current it will allow through its collector and emitter. Basically, the transistor turns more 'on' with more light, and more 'off' with less light. This, like the photoresistor, would require a power source, and the output could be interpreted by an MCU, which would likely require a software formula to calculate and quantify brightness.

3.11.1.4 Optical Sensor IC

Optical Sensor ICs combine the properties of optolectronics with digital circuitry and serial communications, taking the sensing of light one step further and actually quantifying it, and sending a message to an controller via I2C. Rather than manually building a formula in software to interpret the output of the other options, these ICs will send binary values that correspond to actual values of Lux, a measure of illuminance.

3.11.1.5 Technology Comparison - Photosensors

Though many of these technologies are not expensive, comparatively, to each other, ICs would be more expensive than the other options. However, they have the advantage of being able to calculate Lux and send a code to their controller, simplifying measurement. Photoresistors and phototransistors would constantly drain current, but photodiodes would be able to constantly generate current. Even so, the ability to quickly calculate and communicate Lux is well worth the cost, and so the PTC will use an optical sensor IC.

TABLE 3.28: External Light Sensor Technology Comparison

Name	**photoresistor**	**photodiode**	**phototransistor**	**Optical Sensor IC**
Efficiency	Poor	Excellent	Satisfactory	Excellent
Easy to Interpret	No	No	No	Yes
Cost	Less Expensive	Less Expensive	Expensive	More Expensive

3.11.2 Component Selection - External Light Sensor

3.11.2.1 TSL25403

The TSL25403 is a high-sensitivity light-to-digital converter that uses a visible light channel and an infrared light channel together to accurately calculate ambient light. It also includes an interrupt pin that can be used to trigger firmware interrupts in hardware controllers. Additionally, it requires a 1.8 V supply, and costs $0.97 per unit [53].

3.11.2.2 LTR-303ALS-01

The LTR-303ALS-01 is another light-to-digital converter, covering a range from 0.01 to 64 thousand lux, also utilizing both a visible light channel and and infrared light channel together to produce accurate results. It also has an interrupt pin, and can handle a supply voltage of up to 3.6 V, and costs $0.70 per unit citeltr-303als-datasheet.

3.11.2.3 LTR-329ALS-01

The LTR-329ALS-01 light-to-digital converter, like the LTR-303ALS-01, also covers a wide range from 0.01 to 64 thousand lux, and can handle a supply voltage up to 3.6 V. Unlike the other version though, the LTR-329ALS-01 does not include an interrupt pin, and so costs $0.39 per unit citelte-329als-datasheet.

3.11.2.4 Component Comparison - External Light Sensor

All three components in question have excellent resolution, but the interrupt pins would not be necessary for this application, and using a supply voltage of 1.8 V would likely necessitate the use of a voltage regulator. With a maximum 3.6 V supply voltage, the LTR family sensors could be powered directly by the MCU or the voltage regulator powering it. Therefore, the LTR-329-ALS-01 is the optimal choice for its price and higher input supply voltage.

TABLE 3.29: External Light Sensor Component Comparison

Name	**TSL25403**	**LTR-303ALS-01**	**LTR-329ALS-01**
Interrupt Pin	Yes	Yes	No
Supply Voltage	1.8V	3.6V	3.6V
Cost per Unit	$0.97	$0.70	$0.39

3.12 Power Technology

3.12.1 Power Distribution Requirements

To aptly select power distribution technologies and components, it is necessary to analyze the power requirements of each active component. The components listed below will be the main components actively drawing power from the supply, and vary rather widely in maximum input voltage and the current they may require. The MCU, cameras, and ambient light sensor can all operate on a 3.3 V supply, and could all be powered by a single regulator, one that could provide about at least 1100 mA. The Line laser, display screen, and Raspberry Pi can all be powered by a 5 V source, but such a source would need to supply at least 5185 mA. The op-amp can handle up to a 30 V supply voltage, but does not necessarily require 30 V. The IR LEDs, as part of the strobe circuit, will be powered by a capacitor bank, allowing energy to accumulate and be stored over time

so that it can be instantly released as a high current signal that can drive the LEDs. This removes the strain of powering the strobe circuit from the main power supply, and places it on a capacitor bank and switching circuit. In total, the PTC would require at least 6.3 A at 5 V. Furthermore, to drive the IR LEDs at 1 A, each LED would exhibit a voltage drop of about 3.2 V citeAC353A-YRR-1000mA-IR-LED-Datasheet. Of course, if each LED was placed in its own parallel circuit, that would place an extra burden of about 30 A on the PTC, likely on 30 separate capacitor banks. Putting all of the LEDs in series would require a voltage of more than 90 V. But, by placing 5 LEDs in series, in 6 parallel circuits, a common voltage of 24 V can be used to power each parallel strobe circuit, and this would greatly reduce the total amount of current needed during strobing, and would also reduce the number of capacitor banks necessary to supply said current.

TABLE 3.30: Component Power Requirements

Component	**Max Supply Voltage**	**Max Supply Current**
Adafruit 1057 Line Laser	5.2V	25 mA
Raspberry Pi Global Shutter Camera	3.3V	N/A
AC3535A-YRR-1000MA IR LED	N/A	1.5 A
RP2354A MCU	3.3V	1.0 A
Raspberry Pi Model 5	5.0V	5.0 A
Waveshare 2.8-inch Screen	5.0V	160 mA
TL084 Op-Amp	30V	N/A
LTR-329ALS-01 Ambient Light Sensor	3.6V	N/A

3.12.2 Power Source Technology

3.12.2.1 Battery

Batteries would allow our device to work in any suitable environment without the need for a power receptacle nearby. In order to use a battery, however, we must conform to project requirements, and cannot implement a charging circuit. Batteries can also become unreliable at colder temperatures, or dangerous at higher temperatures, necessitating strict safety standards. However, in ideal circumstances achieving 24 V would be as simple as wiring a few batteries in series, and supplying 6.3 to 6.5 A should be simple, as many batteries are capable of safely discharging large currents.

3.12.2.2 Switched-Mode Power Supply

Switched-mode power supplies, much like switched converters, use a switching action to modulate the input voltage into a square wave. The duty cycle of this square wave is controlled such that the average value is the desired output voltage, and then this signal is filtered to produce a smooth output with minimal power loss, although there will be a slight ripple. Typically, switched power supplies are more efficient when switching a

high voltage to a lower voltage. Finding one that could output at least 6.3 A at 24 V would not be hard.

3.12.2.3 Linear Power Supply

A linearly regulated power supply uses a mixture of passive and active components to control the output voltage level, restricting the flow from input to output using a feedback loop. Typically, linear regulators are most efficient when the input voltage is within a small range above the output voltage, and handling greater voltage differences between the two will cause excessive heat dissipation and power loss. Finding one that could supply more than 6.3 A at 24 V may be difficult.

3.12.2.4 Transformer

A transformer will be able to directly convert one voltage level to another via induction, providing efficient conversion that maintains the same power and isolates the lower voltage output from the higher voltage input. Transformers maintain high efficiency, and don't create a ripple effect on the output like a switched power supply would. Additionally, by stepping down the voltage between input and output but still maintaining the same power, significantly higher currents can be safely delivered without overloading or damaging the supply.

3.12.2.5 Technology Comparison

Although batteries would be very portable and less expensive than some other options, they constrain how long the PTC can operate for, as batteries would need to be recharged. A linear supply, while usually the cheaper option, would be far more expensive here due to the difficulty of safely and efficiently regulating power with linear technology at such high voltage and currents. A switched mode supply is a good option, but may struggle to supply high currents efficiently. Therefore, a transformer would be the best option because it would be able to take wall power and efficiently convert it into a high current supply at 24 V.

TABLE 3.31: Power Source Technology Comparison

Name	**Switched-Mode Supply**	**Battery**	**Linear Supply**	**Transformer**
Stability	Reliable	Less Reliable	Reliable	Reliable
Efficiency	Efficient	Efficient	Least Efficient	Most Efficient
Portability	Less Portable	Most Portable	Less Portable	Less Portable
Cost	Least Expensive	Less Expensive	Most Expensive	More Expensive

3.12.3 Component Selection - Power Supply

3.12.3.1 Alitove 192W Transformer

The Alitove 192W transformer is capable of delivering 8 A at 24 V, far exceeding the minimum projected current necessary to run all components of the PTC. The Alitove also has the added benefit of being housed inside a package much like that of a laptop charger, with plugs included, allowing our members to safely use wall power and the transformed output without directly connecting bare wires to screw terminals. It is priced at $39.99 [54].

3.12.3.2 LRS-150-24

The LRS-150-24 is a 156 watt power supply capable of delivering up to 6.5 A at 24 V. Its honeycomb shell allows ample airflow to keep itself cool, and it has two output channels. While certainly capable of supplying the PTC, its near open-air construction brings some danger into its use, as 120-240V wall power and the 24 V, 6.5 A output must be connected to the LRS-150-24 via screw terminals. Using this power supply would necessitate clear caution to prevent risk of and harm from electrocution. It is priced at $18.40 per unit [55].

3.12.3.3 Bingotec 150W Transformer

The Bingotec 150W transformer shares similar specifications to the LRS-150-24, with 24 V output up to 6.5 A, and rated for 150 W. At a price of $18.85, it also delivers a slightly smaller form factor. However, its design is much like that of the LRS-150-24, and uses screw terminals for all connections and a honeycomb casing for airflow, increasing the risk of shock. Using this component would also require clear caution to prevent such circumstances [56].

3.12.3.4 Component Comparison - Power Supply

Due to safety concerns and convenience, there is one clear winner. Both the LRS-150-24 and the Bingotec 150W would require members of this project to handle bare 120VAC wires in order to wire them into the devices' screw terminals, as well as handle 24 V, 6.5 A wires on the output. The Alitove eliminates these concerns as it not only comes in a housing that does not leave inner components exposed, but also comes with plug terminations so that bare wires do not have to be handled. In this case, the Alitove transformer is worth the cost.

TABLE 3.32: Power Source Component Comparison

Name	**Alitove**	**LRS-150-24**	**Bingotec**
Max Current (A)	8.0	6.5	6.5
Housing	Enclosed	Exposed	Exposed
Terminations	Plugs	Screw Terminals	Screw Terminals
Cost per Unit	$39.99	$18.40	$18.85

3.12.4 Voltage Regulation

To ensure components are powered properly and that their requirements and specifications are met and not exceeded, voltage regulation is necessary. This can be achieved with a small variety of technologies. With the strobe LEDs in mind and the large voltage drop across them necessary in order to power them, it is likely that we will need a power supply that supplies a larger voltage, and as such, we should select our voltage regulation technologies accordingly.

3.12.4.1 Switching Voltage Regulator

Switching voltage regulators employ high frequency switching to interrupt the source voltage and produce an intermittent waveform. This waveform is like a square wave, and its duty cycle determines the voltage of the regulator's output. In order to achieve a stable output, the square wave is filtered into a smooth DC output. Due to the switching action inside, a small ripple will still appear on the output, varying the output voltage by a small amount. Switching regulators often have higher efficiencies because of their switching, and are often suitable for regulating large voltages down to smaller ones without significant power loss and heat dissipation. Switching regulators often require external components to function, as their timing circuits may require a large inductance, and inductors are typically too large to fit into a small integrated circuit.

3.12.4.2 Linear Voltage Regulator

In a linear regulator, the output voltage is controlled linearly by an Op-Amp and transistor switch, constantly adjusting how much current is let through the device based on the voltage across the load. Since linear regulators control the output voltage by controlling how 'on' a transistor is, they are typically less efficient, and dissipate more heat as a result. Linear regulators are typically unsuitable for regulating large voltages down to small ones, as a larger difference directly correlates to lower efficiency. Linear regulators also exhibit dropout behavior, where if the input voltage falls too close to the output voltage, the output voltage can no longer be correctly maintained, and falls as a result.

3.12.4.3 Low Dropout Voltage Regulator

Low dropout regulators are just like standard linear voltage regulators, but with optimized circuitry inside to minimize the effects of dropout behavior, greatly increasing the available margin available between input and output voltage before a decrease on the input causes the regulation of the output to fail. This property allows LDOs to be used in circuits where the input voltage has to be close to the regulated output voltage.

3.12.4.4 Zener Diode

A Zener diode is a type of diode that has a controlled avalanche breakdown, and is meant to operate in such a state. The breakdown voltage can be manufactured, and this allows us to create a rudimentary voltage regulation system by placing a Zener diode in reverse bias from the power rail to ground. Voltages above the Zener voltage will cause the diode to start conducting, causing the voltage of the power rail to stay under a certain limit.

This accounts for over-voltage conditions by essentially causing a short, which may not be ideal if such a condition persists. In order to maintain a voltage above a certain level, another diode would need to be used.

3.12.4.5 Technology Comparison - Voltage Regulator

There are many components that will be used to make the PTC function, and as such it would be prudent to select more than one of these technologies. Switching regulators are generally more expensive than linear regulators, but usually are very efficient because of their switching action. Low-dropout regulators are less efficient, but simpler to implement. This makes switching regulators an ideal choice for any instances where we would need to convert a higher voltage, and LDOs an ideal choice for conversions where the voltage difference between input and output is not so large. Regular linear regulators simply don't have the extra advantage of a low-dropout design, and Zener diodes cannot sufficiently maintain a stable voltage, only clamp it. So, the PTC will use a combination of switching regulators and LDOs.

TABLE 3.33: Voltage Regulator Technology Comparison

Name	**Switched Regulator**	**Linear Regulator**	**Low-Dropout Regulator**	**Zener Diode**
Efficiency	High	Medium	Medium	Low
Adjustable	Yes	Optional	Optional	No
Output Ripple	Yes	No	No	No
Adaptable to changes in Input	Yes	Yes	Yes	Somewhat
Cost	More Expensive	Cheap	Less Expensive	Cheap

3.12.5 Component Selection - Voltage Regulators - 3.3V

3.12.5.1 AP5100

The AP5100 is suitable for our needs, being able to handle a rated input voltage anywhere from 4.75 V to 24 V, while supplying 1.2 A. The Raspberry Pi RP2350 Hardware Design guide suggests a 1 A supply current to be suitable for computationally intensive projects, so the AP5100's current capabilities are suitable for both the RP2354A we will be using as well as other components that will not draw much current. The AP5100 is adjustable, and its datasheet also supplies an example application circuit that perfectly suits our needs, a 3.3 V regulator. The AP5100 costs $0.60 per unit, and comes with built-in protections for the regulator [57].

3.12.5.2 LM2575-3.3WT

The LM2575-3.3WT is a fixed output switched buck regulator, capable of taking in inputs varied from 4 to 40 V, well exceeding our 24 V supply. It has an enable pin to

be turned on or off, and, additionally, it can supply up to 1 A of current to its load. The example circuits in the datasheet show applications with an adjustable version, or the 5 V fixed output version. While sufficient, this regulator may not be able to handle both the RP2354A and other 3.3 V components at the same time during worse-case peak-current consumption. It costs $1.72 per unit, but comes in a larger and easier to handle package citeLM2575 datasheet.

3.12.5.3 LM2594M-3.3-EV

The LM2594M-3.3-EV is also capable of managing inputs from 4 V to 40 V, and can supply up to 1.5 A of constant current. It also has an enable pin to turn it on or off with a signal. It comes with built-in protections, and costs $1.27 per unit. With 1.5 A of supply current capability, this component should have no issues powering the RP2354A as well as other 3.3 V components citeLM2594 datasheet.

3.12.5.4 Component Comparison - 3.3V Regulators

The LM2575-3.3WT does not seem to come with internal protections, and is also the most expensive of the options. Not having internal protections could leave the PTC's components open to danger. Meanwhile, the AP5100 will be able to support our MCU and other devices with 1.2 A, has internal protections, and is the most cost-effective option.

TABLE 3.34: 3.3V Regulator Component Comparison

Name	**AP5100**	**LM2575-3.3WT**	**LM2594M-3.3-EV**
Max Current (A)	1.2	1.0	1.5
Built-In Protections	Yes	No	Yes
Adjustable	Yes	No	No
Cost per Unit	$0.60	$1.72	$1.27

3.12.6 Component Selection - Voltage Regulators - 5.0V

3.12.6.1 R78K5.0-0.5

Like other switching regulators, the R78K5.0-0.5 can efficiently regulate voltages, and this one would also be able to handle a 24 V source and supply 500 mA of current. The R78K5.0-05 will regulate to a 5.0 V output, suitable for our line laser and display screen. The added advantage of this regulator is that it is completely housed within a single package, requiring no external components. As is, this regulator can be directly connected to a circuit. It costs $2.79 per unit [58].

3.12.6.2 MPQ4341GLE-5-AEC1-Z

The MPQ4341GLE-5-AEC1-Z is capable of taking an input voltage ranging from 3.3 V to 36 V, and regulating to 5 V while supplying 5 A of continuous current. This is sufficient to power our Raspberry Pi Model 5 while also powering other devices like our display screen. Its switching frequency is also configurable from 350 kHz to 2.5 MHz, allowing us to choose a higher frequency to minimize ripple. It costs $5.77 per unit citeMPQ4341GLE.

3.12.6.3 AP1501A-50K5G-13

The AP1501A-50KG-13 is a buck DC/DC converter capable of handling inputs ranging up to 40V, and is able to supply 5A of load current. It also has internal protection circuitry to limit thermal runaway and excessive currents. While seemingly perfectly suitable to power our Image Processor, the datasheet for this device is watermarked with a warning that it is not suitable for new designs. It costs $3.85 per unit citeAP1501AA.

3.12.6.4 Component Comparison - 5.0V Regulators

Here, it is worth getting two options. The Raspberry Pi 5 will consume 5 A, so it would be reasonable to have another, easy to use regulator available to provide power to other 5 V components that will not require as much current. It is also well worth heeding the manufacturer's warning for the AP1501A, so we have chosen the R78K5.0-0.5 and MPQ4341GLE-5-AEC1-Z.

TABLE 3.35: 5.0V Regulator Component Comparison

Name	**R78K5.0-0.5**	**MPQ4341GLE**	**AP1501**
Max Current	500 mA	5.0A	5.0A
Built-In Protections	Yes	No	Yes
Adjustable	No	No	No
Cost per Unit	$2.79	$5.77	$3.85

3.12.7 Component Selection - Voltage Regulators - Strobe

For the strobe circuit, since it will be powered by a local capacitor bank instead of our main supply, it is necessary to maintain a stable voltage so that all LEDs in series have a similar brightness. As the LEDs draw about an amp, a Low-Dropout regulator can pull current from a local capacitor bank to supply the LEDs while maintaining a constant output voltage. Low-Dropout regulators are especially advantageous here because, as the capacitor bank decreases in voltage while discharging, LDOs will be able to maintain their output longer than standard linear regulators.

3.12.7.1 LD1086DTTRY

The LD1086DTTRY is an automotive grade low-dropout regulator that is a reasonably priced option at $1.76 per unit. It can handle inputs up to 36V and will allow us to deliver

large currents of up to 1.5 A and a dropout voltage of 1.2V at maximum output current. Additionally, the output is adjustable with a simple formula that can be calculated by hand, allowing us to change how many LEDs we power in a strobe circuit if necessary [59].

3.12.7.2 LT1086

The LT1086 is a more expensive option, at $6.42 per unit, but it can handle inputs up to 36V and will also allow us to deliver 1.5A of current with a 1V dropout. It is also adjustable, although the formula to do so is slightly more complex then the simple formulas provided for the LD1086 and the LT3080. The LT1086 has the added benefit of having a spice model in LTspice to simulate with citeLT1086.

3.12.7.3 LT3080

The LT3080 is another option that has the added advantage of having a spice model in LTspice, and priced at $6.05 per unit, it can handle inputs up to 36V and is capable of outputting 1.1A. It is also adjustable with a simple formula like that of the LD1086DTTRY it also has a very low dropout voltage of 350mV citeLT3080.

3.12.7.4 Component Comparison - Strobe Voltage Regulators

While it is helpful that the LT1086 and the LT3080 have spice models in LTspice, the LD1086DTTRY is an LDO with similar performance ratings, and far cheaper than the other options. It also is adjustable with the same simple formula as the LD3080, but with more current delivery. Furthermore, we need one of these for every parallel strobe circuit, and so our choice is the LD1086DTTRY.

TABLE 3.36: 5.0V Regulator Component Comparison

Name	**LD1086DTTRY**	**LT3080**	**LT1086**
Max Current (A)	1.5	1.1	1.5
Spice Model Available	No	Yes	Yes
Adjustable	No	No	No
Cost per Unit	$1.76	$6.05	$6.42

3.12.8 Inrush Current Regulation

The PTC may employ a custom strobe circuit in order to capture images of projectiles moving through the system, and as such may require methods of current regulation to protect the system and its components from overheating or straining the power supply. In this, there is one main area of concern: the capacitor banks that will supply each strobe circuit. Due to their potentially large capacitance, the capacitor banks will act as short circuits until they charge up, potentially causing surges of several amps worth of electricity, and no doubt damaging several components along the way. Below are a few technologies that could manage inrush current.

3.12.8.1 Soft Start Circuit

Using a MOSFET, it is possible to implement a soft start circuit that, upon startup, will limit current, and slowly reduce resistance as the gate of the MOSFET charges and it turns 'on'. The inrush current will then have settled, effectively protecting components from massive current surges due to capacitors acting like short circuits when empty. This circuit also has the added benefit of not causing much of a voltage drop, allowing its output to remain at voltage close to that of its input. This approach requires more components than a simple resistive element, but would reduce power loss and heat dissipation.

3.12.8.2 NTC Thermistor

A thermistor changes resistance based on its temperature. With a Negative Thermal Coefficient thermistor, it would be possible to limit inrush current in a simple manner: the thermistor would exhibit high resistance at first, but then slowly become less resistive as it warmed up, preventing massive surge currents. However, because this component is still resistive, it would have a voltage drop across itself, reducing the available voltage at its output compared to the input. Thermistors also often have long 'reset' times, on the order of several seconds to tens of seconds, and will take a while to return to their base state and resistance. This means that inrush currents may not be effectively dealt with if the system is shut off, the capacitors are drained, and then the system is immediately turned on again, as the thermistor may still be at a temperature that correlates with low resistance, potentially leading to component damage.

3.12.8.3 Typical Resistor

While simple, a typical resistor must be able to handle the power it absorbs, and this could mean that this resistor ends up being larger than others in the circuit, and depending on how much power is expended, could necessitate a heat sink or other method of dispersing heat due to lost power. If used in a section of power delivery, the output voltage could be greatly decreased based on the resistance value of the resistor, reducing efficiency and generating heat.

3.12.8.4 Technology Comparison - Inrush Current Regulation

Since we intend to rely on a large capacitor bank to locally provide current to the strobe circuit, which houses sensitive LEDs, managing inrush current in a timely manner and without exposing the circuitry to unnecessary danger is important. Because of this, it makes sense to use a gate-driven soft-start circuit. This will prevent damage to the PTC if it is turned off and on again, whereas an NTC thermistor may not be able to protect the circuitry in such circumstances. Additionally, a soft-start circuit will allow the capacitor bank to be charged at max voltage, as there won't be a significant voltage drop between the supply and the capacitor bank due to a resistive element.

TABLE 3.37: Current Regulation Technology Comparison

Name	**Soft-Start Circuit**	**NTC Thermistor**	**Typical Resistor**
Efficiency	High	Medium	Low

Complexity	Medium	Low	Low
Safety	Safe	Less Safe	Safe
Cost	More Expensive	Less Expensive	Cheap

3.12.9 Component Selection - Inrush Current Regulators

Inrush-current limiting circuits are often found in voltage converters, and their construction is rather simple. As such, it has been difficult to find a single component that could accomplish this task in an IC, and so, because the design is simple, we have elected to build such a circuit ourselves. By relying on the gate capacitance of MOSFETs and the delay it can cause, a gate-driven inrush-current limiting circuit can be devised with a single P-Channel MOSFET and some extra capacitors and resistors. Below are a few suitable options.

3.12.9.1 AONS21303C

The AONS21303C is a P-Channel MOSFET that can withstand 180 A of continuous current, a Drain-Source voltage of up to -30 V, and a Gate-Source voltage of up to plus or minus 20 V. This option also has the added benefit of being available to simulate in LTspice, making general analysis of circuit behavior easier. Due to its extremely high current carrying capabilities, however, this component is priced at $2.09 per unit. [60]

3.12.9.2 TPS25940 eFuse

The TPS25940 is an integrated eFuse device that is designed for inrush current limiting in low-voltage systems. This device operates over an input voltage range of 2.7 V to 18 V and can support a continuous current of up to 5 A. It incorporates an internal MOSFET with programmable current limiting. The current limit threshold can be set using an external resistor, and the soft-start functionality of this device limits the rate of rise of the output voltage. This device also provides integrated protection features such as overcurrent and overvoltage protections. The TPS25940 has a fast response time on the order of microseconds, and can cost approximately $1.00 per unit. [61]

3.12.9.3 MF73T-1

The MF73T-1 NTC is a thermistor that provides inrush current suppression for electronics. This device operates at an initial resistance of about 5 Ω at room temperature, limiting the inrush current during power initialization. As current flows though the thermistor, it will heat up and the resistance will drop to below 0.5 Ω typically. This device is rated for a steady-state current of appoximately 2-3 A. The MF73T-1 has a thermal time constant on the order of seconds, making it much slower compared to the other components described. However, the thermistor is a very low-cost solution, typically being around %0.30 per part. [62]

3.12.9.4 Component Comparison - Inrush Current Regulators

The AONS21303C provides extremely high current handling capabilities and can be easily modeled in simulation software, making this the ideal component for the PTC.

TABLE 3.38: Inrush Current Limiter Comparison

Characteristic	**AONS21303C**	**TPS25940**	**MF73T-1**
Type	P-MOSFET	eFuse (Active IC)	NTC Thermistor
Voltage Range	Up to 30 V	2.7–18 V	N/A
Current Capability	∼180 A	Up to 5 A	2–3 A
Inrush Control Method	External circuit	Programmable current limit	Passive resistance drop
Response Time	Fast	μs range	Slow (thermal)
Protection Features	None	OCP, OVP, UVLO, thermal	None
Complexity	High	Moderate	Very low
Repeatability	High	Very high	Low
Cost ($)	2.09	∼1.80	∼0.30

3.12.10 Transistor Switches

The PTC must be capable of capturing projectiles at different points in its journey through the PTC in order to calculate trajectory information. For this, it is imperative that the strobe circuits in use can be switched on and off in a timely manner.

3.12.10.1 BJT

Bipolar Junction Transistors (BJTs) are current controlled semiconductor devices known for their fast switching and simplicity. The downside is that they are current controlled, and currents from the source and load can flow into and out of the base, leading to current draw on the MCU, or potentially backflow, if voltages or currents exceed the transistor's specifications. BJTs are often limited in the amperage they can switch, and are therefore not used often for high-power switching tasks.

3.12.10.2 JFET

Junction Field Effect Transistors (JFETs) share an insulated Gate with other transistor technologies, and have the added advantage of being able to be used with AC signals without extra considerations due to the absence of a 'body diode'. This means that the drain and source pins are considered interchangeable, and the orientation of a JFET does not matter. But, JFETs are relatively uncommon and expensive. Furthermore, the operation of their gate works differently than other transistors, as for a P-Channel JFET, a positive voltage on the gate will turn it 'off', a voltage of 0 V will turn it completely 'on', and a negative voltage will likely cause damage to the device, and within specifications, this happens regardless of the voltages on the drain and gate. This is different from

something like a PNP BJT, where the emitter must be at a higher voltage than the base in order for the BJT to conduct, and won't do so otherwise.

3.12.10.3 MOSFET

Metal-Oxide Semiconductor Field Effect Transistors (MOSFETs) don't often use a metal oxide anymore, but the name stuck around. These transistors have an insulated gate, preventing any current draw from or backflow into the MCU. MOSFETs often require driver circuits to ensure they are properly powered and switch fast enough for their applications due to gate capacitance, but these capacitances also allow them to be used in soft-start circuits. The orientation of a MOSFET must be taken into consideration, as the wrong orientation can cause forward-biasing of an internal 'body-diode' and cause unwanted conducting, invalidating the purpose of the switch. MOSFETs can have very low drain source on resistance, allowing for high current and high power applications. One weakness of MOSFETs is the susceptibility to static shocks or transient voltage spikes, which can burn through the insulating oxide layer of the gate. We will use a MOSFET for switching the current to the strobe lights.

3.12.10.4 Enhancement Mode / Depletion Mode

Field Effect Transistors come in either Enhancement Mode or Depletion Mode. Depletion mode transistors are far less common than enhancement mode transistors, and therefore more expensive. This project does not require the unique properties of a depletion mode transistor, so we will be using enhancement mode transistors.

3.12.10.5 NPN/PNP/N-Channel/P-Channel

BJTs come in either NPN or PNP variants, and FETs come in either N-channel or P-channel variants as well. NPN and N-channel transistors turn 'on' with a positive voltage applied to their base or gate relative to the emitter or source pins, and PNP and P-channel transistors turn 'on' with a negative voltage applied to their base or gate relative to the emitter or source pins.

3.12.10.6 Technology Comparison - Transistor Switches

MOSFETs were selected for the PTC project due to their efficient, high-current switching capabilities while maintaining safe and reliable interfacing with the microcontroller.

TABLE 3.39: Transistor Comparison

Feature	**MOSFET**	**BJT**	**JFET**
Power Capability	High	Low	High
Ubiquity	High	High	Low
Control Method	Voltage	Current	Voltage
Ease of use	Moderate	Easy	Moderate
Cost	Low	Low	High

3.12.11 Component Selection - MOSFETs

3.12.11.1 Component Comparison - MOSFETs

There are many different MOSFETs, each for a different purpose. For switching the strobe LEDs, we will need a low $R_{DS(on)}$ or the resistance between drain and source while it is on. Our strobe circuit will have 1 or 2 ohms of resistance, and drive around 1 amp of current. Therefore we want $R_{DS(on)}$ to be less than 100 milliohms to avoid changing the series resistance of the strobe circuit by more than 10%. The strobe circuit needs to turn on for 100 μs, so the turn-on time needs to be less than 10 μs. We will be switching a voltage of up to 24 V, so the MOSFET needs to have a breakdown voltage of higher than 24 V. Since we plan to use a MOSFET driver, we don't need to have the gate work at logic level voltages like 3.3 V, instead we can drive it with 12 V - 24 V, which expands the selection of components considerably. One common "jellybean" MOSFET is the 2N7000, however the $R_{DS(on)} = 5\ \Omega$, which is much worse than what we need. The PJA3406_R1 MOSFET by Panjit is a good fit, with a voltage, rise time, and other specifications that meet our needs. The PJA3404A_R1 MOSFET is slightly better due to the lower $R_{DS(on)}$, so that is what we chose. The MMFTN3402 by Diotec Semiconductor would switch faster, but has a higher $R_{DS(on)}$, and since we only need under 1 to 10 microseconds of rise time, the faster switching speed of nanoseconds is unimportant.

TABLE 3.40: MOSFET Comparison

Feature	**2N7000**	**PJA3404A**	**PJA3406**	**MMFTN3402**
Manufacturer	Microchip	Panjit	Panjit	Diotec Semiconductor
$R_{DS(on)}$	5 [Ω]	23 [mΩ]	35 [mΩ]	55 [mΩ]
Gate Charge Q_g	200 [pC]	12.8 [nC]	5.8 [nC]	5.9 [nC]
Rise Time	10 [ns]	34 [ns]	39 [ns]	3.3 [ns]
Drain Current	0.2 [A]	5.6 [A]	4.4 [A]	1.9 [A]
Drain-Source Voltage	60 [V]	30[V]	30[V]	30[V]
Gate Threshold Voltage	3 [V]	2.1 [V]	2.1 [V]	1.4 [V]

3.12.12 MOSFET Driver

MOSFETs work by charging or discharging the gate capacitor to turn the transistor on or off. The capacitance of the gate means that the transistor takes time to turn on or off, limited by the current into or out of the gate. While the simplest way to control a MOSFET is to use a pull-up or pull-down resistor combined with a resistor on a GPIO pin of the microcontroller, this leads to slow switching times, and is limited by the microcontroller voltage in how high the gate voltage can be driven. A MOSFET driver is a microchip that can quickly source and sink large amounts of current in order to charge and discharge the gate of a MOSFET, turning the MOSFET on or off rapidly. Often using BJTs to amplify both the voltage of the microcontroller and the current, these can take the low voltage low current GPIO of the microcontroller and provide a high current and higher voltage signal to the MOSFET gate.

3.12.13 Component Selection - MOSFET Driver

3.12.13.1 Component Comparison - MOSFET Driver

A 4.5 amp high-speed power MOSFET driver, the MCP1403/4/5 is a solid general purpose MOSFET driver. The MCP1406/07 is a higher current version, capable of 6 amps instead of just 4.5 amps. The IXDN602 has specifications similar enough for our purposes, but only 2 amps of current. The ZXGD3009DY was a far less expensive option, yet it still meets the specifications that we required.

TABLE 3.41: MOSFET Comparison

Feature	**ZXGD3009DY**	**MCP1404**	**MCP1407**	**IXDN602**
Manufacturer	Diodes	Microchip	Microchip	IXYS
Peak Source/Sink Current [A]	2	4.5	6	2
Delay [ns]	40	40	40	38
Max Supply Voltage [V]	40	20	20	40
Cost	$0.19	$2.62	$1.17	$1.39

3.13 Communication Technologies

Dedicated sensors and external ICs can accomplish various tasks, and perhaps to a greater degree of effectiveness than the MCU, without using up much of the MCU's processing power. Although, in order to use these external sensors and IC's, they must be able to communicate with the MCU and trade information. This process can be facilitated through a variety of communication protocols that are common in industry. Additionally, our MCU and Image Processor must be able to communicate with each other, and this may necessitate using another protocol.

3.13.1 Communication Protocols

3.13.1.1 I2C

I2C, which stands for Inter-Integrated Circuit, is a synchronous communication protocol that utilizes two wires from the MCU as a bus to send and receive information from multiple devices. I2C is able to facilitate communication between many devices with a total of 2 wires: a clock wire driven by the master chip, and a data wire that is used intermittently based on which device the master chip wishes to receive information from. This allows us to control when information is queried from sensors, partially allows us to control how fast this process happens, and saves space on our board and conserves our GPIO pins by limiting the amount of wires in use.

3.13.1.2 SPI

SPI stands for Serial Peripheral Interface, and is another communication protocol that is commonly used, especially for small displays. Unlike I2C however, it is not thoroughly standardized, and there can be small differences in how different pins used for SPI may be named. SPI is also a synchronous protocol, but in order to use multiple devices with the SPI protocol, they must be daisy-chained. In a daisy-chain, data that sent to a device further down the chain must first pass through all previous devices, which can complicate software and slow down the transfer of information. SPI also typically involves 4 data pins, more than I2C or UART.

3.13.1.3 UART

UART is an often used asynchronous protocol, and stands for Universal Asynchronous Receiver-Transmitter. UART is simple and especially useful for communications between devices that operate at different clock speeds, and can be configured as such. UART typically requires two data lines and a ground connection. The PTC will likely require an asynchronous connection between the MCU and the Image Processor, since the Image processor has a higher clock speed. This is facilitated by using an agreed upon Baud rate, or rate of transmission. By doing so, both devices can send and receive data at the same speed, while their processors operate at different speeds.

3.13.1.4 Technology Comparison

The PTC project requires a user-interface and communication between two computational devices. The user-interface will incorporate a screen display, and this will likely be driven by the SPI protocol. As for communication between the MCU and Image Processor, it makes sense to use UART, since speed of computation and data transmission is of less importance after the superimposed images of the projectile have been taken and processed. If extra sensors are added to improve calculations and extrapolation of results, these will likely use I2C. Therefore, all of these technologies are extremely useful and will be beneficial to the implementation of the PTC.

TABLE 3.42: Comparison of Communication Protocols

Protocol	**Type / Key Feature**	**Notes / Advantages**
I2C	Synchronous, 2-wire bus	Facilitates communication with multiple devices using a clock and data line; conserves GPIO pins; allows controlled querying speed
SPI	Synchronous, daisy-chain	Commonly used for displays; may require 4 pins; devices must be daisy-chained, which can complicate software and slow transfer

UART	Asynchronous	Useful for devices with different clock speeds; uses 2 data lines + ground; baud rate ensures synchronized transmission

3.14 Software Technologies

The software component of the Projectile Tracking Chronograph (PTC) is responsible for real-time hardware control, high-speed image acquisition and processing, trajectory computation, and user interface rendering on the Raspberry Pi 5. Selection of appropriate development tools, programming languages, and libraries was critical to achieve reliable real-time performance within the constraints of an embedded platform.

Because the system must react quickly to projectile detection events, the software must be efficient, modular, and capable of handling multiple tasks such as sensor monitoring, image processing, and data visualization simultaneously. This requires careful consideration of both performance and maintainability during development.

In addition, version control is essential for tracking changes, enabling collaboration, and maintaining code integrity in a multi-module project. Git was easily selected as the version control system due to its widespread adoption, excellent branching capabilities, and seamless integration with Visual Studio Code and GitHub. It allows the team to work concurrently on different subsystems while maintaining a stable main branch. [63]

3.14.1 Integrated Development Environments

Several Integrated Development Environments (IDEs) were evaluated for Python-based development and debugging on the Raspberry Pi 5. The primary options considered were Thonny, PyCharm Community Edition, and Visual Studio Code.

3.14.1.1 Thonny

Thonny is a lightweight IDE that comes pre-installed on Raspberry Pi OS and is designed primarily for beginner-level Python development. It provides a simple and intuitive interface along with built-in debugging tools that allow users to step through code and inspect variables in real time. This makes it useful for quick testing, prototyping, and verifying small portions of the PTC software, such as sensor input or basic image capture routines. However, Thonny lacks advanced features such as strong Git integration, project-level organization, and support for large multi-file systems. Since the PTC software consists of multiple interacting modules, including image processing, hardware control, and user interface components, these limitations reduce its effectiveness for full system development. [64]

3.14.1.2 PyCharm

PyCharm Community Edition is a professional-grade IDE that offers advanced code analysis, intelligent autocompletion, and powerful debugging tools. It is well-suited for large-scale software projects where maintainability and code structure are important. In the context of the PTC system, PyCharm provides strong support for managing complex logic such as trajectory computation and multi-module integration. However, one major drawback is its high resource usage. When running on the Raspberry Pi 5, PyCharm can consume significant system memory and processing power, which may interfere with real-time tasks such as image acquisition and processing. This makes it less ideal for development directly on the embedded platform. [65]

3.14.1.3 VS Code

Visual Studio Code is a lightweight and highly extensible code editor that provides a strong balance between performance and functionality. It supports Python development through extensions that enable debugging, linting, and intelligent code completion. Additionally, it includes built-in Git integration, allowing seamless version control and collaboration within the development team. For the PTC system, VS Code is particularly advantageous because it runs efficiently on the Raspberry Pi 5 while still supporting a modular project structure. Its ability to handle multiple files, integrate extensions, and maintain responsiveness during development makes it well-suited for managing the combined hardware and software complexity of the system. [66]

3.14.1.4 Technology Comparison - IDEs

Visual Studio Code was selected as the primary IDE for the PTC project. While Thonny offers simplicity and PyCharm provides advanced features, VS Code achieves the best balance between performance, flexibility, and usability on the Raspberry Pi 5. Its efficient resource usage ensures that development can occur alongside real-time system testing, while its extensibility supports the complex, multi-module nature of the PTC software.

TABLE 3.43: Integrated Development Environment Comparison

Characteristic	**Visual Studio Code**	**Thonny**	**PyCharm**
Resource Usage on RPi 5	Low	Very Low	High
Python Support	Excellent	Good	Excellent
Debugging Capabilities	Excellent	Excellent	Excellent
Git Integration	Excellent	Basic	Good
Extension Ecosystem	Extensive	Limited	Moderate
Ease of Use on Embedded Hardware	High	Very High	Moderate
Suitability for PTC Project	**High**	Moderate	Low

3.14.2 Available Languages

The choice of programming language directly affects development speed, execution performance, and maintainability on the Raspberry Pi 5. The two main candidates evaluated were Python and C++.

3.14.2.1 Python

Python is a high-level, interpreted programming language known for its simplicity and rapid development capabilities. It is widely used in embedded Linux environments such as the Raspberry Pi and provides extensive support for scientific computing and computer vision through libraries such as OpenCV and NumPy. For the PTC system, Python enables quick implementation of image processing algorithms, trajectory calculations, and hardware interfacing. Its readability also improves maintainability, which is important for a multi-developer project. Although Python is slower than compiled languages, its performance is sufficient for the PTC system when combined with optimized libraries and efficient coding practices. [67]

3.14.2.2 C++

C++ is a compiled programming language that offers high execution speed and precise control over system resources. It is commonly used in performance-critical applications and embedded systems requiring low-level hardware interaction. In the PTC system, C++ could provide faster execution for image processing and timing-sensitive operations. However, it introduces increased development complexity, longer implementation time, and more difficult debugging. Given the project timeline and the availability of optimized Python libraries, the performance benefits of C++ do not outweigh the added complexity for this application. [68]

3.14.2.3 MicroPython

MicroPython is a lightweight implementation of Python designed for microcontrollers and resource-constrained embedded systems. It provides a simplified version of Python that can run on devices with limited memory and processing power. In the context of the PTC system, MicroPython is not suitable for the Raspberry Pi's image processing tasks due to its limited library support and lack of compatibility with OpenCV. However, it is well-suited for use on microcontroller-based subsystems, where it can be used to handle simpler tasks such as sensor interfacing or signal triggering. This allows for consistency in programming style across different parts of the system. [69]

3.14.2.4 Technology Comparison - Available Languages

Python was selected as the primary programming language for the PTC system. While C++ offers higher performance, Python provides significantly faster development, better library support, and easier integration with the Raspberry Pi hardware. These advantages are critical for implementing real-time image processing and system control within the project timeline. MicroPython may still be used in supporting subsystems, but Python remains the most suitable choice for the main processing unit.

TABLE 3.44: Programming Language Comparison

Characteristic	**Python**	**C++**	**MicroPython**
Development Speed	High	Low	High
Execution Speed	Moderate (adequate with optimization)	High	Moderate
Library Support (Computer Vision)	Excellent (OpenCV, NumPy)	Good	Limited
Learning Curve	Low	High	Low
Real-time Suitability on RPi 5	High (with optimization)	Excellent	Moderate
Suitability for PTC Project	High	Moderate	Moderate

3.14.3 Image Processing Software

To determine the most effective platform for projectile detection and trajectory estimation, three approaches were evaluated: OpenCV, MATLAB Image Processing Toolbox, and Scikit-image.

3.14.3.1 OpenCV

OpenCV is an open-source computer vision library specifically designed for real-time image processing applications. It provides highly optimized implementations of algorithms such as frame differencing, edge detection, contour extraction, and centroid calculation. These features are essential for detecting and tracking high-speed projectiles within sequential image frames. OpenCV also includes tools for camera calibration and coordinate transformation, which are necessary for converting pixel data into real-world measurements in the PTC system. Additionally, it integrates seamlessly with Python and can efficiently utilize the processing capabilities of the Raspberry Pi 5, making it ideal for embedded real-time applications. [70]

3.14.3.2 MATLAB

MATLAB Image Processing Toolbox provides a structured and mathematically rigorous environment for developing and analyzing image processing algorithms. It offers strong visualization tools, making it useful for understanding system behavior and validating algorithm performance during early development stages. However, MATLAB requires paid licensing and is not optimized for deployment on embedded platforms such as the Raspberry Pi 5. Its higher resource requirements and limited hardware integration capabilities make it unsuitable for real-time execution in the final PTC system. [71]

3.14.3.3 Scikit-image

Scikit-image is a Python-based image processing library built on top of NumPy and SciPy. It provides a wide range of image analysis functions and is easy to use for algorithm development and testing. However, Scikit-image is not optimized for real-time performance and lacks direct support for high-speed video processing and hardware-level camera interfacing. This limits its effectiveness in applications like the PTC system, where rapid frame processing and accurate timing are critical. [72]

3.14.3.4 Technology Comparison - Image Processing Software

OpenCV was selected as the primary image processing framework. Its strong compatibility with embedded systems, excellent real-time performance, zero licensing cost, and seamless integration with Python make it the optimal choice for the PTC project. MATLAB was used only for early conceptual validation, while Scikit-image served as a secondary reference during algorithm development.

TABLE 3.45: Image Processing Software Comparison

Characteristic	**OpenCV**	**MATLAB Toolbox**	**Scikit-image**
Real-time Performance	Excellent	Moderate	Moderate
Embedded (RPi 5) Compatibility	Excellent	Poor	Good
Cost	Free	High (license required)	Free
Integration with Python	Seamless	Limited	Excellent
Speed	High	Moderate	Moderate
Suitability for PTC	High	Low	Moderate

4 Standards and Design Constraints

The PTC system was designed according to established engineering standards to ensure safe operation, reliable measurements, and maintainable system development. Since the project integrates lasers, optical components, embedded electronics, and real-time software processing, multiple categories of standards were considered during both the design and implementation phases. [73]

4.1 Engineering Standards

While a variety of standards are in use, there are a few in particular that we absolutely must understand and adhere to in order to successfully implement our design.

4.1.1 Communication Standards

In order to program certain devices, and for the end user to receive trajectory information from the PTC, certain devices must be able to communicate with each other and transfer data. To power the entire project, our MCU and Image Processor must first be programmed with one communication standard. These two devices must also communicate with each other during operation with another, and, additionally, the display screen must be driven with yet another standard.

4.1.1.1 USB

The USB, or Universal Serial Bus, is an industry standard maintained by the USB-IF (USB - Implementers Forum), a non-profit corporation founded by the companies that originally worked together to create the USB protocol, companies such as Apple, Microsoft, and Texas Instruments. The USB-IF is thus solely capable of defining what criteria a device must meet in order to be considered a proper USB device. The USB 2.0 Specification, originally released on April 27, 2000, serves as the governing document for USB 2.0 compliance. [74]

According to the USB 2.0 standard, in section 4.1, a USB system is made up of 3 categories: the USB interconnect, the USB devices, and the USB host. Every USB system only has one host in a tiered system, and, according to the USB 2.0 standard, the host can only maintain a connection with a maximum of six lower tiers, with a hub at the center of all tiers except the last one. This means that should the PTC project require it, we could build a USB connection several tiers deep (From host to hub, hub to hub, etc.) without violating the standard, ensuring proper functionality. However, our use case is limited to just programming our devices. [74]

The USB 2.0 Specification requires that all high-speed capable devices support specified test modes for their ports, ensuring that there is interoperability across various implementations [75]. This specification defines three signaling rates to accommodate different device requirements. For our application, the low-speed data rate configuration exceeds our requirements for programming and system configuration:

- **High-speed:** Defined at 480 Mb/s $\pm$ 0.05% in section 7.1.11 of the specification.
- **Full-speed:** 12 Mb/s
- **Low-speed:** 1.5 Mb/s

The USB 2.0 Electrical Complicance Test Specification Version 1.07 defines a set of electrical test criteria for USB 2.0 products. For example, for upstream facing ports, the specification requires testing for low-speed and full-speed signal quality, as well as inrush current and back voltage. Additionally, low-speed and full-speed signal quality testing as well as drop/droop characteristics must be categorized for downstream facing ports [75]. Section 7.1.2.2 of the base specification defines template waveform requirements measured at specific test points. A USB 2.0 downstream facing port must meet the

Template 1 transform wave requirements measured at TP2 (each hub downstream port) and a USB 2.0 upstream facing port (without a captive cable) must meet the Template 2 transform wave requirements measured at TP3. [76]

It is our responsibility as engineers to ensure the products we use conform to the USB-IF's specifications like the USB 2.0 Specification, released in the year 2000 and continually updated to optimize performance across a variety of technologies. Other specifications exist for newer USB versions. In order to do this, we can simply check that the manufacturer of a so-called USB device is in fact recognized by the USB-IF as a Standard Adopter, meaning that the manufacturer has proven that their devices meet the USB-IF's specifications. The USB 2.0 specification itself and the list of standard adopters can both be found on the USB-IF's website, USB.org. [74]

4.1.1.2 UART

The Universal Asynchronous Receiver-Transmitter (UART) protocol is a hardware communication protocol that used asynchronous serial communication with a configurable speed. UART communication only need two wires to connect devices, a TX (transmit wire) and RX (receiver wire). UART does not require a clock signal to work, instead relying on both devices being configured to have the same baud rate. A common baud rate for this protocol is 9600 bits per second, with higher rates supporting faster data transfer for time-sensitive applications. For the PTC system, UART can facilitate communication between the microcontroller and other peripherals where simple serial communication is sufficient. [77]

4.1.1.3 SPI

The Serial Peripheral Interface (SPI) protocol is a synchronous serial communication protocol is a protocol that enables high-speed, full-duplex data exchange between a master device and one (or more) slave device(s) [78]. SPI operates using a four-wire interface:

- **Slave Select (SS):** Here, each device contains a dedicated SS pin if it is a slave device. Multiple slave devices can be shared with the same MOSI, MISO, and SCK lines but must have separated SS lines. [78]
- **Master Out Slave In (MOSI):** In this protocol, data is shared from the master device to other slave devices.[78]
- **Master In Slave Out (MISO):** In this protocol, data is shared from the slave device with the master device. [78]
- **Serial Clock (SCK):** This is the clock signal used by the master and slave devices for coordinating the timings for data transfer. [78]

In the PTC system, SPI can be used for communication with the more high-speed peripherals where timing predictability is crucial for system success.

4.1.1.4 CAN

Controller Area Network (CAN) is a serial communication protocol originally developed for industrial and automotive applications. This is a CSMA/CD protocol, which means that each node on the bus can detect collisions and stop trying to transmit the data for a certain amount of time before trying to then retransmit. CAN makes use of differential signaling on a two-wire bus (CAN_H and CAN_L), where the differential voltage corresponds to the dominant and recessive states of the CAN signals. While this protocol is very reliable for applications requiring high safety regulations, the current plan for the PTC does not include the usage of CAN. [79]

4.1.2 Laser Safety Standards

Laser safety is the discipline of identifying, evaluating, and controlling the hazards associated with the usage of lasers. The primary concern with these devices is the tightly concentrated beam of optical power that the laser produces. A 1 mW laser operating the visible spectrum produces an irradiance at the retina approximately 100,000 times greater than looking directly at the sun. This is because the laser is a coherent source with a low-divergence, which can cause great damage to the eye of a person. Higher-power lasers that are used in fields such as medicine, research, and manufacturing cause injuries that are often permanent and severe. [80]

Laser hazards are often categorized into two broad categories: beam hazards and non-beam hazards. Beam hazards are associated with the emitted radiation from the laser itself either through direct, reflected, or scattered contact with the eye or skin. Non-beam hazards come from interactions that are typically caused by laser usage, such as electrical shock from power supplies, fire ignition by high-powered lasers, or toxic fumes generated by certain laser-material interactions. [80]

4.1.2.1 Regulatory Framework

There are three principle documents for laser safety that govern the manufacturing, sale, and workplace use of lasers:

- **FDA/CDRH Regulations (21 CFR 1040):** The FDA's Center for Devices and Radiological Health (CDRH) regulates the manufacturing and sale of laser products under the Federal Laser Product Performance Standard (FLPPS) in the United States. This standard establishes requirements for laser product classification, labeling, and other safety features.

- **ANSI Z136.1:** The American National Standards Institute (ANSI) publishes the ANSI Z136.1, which is documentation on how lasers should be used in the workplace. This standard provides many details on training requirements, control measures, program administration, and other workplace-related protocols.

- **IEC 60825-1:** The International Electrotechnical Comission (IEC) publishes the IEC 60825-1 document, which defines safety requirements and hazard classifications

internationally. This standard shares many similarities with the U.S. standards, but there are differences in labeling requirements and certain numerical limits.

Commercial lasers that are purchased with a hazard classification provided by the manufacturer must conform to the classification requirements set by the FLPPS. These standards all share a common technical foundation, but differ in qualities such as administrative detail and certain requirements/limits. [81]

4.1.2.2 Laser Hazard Classification System

The laser hazard classification is a system that ranks every type of laser product based on its potential to cause biological harm. The classification is determined by comparing the laser's accessible emission to the Accessible Emission Limit (AEL) for each class, with characteristics such as maximum power output, wavelength being determining factors for this classification. The different classes of lasers are as follows:

- **Class 1 Lasers:** These lasers emit radiation that may be visible or invisible, but are incapable of producing radiation levels that would be damaging under normal operating conditions. These lasers may have an accessible beam with very low intensities or even potentially fully enclosed. Class 1 laser are not required to follow personal protective equipment (PPE) or control measure requirements due to their low-level of potential harm. [81]

- **Class 1M Lasers:** These types of lasers are safe for viewing to the naked eye, but could potentially cause harm if viewed through optical instruments such as magnifying lenses. These lasers are still low-power, but could cause harm, so they should have special consideration when optical components are present in the same environment. [81]

- **Class 2 Lasers:** Class 2 lasers are categorized by their emission in the visible spectrum (400-700 nm) and their higher intensity beams. These lasers trigger the eye's natural aversion response, but are generally safe under normal circumstances. [81]

- **Class 3R Lasers:** These lasers are still considered "low-risk" for causing injury to the eye, but exceed the limits of Class 2 lasers. Direct beam viewing could be dangerous, especially for prolonged exposure, so these lasers require basic administrative controls. [81]

- **Class 3B Lasers:** Class 3B lasers are more hazardous to the eye from direct beam viewing as well as reflections. These lasers can also present a minor hazard to biological tissue under prolonged exposure conditions. These lasers require administrative and engineer controls, and necessitate protective eyewear to be worn from exposed individuals. [81]

- **Class 4 Lasers:** These are the lasers that present the highest hazard classification. They are capable of causing significant eye injury from direct, reflected, or scattered

radiation. They also present burning hazards to skin or fire hazards to combustible materials. These laser require the most strict administrative controls and procedures as well as mandatory PPE for personnel. [81]

4.1.3 Electrical and Hardware Safety Standards

Electrical safety standards guided the hardware architecture to prevent shock hazards, unsafe operating conditions, and damage to components. The system design aligns with principles described in IEC 61010, which governs safety requirements for laboratory equipment and measurements. Since the Projectile Tracking Chronograph (PTC) operates as a precision measurement device involving optical sensors, high-intensity strobe lighting, and external power sources, these guidelines directly influence decisions related to grounding, insulation, and safe power distribution. Proper grounding techniques are used to prevent unintended current paths and ensure stable operation during high-speed measurements. [82]

In addition to grounding and power distribution, protection against transient voltage spikes and improper GPIO usage was carefully considered. The Raspberry Pi 5 interfaces with photodiodes, laser modules, and triggering circuitry, making it vulnerable to overvoltage conditions. To address this, voltage regulation and level shifting techniques are incorporated to maintain safe operating levels between components. Current-limiting resistors and protective elements such as diodes are also included to reduce the risk of short circuits, component damage, or failure during repeated testing cycles. These protections are especially important due to the rapid switching behavior of the detection system. [83]

Another important consideration was electromagnetic interference due to the presence of high-speed switching elements such as strobe lighting and digital communication signals. These elements can introduce noise that affects sensor readings and image capture accuracy. To mitigate this, careful PCB layout practices, proper wire routing, and common grounding references are used to reduce interference. Shielding and separation of analog and digital signals are also considered where necessary. These design practices ensure reliable signal integrity and improve the consistency of projectile detection and trajectory calculations under real operating conditions. [73].

Finally, system-level safety was considered in relation to the energy stored within the strobe system. Since the strobe requires rapid discharge of energy from capacitors to produce high-intensity light, proper component ratings and discharge paths are necessary to prevent overheating or component stress. Safety margins were applied to voltage and current ratings, and controlled discharge mechanisms were considered to ensure safe and repeatable operation. These considerations are critical for maintaining both user safety and long-term system reliability in the PTC design.

4.1.4 Data and User Interface Standards

User interaction design for the PTC system follows established human-computer interaction and usability engineering standards to ensure clarity, consistency, and reduction

of operational errors. In particular, the interface design is aligned with ISO 9241, which defines ergonomic requirements for human-system interaction, and ISO/IEC 25010, which outlines software quality models including usability and user satisfaction.

Measurement results are displayed using standardized formatting so users can easily interpret velocity outputs and trajectory information. The interface provides consistent feedback during operation, including system readiness, detection events, and processing status. These design decisions follow ISO 9241 guidelines, which emphasize clarity of information presentation and user control during interactive processes. In addition to clarity and consistency, the interface design incorporates principles of visual hierarchy and minimal cognitive load as described in usability heuristics such as Nielsen's usability principles. Important system states such as "charging," "idle," and "detection complete" are visually distinct to prevent user confusion during operation and to support rapid decision-making in time-sensitive scenarios.

Color coding and text indicators are used together to ensure accessibility and reduce ambiguity in interpretation. This approach aligns with ISO 9241 recommendations for multimodal feedback, ensuring that users can interpret system status reliably even under varying environmental or viewing conditions. For example, warning or non-ready states are visually differentiated from operational states to clearly communicate system readiness and prevent incorrect usage. Furthermore, response latency in the user interface is minimized to ensure that feedback appears effectively instantaneous to the user. According to human factors guidelines, system response times below perceptual thresholds are critical for maintaining user engagement and trust in real-time systems. This is particularly important in measurement applications where delays could negatively impact usability and perceived system reliability.

4.1.5 Software and Coding Standards

Software development for the PTC system follows structured and modular programming practices to ensure maintainability, scalability, and reliability. The system is implemented in Python due to its flexibility, readability, and strong support for image processing libraries such as OpenCV. Code organization is divided into functional modules, including image acquisition, projectile detection, trajectory computation, and user interface control. This modular design allows individual components to be development, tested, and debugged independently. To maintain consistency across the development team, coding practices follow general guidelines inspired by the Python PEP8 standard. These include consistent naming convention, indentation rules, and code formatting to improve readability and collaboration. [84]

Additionally, inline comments and documentation are used to explain complex logic, particularly in image processing routines and mathematical computations. Error handling is also a critical component of the software design. The system must handle potential failures such as camera disconnection, improper image capture, or unexpected input values. Instead of allowing the system to crash, an exception handling mechanism will be implemented to ensure recovery and continued operations. These software standards contribute to a stable and maintainable codebase that supports real-time projectile tracking

and future system improvements. [85]

4.2 Design Constraints

Engineering constraints significantly influenced the PTC system architecture by limiting available design choices and forcing tradeoff decisions among safety, performance, cost, and usability considerations.

4.2.1 Safety Constraints

Safety constraints are the most influential factors in system development. Laser exposure risk requires strict beam control and enclosure design to maintain eye-safe operation. Since the projectiles we will measure travel at a higher velocity, projectile containment and controlled firing direction are necessary to prevent unintended impacts. Indoor operation requirements further restricts the power levels for lasers and demands careful management of reflections. Electrical isolation between subsystems will be required to prevent shock hazards as well as protecting sensitive electronics.

4.2.1.1 Laser Exposure Risk

Laser exposure risk requires strict beam control and eclosure design in order to maintain operation safe to the human eye. The laser used for the light gate must be positioned in such a way that the beam path does not intersect with any areas where users may be present. The housing of the PTC system will incorporate beam stops and other additional light collecting methods to capture or block and stray reflections that could potentially redirect laser light toward users. The choice of laser directly impacts safety requirements, and by selecting a Class 1 laser, we minimize the required safety infrastructure while still achieving adequate detection sensitivity. If we were to use a higher-class laser, additional safety procedures would have to be implemented such as PPE and engineering controls for the user. For this project, the PTC will be demonstrated in an indoor environment, which can introduce more reflective surfaces around the system. This further drives the need for additional beam blocking controls in order to minimize the risk of uncontrolled beam paths hitting a user or bystander.

4.2.1.2 Projectile Containment and Electrical Isolation

Since the projectiles measured by the PTC travel at high velocities, projectile containment and controlled firing direction are necessary constraints to consider in order to prevent unintended impacts. The PTC must be positioned within a controlled range environment where projectiles can be captured by appropriate backstops, as opposed to capturing the projectile within the device itself. This system was designed with the intention to be operated in environments like shooting ranges (for higher velocity projectiles) as well as indoors (for lower velocity projectiles).

Electrical isolation between subsystems is required to prevent shock hazards and protective sensitive electronics within the device. Adding safety features such as level shifters or

other isolating circuits can prevent ground loops and protect against transient events that could damage components or pose safety risks. Power supply designs are also important, as appropriate fusing and overcurrent protection can prevents hazards like electrical fires.

4.2.1.3 Fail-Safe System Behavior

Another key consideration is fail-safe system behavior. In the event of power loss, sensor failure, or unexpected software errors, the system must default to a safe state. This includes disabling laser outputs and preventing further triggering until the issue is resolved. These safety measures reduce the likelihood of accidental exposure or incorrect system operation. User training and clear operational procedures are also considered part of the safety constraints. Even with engineered protections, improper use can still introduce risk. Therefore, the system design emphasizes intuitive operation and clear instructions to minimize user error. [82]

4.2.2 Performance Constraints

Performance limitations were defined by measurable system requirements such as camera frame rate and processing latency. Accurate velocity measurement requires precise synchronization between projectile detection, strobe activation, image capture, and computational analysis.

4.2.2.1 Timing Requirements

The system must detect a projectile, trigger a strobe event, capture sequential images, and compute velocity within strict timing constraints. Camera hardware limits the frame acquisition speed while the software processing time introduces latency that must remain low enough to maintain measurement accuracy. For a projectile traveling at 300 m/s through a 30 cm measurement area, the transit time is about 1 millisecond. Within this window, the system must capture at least two distinct images to find the positional and other ballistic data of the projectile.

Camera hardware limits the frame acquisition speed greatly. The Raspberry Pi camera module is able to support various frame rates depending on the resolution, with their being a significant tradeoff between image resolution and frame rate. This tradeoff directly impacts the measurements of the strobe imaging subsystem, where higher resolution provides more precise positional determination, but lower frame rates may miss the faster moving projectiles entirely. [70]

4.2.2.2 Synchronization Accuracy

One critical performance consideration is system synchronization accuracy. The alignment between the laser detection subsystem, strobe illumination, and camera capture must occur within extremely small time windows to ensure valid measurements. Even minor timing mismatches can introduce significant errors in trajectory estimation. The Raspberry Pi's operating system introduces non-deterministic scheduling, meaning that precise timing control can be difficult. This limitation requires the software design to

implement strategies to minimize timing variability, such as using hardware-triggered capture modes like the light gate subsystem. [86]

4.2.2.3 Computational Performance

Computational performance must be balanced with energy consumption and thermal limitations of the Raspberry Pi 5. Prolonged high CPU usage can lead to thermal throttling, reducing processing speed and affecting real-time performance. As a result, efficient algorithms and optimized processing pipelines are necessary to maintain consistent system operation. Open CV operations must be optimized in order to reduce processing time and minimize unnecessary computations that will affect performance. The tradeoff between computation complexity and real-time responsiveness directly influences the measurement rates of the PTC system.

4.2.3 Economic Constraints

Economic limitations strongly influenced the component selection. This project was not sponsored, therefore all the funding came from the team itself. After researching the current technologies and what is commercially available, a budget of five-hundred was set. Optical sensing was chosen over radar-based measurement due to lower cost and easier integration. A Raspberry Pi platform was selected instead of an FPGA to reduce development complexity and expense while still meeting the processing requirements. These decisions illustrate the engineering tradeoffs between performance, complexity, and affordability. [73].

4.2.4 Environmental Constraints

Environmental conditions such as indoor lighting variation, temperature changes or vibrations can affect the optical sensing accuracy. The system must maintain reliable operation despite the fluctuations in light and other mechanical disturbances, requiring further controlled lighting and a stable housing structure.

Lighting variability is one of the most significant environmental challenges affecting system performance. Changes in ambient lighting conditions can impact the effectiveness of image differencing and contour detection algorithms. To mitigate this, the system relies on controlled strobe illumination to provide consistent lighting during image capture.

Vibrations and mechanical disturbances can also introduce alignment errors in both the optical and camera systems. Even small deviations in alignment can affect trajectory accuracy. Therefore, rigid mounting structures and vibration-resistant designs are necessary to maintain system stability.

Temperature variations may also impact electronic components and sensor performance. Prolonged operation in elevated temperatures can affect processing efficiency and potentially reduce the lifespan of components. These environmental constraints necessitate careful system design to ensure reliable operation under varying conditions. [73].

4.2.5 Manufacturing Constraints

Manufacturing considerations include the limitations of available fabrication methods, particularly with respect to 3D printing and mechanical assembly. Components must be designed with tolerances that can be reliably achieved using accessible manufacturing techniques. Misalignment in optical components, such as lasers and mirrors, can significantly impact system accuracy, making precision in mounting structures a critical requirement.

Additionally, the availability of standard hardware components influenced the design process. The system was designed to utilize parts 'off the shelf' whenever possible to reduce cost and simplify assembly. This includes standard mounts and enclosure components. These constraints required careful consideration of part compatibility and assembly procedures to ensure that the final system could be constructed reliably without specialized manufacturing equipment.

4.2.6 Maintainability Constraints

The system is to be designed with modular subsystems to allow easy replacement of sensors and simplified troubleshooting. The use of the Raspberry Pi 5 will allow software updates without hardware modification, granting long-term usability.

To further enhance maintainability, the system incorporates clear separation between hardware and software components. This allows updates or replacements to occur without requiring full system redesign. For example, camera modules or sensors can be upgraded independently of the core processing software.

Documentation is also an important aspect of maintainability. Clear code documentation, wiring diagrams, and system architecture descriptions ensure that future developers or users can understand and modify the system effectively. Without proper documentation, even a well-designed system can become difficult to maintain over time.

In addition, modular software design allows for easier troubleshooting by isolating faults within specific subsystems. This reduces debugging time and improves overall system reliability during long-term use. [85]

4.2.7 Usability Constraints

Usability constraints are important in the design of the PTC system because the device must be easy to operate while still producing accurate measurements. The system is intended for users with different levels of technical experience, so setup and operation must be as simple as possible. To address this, the design minimizes manual calibration by using fixed hardware alignment and predefined system parameters. This reduces user error and ensures more consistent results across different uses.

The user interface is designed following principles from ISO 9241, which focuses on effective human-system interaction. A touchscreen interface is used to simplify user input

and eliminate the need for complex controls. The system follows an automated workflow, where after initialization it enters an “idle” state and automatically detects, captures, and processes projectile data. This reduces the number of steps required by the user and improves overall system usability.

Clear feedback is also a key usability constraint. The interface provides visual indicators for system states such as charging, ready, and detection complete, ensuring that users always understand what the system is doing. This aligns with usability principles that emphasize visibility of system status and immediate feedback. However, simplifying the interface introduces a tradeoff by reducing advanced user control. Despite this, the design prioritizes ease of use and reliability to ensure safe and effective operation.

4.2.8 Programming Constraints

Programming constraints are largely influenced by the limitations of the Raspberry Pi platform and the use of Python as the primary development language. While Python provides ease of development and strong library support, it is not optimized for high speed processing compared to lower-level languages such as C or C++. This introduces challenges when processing high-resolution images or performing computations within strict timing constraints.

Additionally, the Linux-based operating system used by the Raspberry Pi introduces non-deterministic scheduling, meaning the precise timing control is more difficult to guarantee. This is particularly important when synchronizing camera capture with strobe lighting and projectile detection events. To mitigate these limitations, efficient coding practices and lightweight processing techniques are used wherever possible.

OpenCV operations must also be carefully optimized to reduce processing overhead. Techniques such as reducing image resolution, limiting region-of-interest processing, and minimizing unnecessary computations are used to maintain acceptable performance. These programming constraints directly influence system design decisions and require tradeoffs between computation complexity and real-time responsiveness.

4.2.9 Time Constraints

Project scheduling constraints between Senior Design 1 and 2 required a prototype first development strategy. Early functional demonstrations are prioritized over optimization to help ensure completion deadlines are met.

Time constraints also influenced the prioritization of system features. Core functionality, such as projectile detection and velocity measurement, was prioritized over advanced features like enhanced visualization or extended analytics. This ensured that a working prototype could be completed within the academic timeline.

Iterative development was adopted as a strategy to manage time effectively. Instead of attempting to implement a fully optimized system from the beginning, the team focused on incremental improvements, validating functionality at each stage before progressing

further. This approach reduces risk and ensures steady progress toward project completion.

Testing and validation time was also limited, requiring efficient debugging and verification methods. As a result, simulation and controlled testing environments were used to accelerate development while still ensuring system accuracy and reliability. [87]

5 Application of ChatGPT and Other Similar Platforms

5.1 Case Study 1 - Image Processing Design Using Large Language Models

Please refer to Appendix C, [1a],[1b],[1c],[1d] for the Case Study 1 prompt and outcomes generated by ChatGPT, Google AI, and DeepSeek respectively.

This case study was conducted to evaluate how effectively large language models (LLMs) assist in the design of a real-time image processing pipeline for the PTC projectile tracking chronograph. Unlike previous prompts focused on hardware comparisons, this study emphasized an engineering problem directly related to the project implementation. The prompt requested a complete workflow capable of detecting a high-speed projectile using synchronized camera images captured through a strobe illumination system, followed by velocity and trajectory computation using OpenCV.

The objective of this comparison was not to determine which model produced the most information, but rather to evaluate the practicality and engineering usefulness of each response when applied to an embedded vision system operating under real hardware constraints.

ChatGPT [1b] provided the most structured response in terms of overall system architecture. The model organized the solution into clear processing stages including image acquisition, motion detection, and trajectory calculation. The suggested techniques such as frame differencing and centroid tracking aligned well with standard OpenCV workflows and were directly applicable to the PTC design. One notable strength was the explanation of why each processing step was necessary, allowing the response to function as both documentation guidance and conceptual design support. However, the response assumes ideal imaging conditions and did not account for limitations such as Raspberry Pi processing latency or lighting variability. Though the architecture was useful, engineering judgment is still required to adapt the recommendations.

Google AI [1c] approached the same prompt differently by emphasizing algorithm descriptions and referencing commonly used computer vision methods with stronger ties to external documentation. The information was technically accurate and often included broader comparisons between detection techniques. The response did not clearly describe how individual stages connected within a real-time embedded system. While useful

for research validation and terminology clarification, an engineering interpretation was necessary to translate this information into an executable workflow.

DeepSeek [1d] demonstrated strengths in proposing solutions for implementation suggesting pseudo code structures and optimization considerations. The model discussed alternative tracking strategies and offered suggestions for improving robustness under noisy image conditions. In several cases, DeepSeek proposed computationally intensive solutions that may be better suited for desktop GPUs rather than embedded processors. This highlighted a recurring limitation observed across models: a lack of awareness of hardware constraints unless explicitly specified within the prompt.

Across all responses, a consistent observation emerged that large language models are highly effective during early designs but require careful validation. ChatGPT proved most beneficial for establishing system architecture and documentation structure, Google AI supported verification through a broader context, and DeepSeek contributed implementation focused ideas that expanded possible choices. None of the models independently produced a fully deployable solution, but proved to benefit in accelerating different stages of the engineering design process.

From an engineering perspective, this case study demonstrates that large language models function best as a collaborative effort rather than a single technical source. The results indicate that AI-assisted development can significantly reduce initial research time while still requiring traditional engineering analysis to ensure accuracy, feasibility, and system reliability.

5.2 Case Study 2 - Document Formatting and Table Generation Efficiency

Please refer to Appendix C, [2a.1],[2b],[2c], for the Case Study 2 prompt and outcomes generated by ChatGPT and Copilot respectively. An additional prompt, [2a.2], was used for the case study.

In this case study, the effectiveness of using an LLM to help create a standardized table format was tested. Specifically, two LLMs were leveraged to organize and refine the tables generated by the different authors of this report into one unified structure. The investigation focused on assessing the model's ability to reduce manual formatting effort and improve uniformity. Additionally, the limitations of this approach were studies, including the need for human oversight to verify the accuracy of the translated tables and to ensure compatibility with the pre-established formatting structure of the document. Through the evaluation of this study, insight was gained into the benefits and trade-offs of using LLMs for documentation editing purposes.

ChatGPT [2b] was structurally consistent with its response when converting tables to the defined "Standard Table Format" (STF). The model maintained the row separation, caption placement, and usage of the longtable command. In particular, this model strictly

maintained the column widths. In the original query, the reference table used had only four columns, with the follow-up query containing a table with 5 columns to be converted to the STF. In ChatGPT's response, it prioritized the table structure provided by the reference table over original distinctions made in the follow-up table. This caused the LLM to merge two columns together to align the column widths set by the STF. The LLM finished its query by noting the changes it made to the columns, and also provided an option for the original table's columns remain intact and dynamically update the size of the columns instead of merging them. Upon further prompting, the LLM was able to produce a table that maintained all of the original information, albeit the column sizes were not sized correctly for the document, leading to manual correction. Even though this LLM maintained most of the original information from the follow-up table, this change in table structure necessitated the need for additional human oversight to confirm if these changes were acceptable.

Copilot [2c] was asked the exact same prompt as ChatGPT and was more to the source table than ChatGPT. This response maintained all of the original columns, maintaining full data fidelity. The only correction to be made was manually adjusting column widths, as they no longer adhered to the standard size of the table set by the STF. Copilot finished its response by asking the user if to use different column widths or standardize captions/labels. This response was much closer to the desired outcome of the query, as the data fidelity is more important than strict structural compliance.

Overall, ChatGPT emphasized structural adherence to the source material over data fidelity, leading to more time spent querying to get a more suitable response, whereas Copilot was more faithful to the data in the tables in generating its response. This shows the nuance in how these two models approach a problem when it is up to them to make a decision. In ChatGPT's case, it chose alter the follow-up tables physical dimensions (rows/columns) to better replicate the exact structure of the STF, while Copilot chose to maintain the physical dimensions of the follow-up table and adapt the structure set by the STF to that table. While both LLMs were able to produce workable tables, this study showed the limitations of the transformative nature of certain AI models for tasks such as standardized table generation.

5.3 Case Study 3 - Metaprompt to use AI for ideas on how to use AI in the project

In this case study, LLMs were evaluated on the capability of determining potential applications of AI for the senior design project. Please refer to [appendix] for the prompt and responses.

Gemini clearly thought the prompt was referring to using AI within the operational phase of the project itself, not in the design or research phase as was desired. Gemini mentioned a number of potentially useful advanced uses of AI, including object detection, replacing the kinematic equations with an AI model. One of the limitations is that deep neural networks would require large amounts of training data, which would be impossible to

get without either simulation or automation, either of which would be a large expenditure of effort.

Gemini mentioned using AI for sensor fusion, in which it could be useful for filtering out noise. One suggestion was to compare a Kalman filter to a Gated Recurrent Unit. Signal denoising was also mentioned for allowing lower-cost hardware and compensating for the higher noise with better filtering with AI. Unfortunately, the economics of this don't work out, since the vast majority of the cost is in development time, and BOM costs are nearly negligible. Since there is only a couple prototypes that will be made, reducing development cost and time at the expense of even $10 in parts is valuable, so spending significant time on developing a use for AI to allow a lower cost part is not beneficial.

Grok interpreted the prompt in a similar way to Gemini. It recommended a number of ways AI can be used in the operational phase of the project. First, Grok recommended using AI for object detection or localizing. Models like YOLO give a bounding box on the objects that are detected within the image.

One previous project involved using YOLO on a security camera video stream. One part of the project involved using the detection of cars to alert the user when there was a car in the driveway, indicating there was a visitor. That way when visitors inevitably knocked as if there was a baby sleeping, such that the user was unable to hear the knock on the door, the alert from the security camera was able to let the user know to go open the door. The bounding box of the detected car was useful, since the camera was pointed at the road in front of the house, so the bounding box of the car detection was used to filter out cars that were simply passing by, or parked in the neighbor's driveway. Another part of this previous project using YOLO was to use the security camera to detect where people were in the yard, and control the outdoor lights to light up only the area of the yard where the user was. This was accomplished using addressable LED strips for the outdoor lighting, so that the location used with the bounding box of the detection was used to select which LEDs lit up. Unfortunately, the positional accuracy of the bounding boxes of models like YOLO leave a lot to be desired. It is a coarse position, and is often not centered on the object. Instead, the bounding boxes of YOLO are useful for things such as determining if a car is within a driveway, like in the aforementioned project, or to get an idea if a person is on the left side, right side, or center of an area. YOLO does not have a bounding box resolution near the resolution of the image, instead it is more coarse. [Cite Ramos and Sappa 2025 YOLO] Since we will be able to take calibration pictures without the projectile, it will be significantly easier to use simple math to determine the location of the projectile rather than using AI for object detection and bounding boxes.

Grok also recommended trajectory prediction, drawing on examples such as using neural networks for trajectory estimation for mortar trajectory estimation in GPS/GNSS denied areas. It also recommended Physics Informed Neural Networks (PINNs) for 3d trajectory estimation from 2d video. It linked to a patent on estimating 3d trajectory using 2d camera images [cite Thornbrue, Cherveny, and Haas patent].

One aspect of the image enhancement and denoising recommendation from Grok that was novel was the recommendation to compare autoencoders or diffusion models versus

classical filters such as Gaussian for improving tracking accuracy. This could be useful for narrowing down the centroid of the projectile in the images for better accuracy of positioning the projectile.

5.4 Case Study 4 - AI Tools in Search Engines

It is apparent that AI has become increasingly powerful and useful, especially in the accumulation of data and recognition of patterns. As such, it has become prevalent in search engines as an automatically implemented tool to try and make research faster and deliver answers quicker. In this case study, the effectiveness of these searching tools were benchmarked with search queries of more obscure topics, engineering standards and protocols in this case.

In my research of the USB standard and protocol, I occasionally struggled to find documents in the USB-IF's archive, and this was sometimes because of a particular document's unique naming scheme, or because the document had been removed due to obsolescence of an old standard. For reference, I was trying to find the USB-IF's standard on USB micro connectors, which debuted in the early 2000's. It was unclear to me whether such connectors conformed to the USB-IF's standards, and if they did, which USB standard they conformed to. The search engine's (DuckDuckGo's) AI search tool was able to help me find the document outlining the USB 1.0 standard by providing me a link to the Wayback machine, an internet archive that preserves such documents, and a link to a Wikipedia article with suitable references. This was the inspiration for this case study, detailed below.

For this case study, the prompt and responses can be viewed in Appendix []. The search results were relevant, and led me to a blog post on the Analog devices website on how to use UART, and a Wikipedia entry. With another search, the AI tool was able to clarify that UART is not a standard like USB, but a communication protocol.

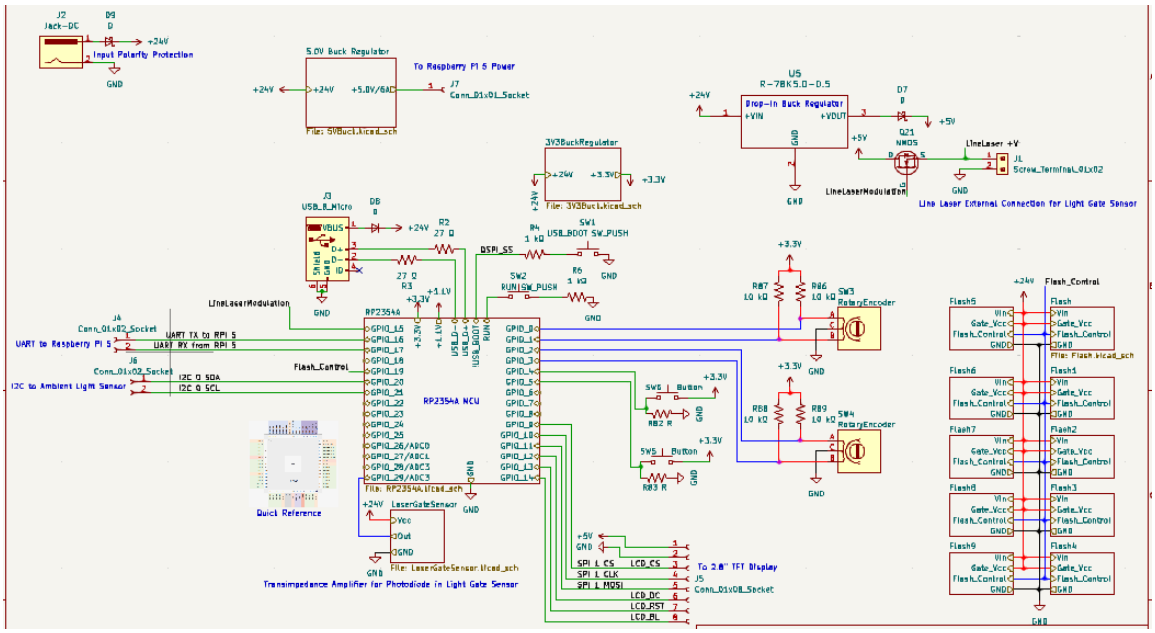
This leads me to believe this tool is useful in matters of research, but may lead me down certain avenues without clarifying my misconceptions. Thus, it is important to always ask multiple questions, and thoroughly read the sources it points me to, so that potential misconceptions don't interfere with our implementation of the PTC.

6 Hardware Design

6.1 Schematic Diagrams

6.1.1 Top-Level Schematic

The overall schematic of the current PTC project is shown below, with essential circuits incorporated in hierarchical blocks to save space on the master schematic sheet.

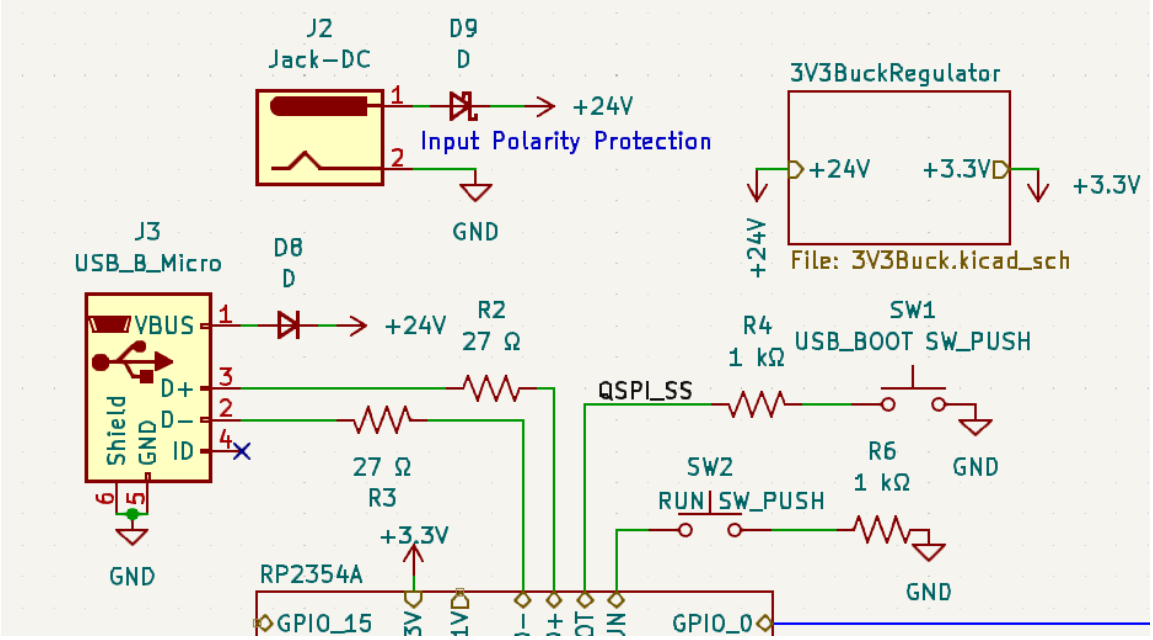


The overall schematic shows power inputs, our central micro-controller, user-interface peripherals, as well as essential circuits like voltage regulators, sensors, and strobe circuits.

6.1.2 Power

We intend to use a wall outlet and 24V/8A DC power supply to power this project, and this device uses a Barrel Jack port. Additionally, the Raspberry Pi Pico 2 RP2354A micro-controller requires a power input at a voltage of 3.3 V. The RP2354A also is programmed via USB 2.0 protocol, and thus requires a USB port the Raspberry Pi 5 will be detached from this PCB, with UART connections to facilitate communication between the MCU and the Image Processor. The Raspberry Pi 5 also requires a 5V/5A supply, and so a regulator is used to provide this.

To provide 3.3V from a 24V input, the AP5100 switching buck regulator is used. This circuit will be described in detail in the schematic subsystems section. Additionally, it is in our best interests to install basic circuit protections to prevent device failure; among which are input polarity, over-voltage, and over-current protections.



To provide input polarity protection for the main power rail, a schottky diode has been placed after the output of the DC barrel jack. The schottky diode will prevent damage to components by effectively opening the circuit when a negative voltage is applied, and is particularly useful in that there will be less of a voltage drop across the diode than if a standard silicon diode was used, preserving most of the 24 V to be used in the main power rail. Other protections will be discussed in later sections.

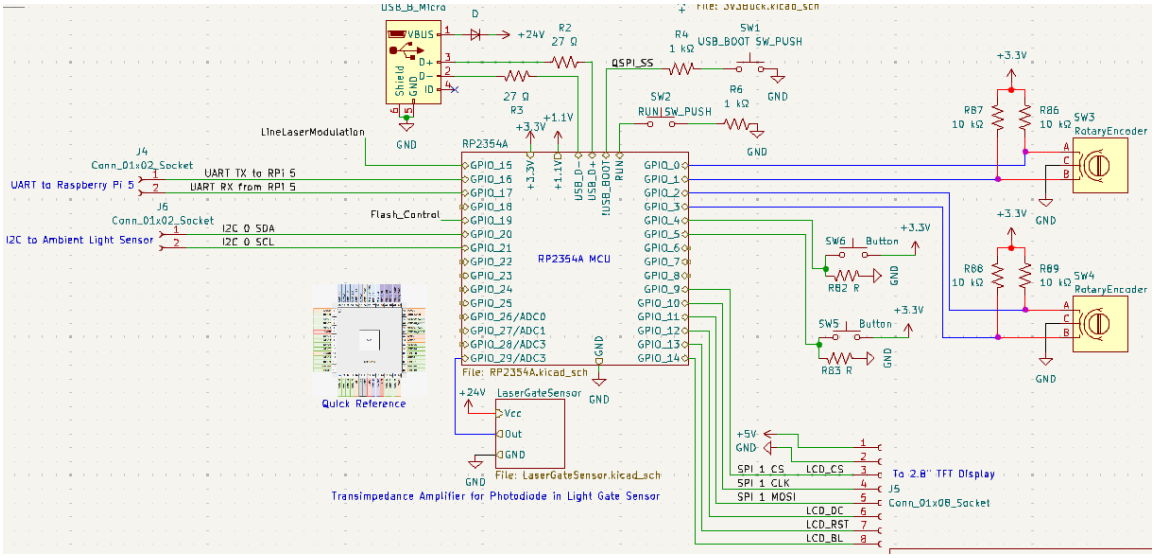
The USB micro B port is simple in construction, and technically we only require the ground pin and the data pins. But, by incorporating a diode on the USB port's VBUS output, it will be possible to power and program the RP2354A without supplying full power to the device, as other systems will prevent excessive current draw, and the AP5100 3.3 V regulator will function properly with an input voltage ranging from 4.75 V to 24 V. Additionally, this diode serves to protect the USB port and its connections from over-voltage should the device be fully powered on, preventing the 24 V main power from reaching the port. The shield pin and ground pin of the USB port are wired together to prevent floating voltages. [57]

6.1.3 Peripherals

To facilitate the functions of the PTC project, a few essential peripheral components and circuits are absolutely necessary, such as sensors, displays, and user-interface components.

Of the user-interface components, we have a couple rotary encoders and a few buttons. Additionally, we have connections to a TFT display, and an Ambient light sensor that will be placed in the imaging chamber. The Ambient light sensor can help the PTC during calibration by determining how bright it is in the imaging chamber, and will communicate with the MCU via I2C. The TFT display can be controlled by the MCU to relay information gathered by the PTC about the projectile in an intuitive manner, as

well as provide on-screen graphics. It will communicate with the MCU via SPI.

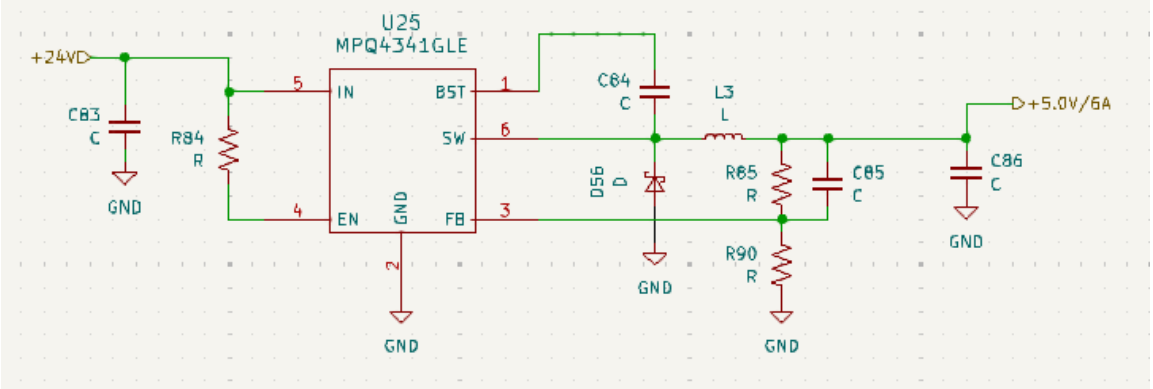


The buttons labeled USB_BOOT and RUN are necessary for a few key functions of the RP2354A, and work in an active-low principle. To enable the end-user to interact with the PTC, 2 buttons and 2 rotary encoders will be in use. This should allow enough controls for an intuitive user-interface while still conserving GPIO pins for other potential use cases, such as a display screen and sensors. For the buttons, an active-high configuration is used by placing the micro-controller's GPIO connections between the button and pull-down resistor. The rotary encoders require signal pins A and B to be held high, so that the rotating contacts inside can ground them in an alternating fashion. The MCU will be able to determine the direction of rotation by reading each signal and comparing them with their previous states.

6.2 Subsystem Schematics

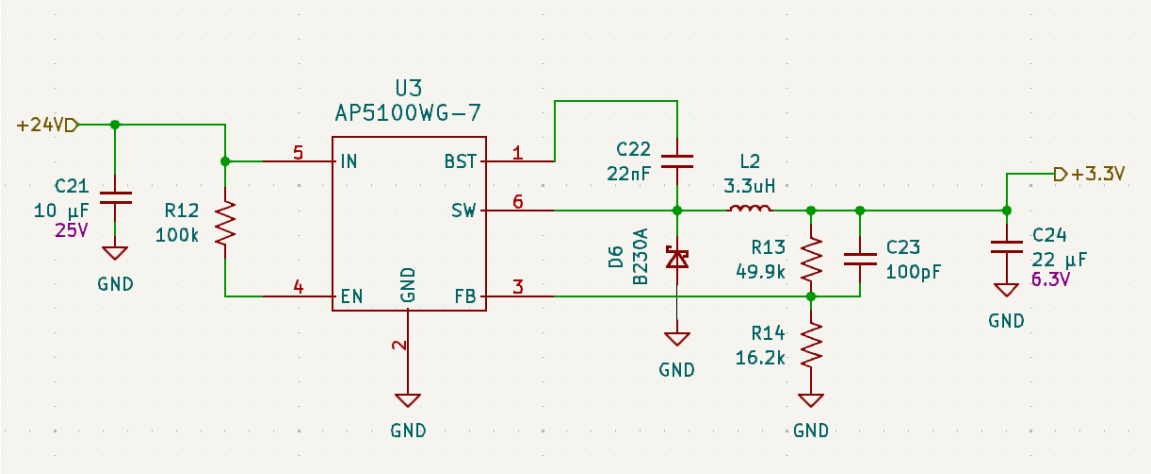
6.3 5V/5A Regulator

The MPQ4341GLE-5-AEC1-Z is the regulator we are using to provide power to the Raspberry Pi 5, as our main supply is at too high of a voltage for it. This converter will be able supply the 5A that the Raspberry Pi 5 requires at 5V, and is not adjustable. Since it is its own converter and we have another 5V converter, this one will only focus on powering the Raspberry Pi 5.



6.3.1 3.3V Regulator

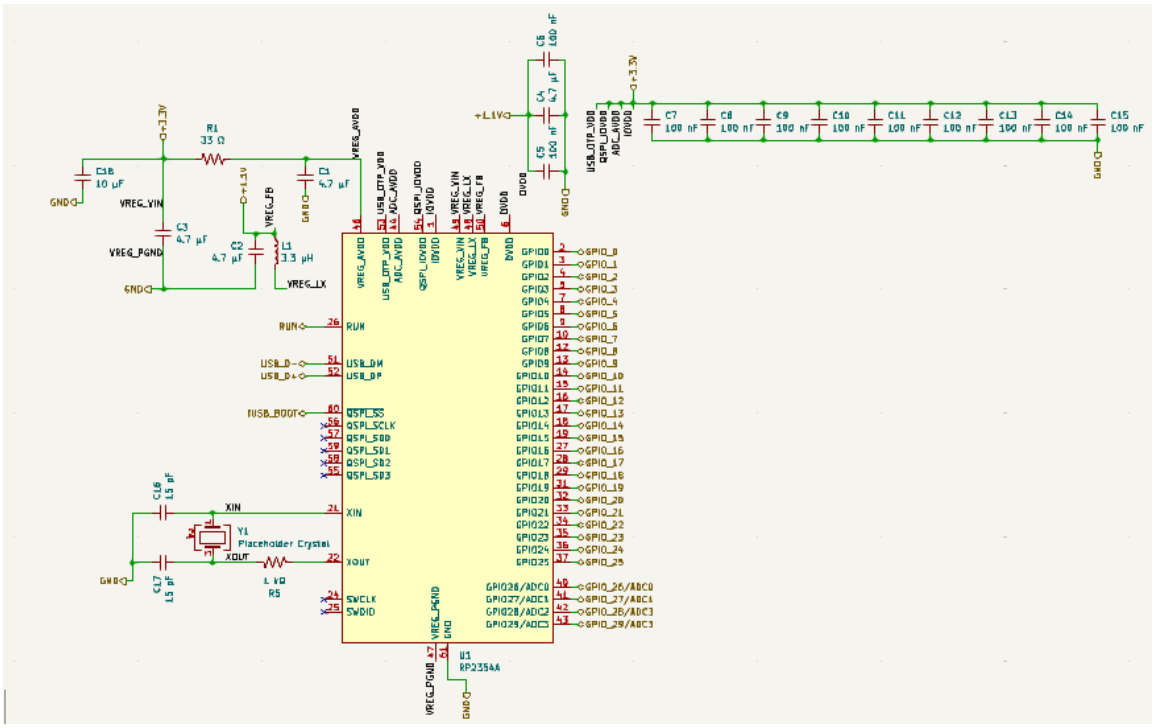
As previously mentioned, the AP5100WG7 is a switched step-down converter, and can handle input voltages from 4.75 V to 24 V at currents up to 1.2 A. This converter does have an adjustable output voltage, but this schematic was assembled with reference to the typical application circuit provided in the device's datasheet, wherein it outputs 3.3 V, which is what we require to supply power to our MCU. [57]



The values for the components are also provided in the example circuit, and we have elected to use the same values to ensure proper functionality.

6.3.2 Micro-controller

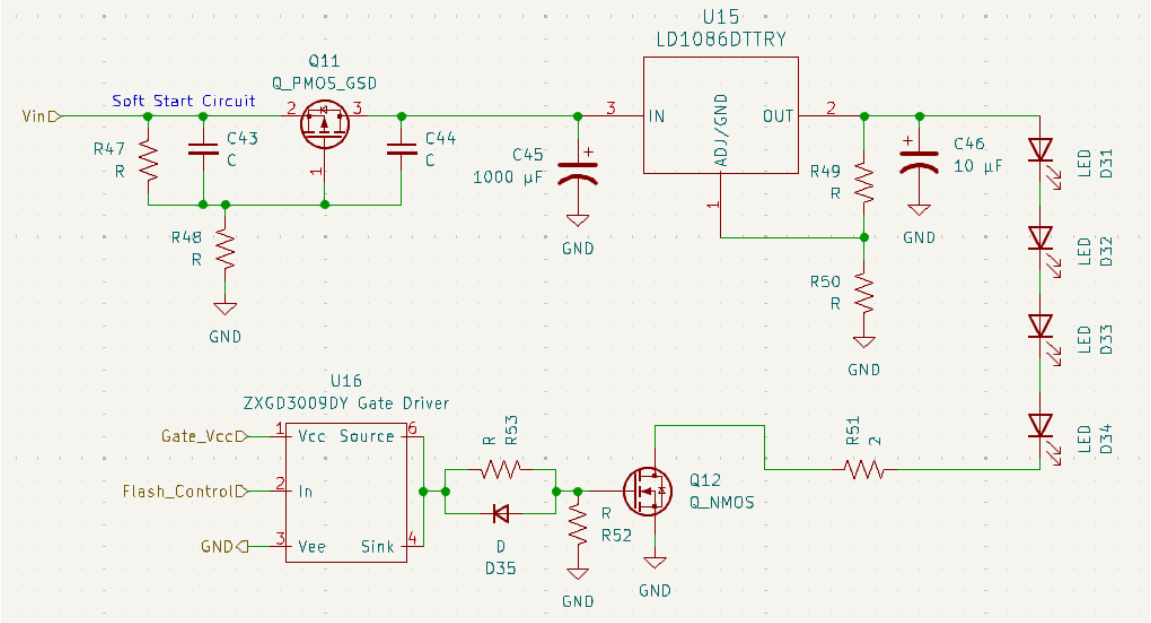
The RP2354A is a complex device, but luckily the team at Raspberry Pi has not only provided a datasheet, but also a simple user-guide for building the minimum circuitry required to run this hardware. [41]



The RP2354A actually not only requires a 3.3 V source, but also a 1.1 V source. The RP2354A does, however, come with an internal regulator to provide this 1.1 V voltage, at the cost of more external components. At the 3.3 V input and 1.1 V output, various capacitors are used to filter the signals and provide stable DC power, while the inductor and capacitor next to it are used to configure the internal regulator. Many small capacitors are also used as coupling capacitors to filter out noise, and a crystal oscillator is used to provide a timing reference to the chip. This crystal oscillator requires a precise capacitance, which the Raspberry Pi team calculated to be 10 pF, but accounting for parasitic capacitances, chose to use two 15 pF capacitors in parallel. [41]

6.3.3 Strobe

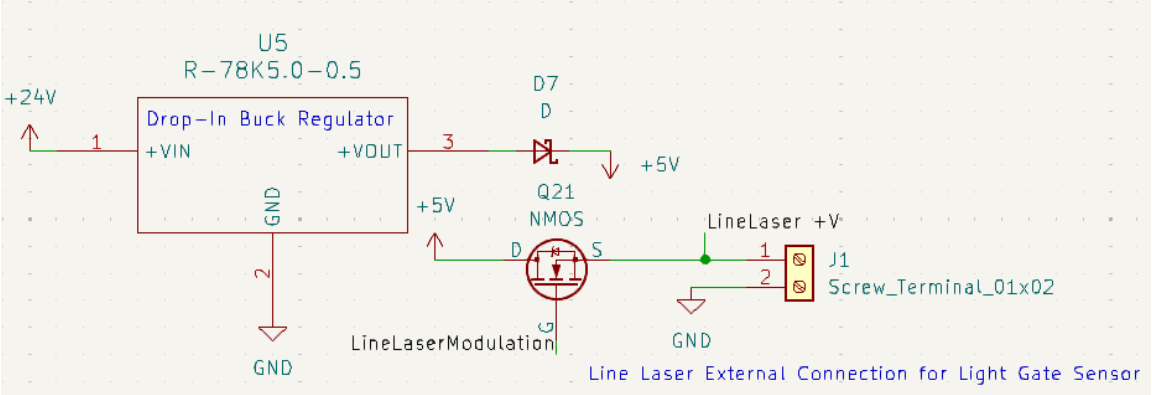
This strobe circuit has been designed by our own team to meet our requirements. We require a very bright flash for the strobe, and so elected to use high power LEDs. These LEDs not only require a larger current, but also have a large voltage drop across them, limiting the amount we can place in series before we exceed our supply voltage. In order to achieve our desired brightness then, it is necessary to build a number of identical circuits that can drive a short series of these LEDs. Additionally, since we now had identical parallel circuits that would all flash at the same time, the total current draw of all the LEDs would now exceed or take a significant portion of the current that our power source could supply, causing issues with our other components. Thus, a capacitor bank is needed. But with a large capacitor bank, now we needed to manage inrush current so that components didn't overheat when the PTC is plugged in.



In the schematic above, we implement a soft start circuit with a P-channel MOSFET that slowly turns on as the large capacitor charges. This circuit topology effectively limits inrush current while still providing a short charge time of at most a few seconds. This allows the strobe circuit to be operational long before a user activates it. The LD1086DTTY low-dropout linear regulator is used to ensure that a constant current is supplied to the strobe LEDs even as the capacitor bank discharges when the circuit is active, ensuring a stable light source with minimal variance in intensity. Since the LEDs have a high forward voltage, putting just a few of them in series creates a large voltage drop that is close to the supply voltage. Thus, a low-dropout regulator is preferred to maximize the voltage headroom the capacitor bank has before it discharges to a level that compromises the regulator's ability to maintain a stable voltage. The feedback resistors are calculated for the LD1086DTTY to supply a voltage of 20 V, allowing about 3 V of headroom. A low impedance resistor is used at the end of the LED chain to limit the maximum current and protect the LEDs. An N-channel MOSFET is placed at the low-side of the chain to activate the strobe LEDs when a signal is received through the ZXGD3009DY Gate Driver from the micro-controller. The Gate Driver is used to ensure fast switching times for the MOSFET, minimizing the delay from the flash signal to the actual flash. With this system, we should be able to reliably provide enough power and sufficient lighting for repeated flashes and image captures when the PTC is in use, with minimal delay, thus preserving the timing accuracy of the system.

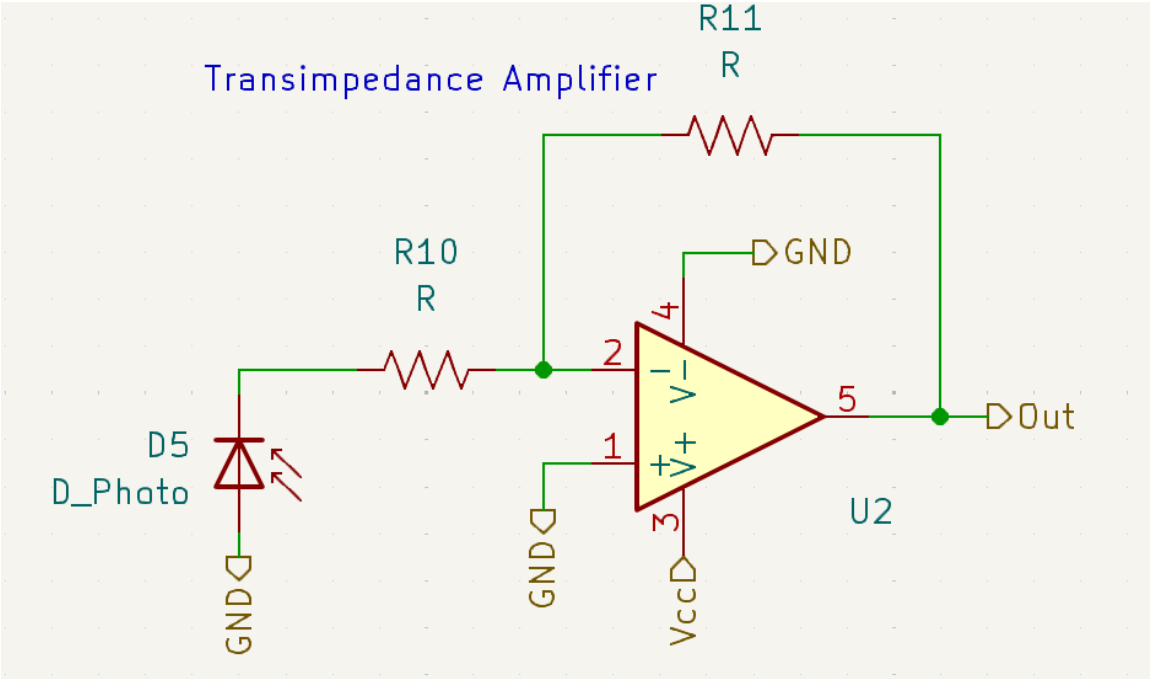
6.3.4 Laser Sensor Gate

The line laser is necessary for the light gate sensor at the beginning of the PTC. It is partly responsible for the generating the signals that the MCU will use to initially calculate the speed of a projectile, as its beam will be interrupted twice when a projectile moves through the PTC, allowing a photodiode's output to fluctuate in correspondence. Below is a zoomed in capture of the circuit that powers the line laser.



The line laser is an external component, and will not be soldered to the final PCB, so a screw terminal connection is used to incorporate it. From the line laser's datasheet, it requires an input voltage within a range of 2.8 V to 5.2 V. Additionally, the laser has an internal driver and only requires an input voltage. Therefore, we have elected to use a drop-in component, a full switching buck regulator within a single confined package in a component called the R-78K5.0-0.5. This regulator is capable of taking our 24 V main power and outputting a stable 5 volts, and a schottky diode is added in series with the output for protection, as suggested by the device's manufacturer in the datasheet. [12], [58]

In order to capture precise images of a projectile as it travels through the PTC, it is necessary to know the speed at which it is traveling before the images are taken. With the laser gate system, this will be possible, but a sensing circuit is necessary. Since the laser gate system will work by using lasers to beam light into photodiodes, transimpedance amplifiers can be used to turn the currents produced by the photodiodes into readable voltage outputs that can be interpreted by the MCU.



Just before a projectile is shot through the chronograph, the MCU can run a calibration to determine the current voltage produced by the transimpedance amplifiers. When the projectile is shot through, the incident light on the photodiodes will be interrupted, briefly reducing the transimpedance amplifier's output voltage, twice, allowing the MCU to capture two distinct signals and measure the time between them to calculate the projectile's speed.

7 Software Design

7.1 Software Design Overview

The software system for the Projectile Tracking Chronograph (PTC) is responsible for integrating sensor inputs, image processing, and user interaction in a unified real-time measurement platform. The system combines data from the laser-based velocity subsystem with camera-based image processing to generate a complete representation of projectile motion. The software is designed for execution on the Raspberry PI 5, serving as the central processing unit for all computational tasks. [88]

The architecture follows a modular design approach, allowing independent development of key subsystems including image acquisition, projectile detection, trajectory estimation and user interface rendering. This structure improves system maintainability and enables future optimization without requiring an entire redesign. [85]

The high-level functional requirements of the system are illustrated in the use case diagram below.

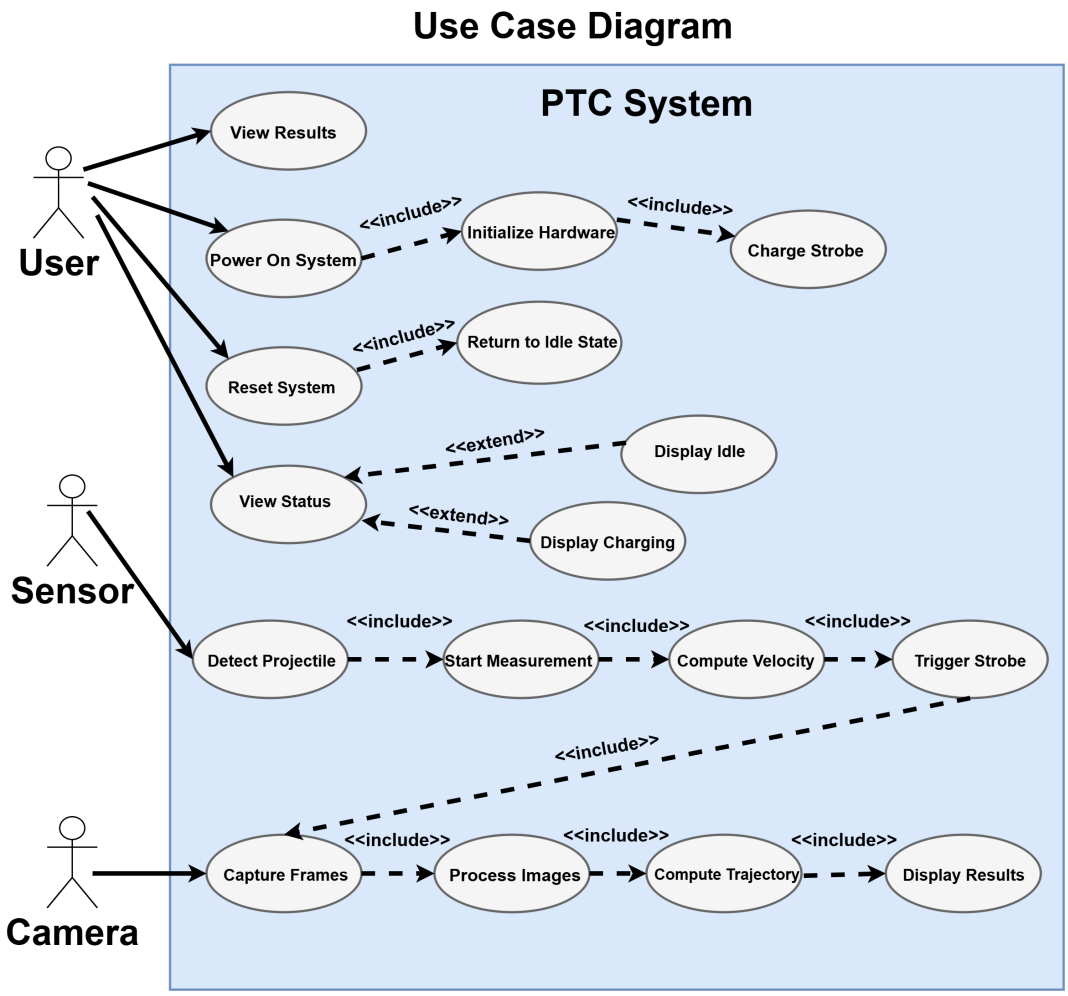


Fig. 7.1: Use case diagram of the Projectile Tracking Chronograph (PTC)

7.2 System Execution Flow

The overall execution flow of the PTC software system follows a sequential yet modular pipeline that ensures real-time performance and accurate data processing. The system begins in an initialization state, where all hardware components including the camera, laser subsystem, and strobe lighting are configured and verified for proper operation.

Once initialization is complete, the system transitions into a standby or “Idle” state. In this mode, the software continuously monitors for trigger conditions indicating that a projectile has entered the detection region. This event-driven design ensures efficient use of computational resources by avoiding unnecessary processing when no activity is present.

Upon detection, the system immediately activates the strobe lighting and captures synchronized image frames. After the projectile passes, additional empty frames are captured for background subtraction and camera calibration. These frames are then passed into the

image processing pipeline, where motion detection and contour extraction are performed. The centroid positions obtained from each frame are forwarded to the trajectory computation module. Simultaneously, the laser subsystem provides an independent velocity measurement, which is integrated with the positional data. The combined dataset is then used to estimate the projectile's trajectory. Finally, the computed results are transmitted to the user interface, and the system resets to the Idle state.

The detailed step-by-step workflow of a complete measurement cycle is illustrated in the activity diagram below.

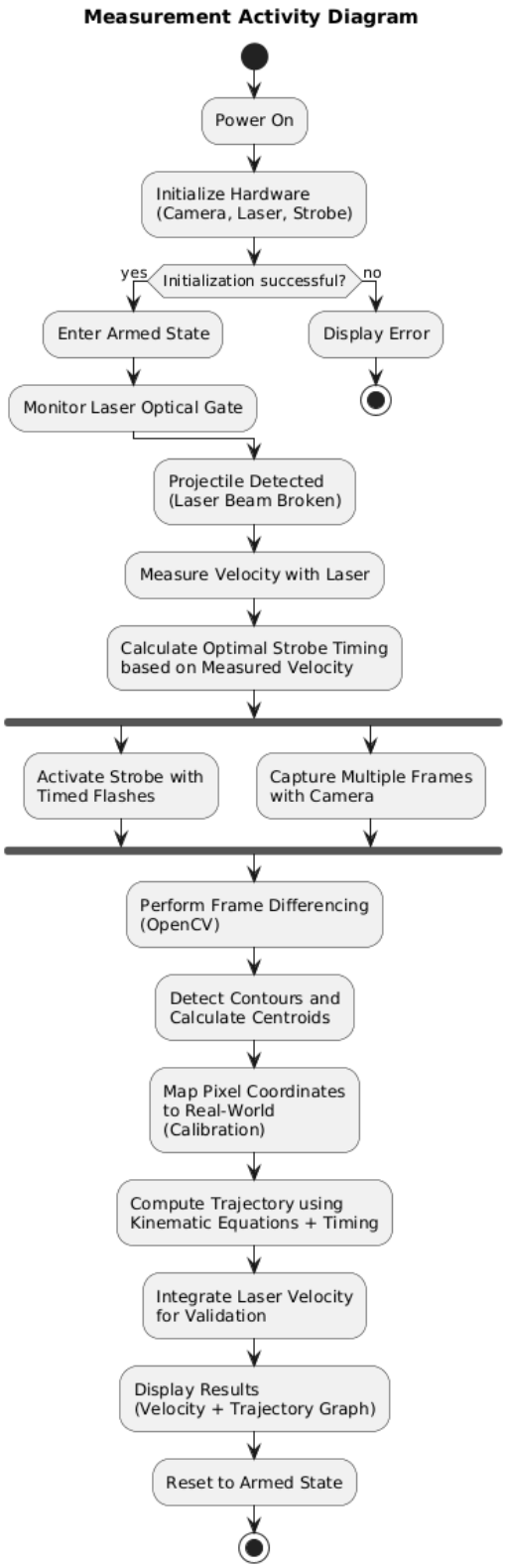


Fig. 7.2: Activity diagram of the Projectile Tracking Chronograph system (PTC)

The various operational states and transitions of the system are further presented in the state machine diagram shown in Figure below.

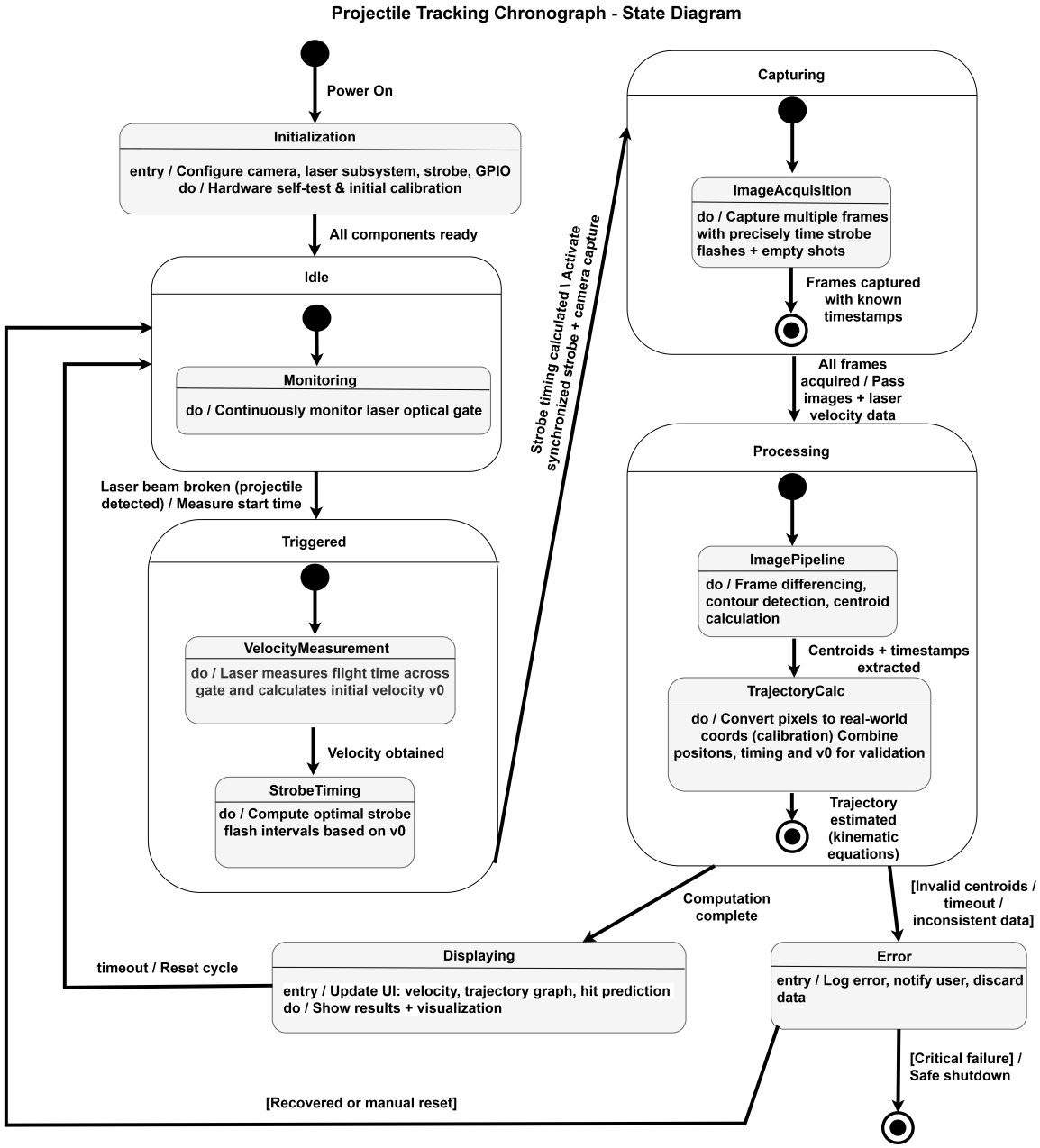


Fig. 7.3: State diagram of the Projectile Tracking Chronograph system (PTC)

This structured execution flow ensures deterministic behavior and efficient coordination between hardware and software components.

7.3 Software Architecture

The software will be implemented in Python due to its ease of development and strong support for frequent prototyping. The system is structured into modular components that handle specific tasks, such as sensor input and handling, image acquisition and processing, data computation and analysis, and user interface control.

This modular approach allows for parallel development, easier debugging, and better maintainability. The static structure of the software, showing the main classes and their relationships, is presented in the class diagram below.

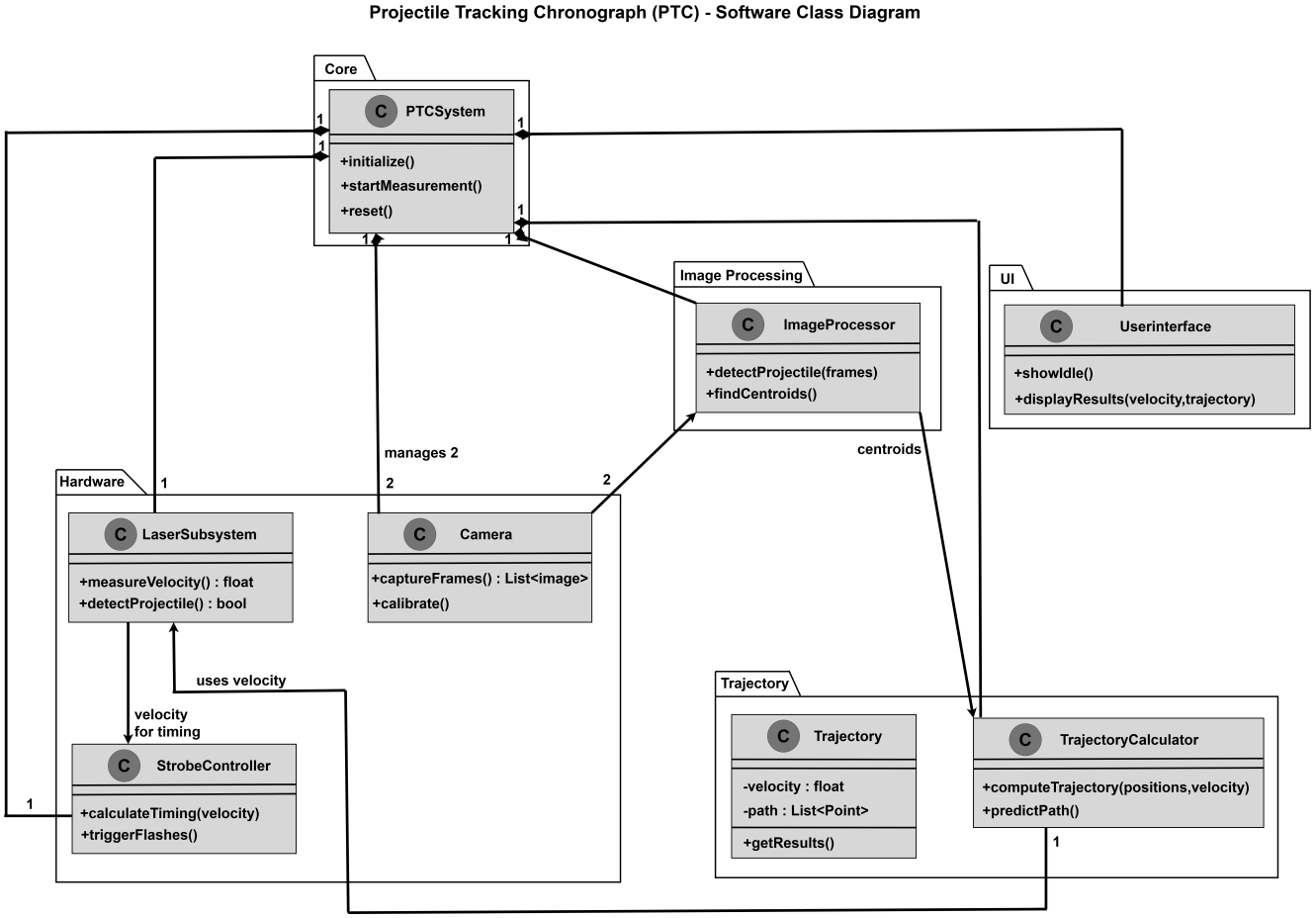


Fig. 7.4: Class diagram of the PTC

The modular architecture also supports concurrent execution of tasks, which is particularly important for maintaining real-time performance. For example, while one module handles the image acquisition, another may process previously captured frames, and a separate module may update the user interface. This level of concurrency can be achieved through multi-threading or asynchronous programming techniques with Python.

Inter-module communication is handled through well-defined data structures and inter-

faces, ensuring that information flows efficiently between different parts of the system. For instance, the images processing module outputs positional data that is directly consumed by the trajectory computation module. This separation of concerns reduces code complexity and improve system reliability.

Version control practices are also integrated into the development process to track changes and maintain code integrity. By using version control systems, the development team can manage updates, revert changes if necessary, and collaborate effectively. This is especially important in projects where multiple members are modifying the code.

7.4 Data Flow Between Subsystems

The PTC system depends on smooth and timely data exchange between its different parts. While each subsystem has its own job, they must work together quickly and efficiently during a measurement. The sequence diagram below shows exactly how data moves through the system during a typical measurement cycle. It illustrates the step-by-step interaction between the main software, laser subsystem, cameras, image processing module, trajectory computation, and user interface.

As shown in the diagram, the process begins when the laser detects a projectile and sends the velocity data to the main controller. The main software then coordinates the cameras to capture both projectile frames and post-shot empty frames. These images are passed to the image processing module, which returns cleaned positional data (centroids). This data, along with the laser velocity, is sent to the trajectory computation module, which calculates the final results. Finally, all key information is forwarded to the user interface for display. This clear data flow helps the system operate in real-time with minimal delays and ensures that each component receives the right information at the right time.

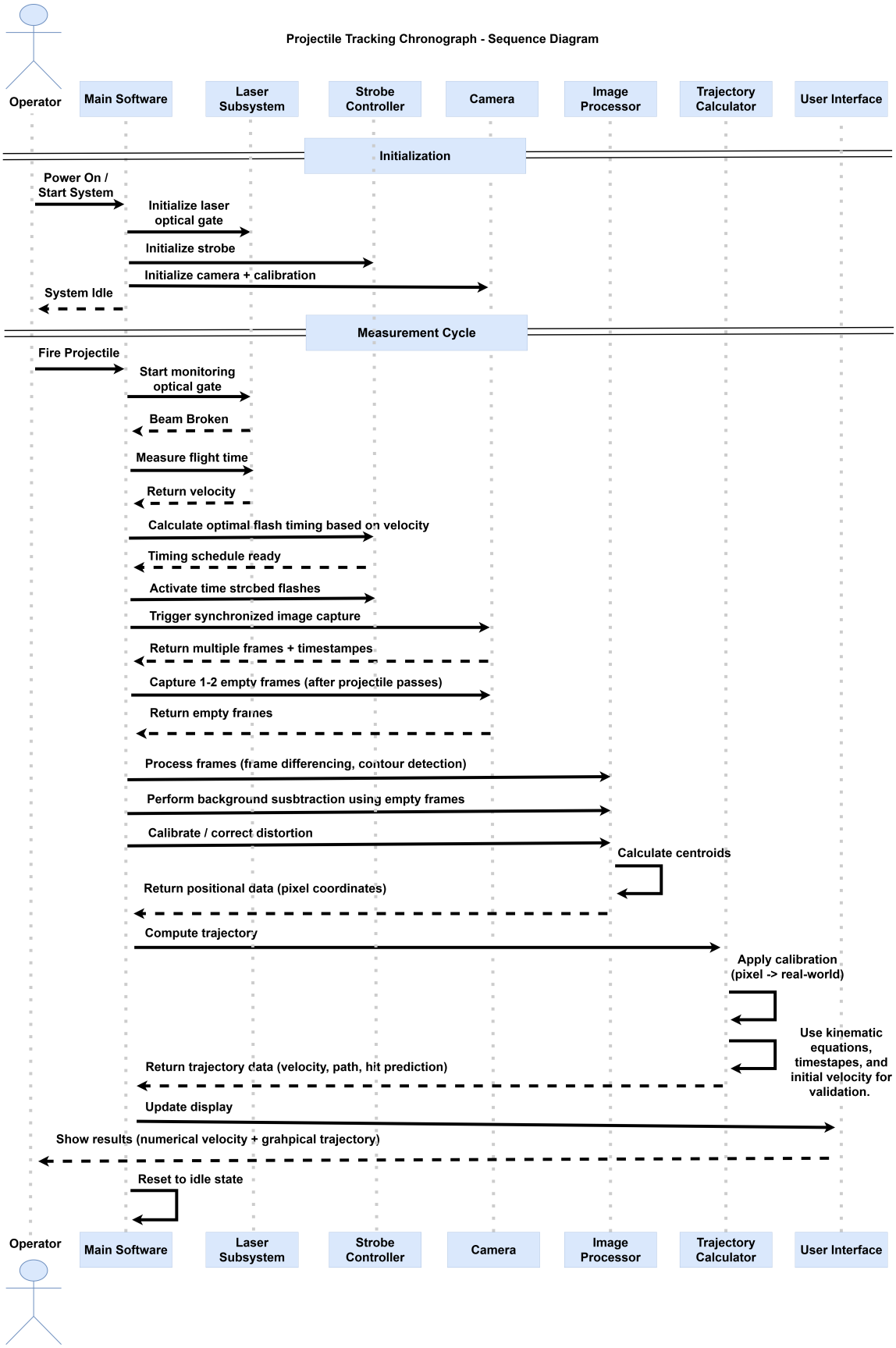


Fig. 7.5: Sequence diagram of the PTC

7.5 Image Processing Pipeline

The image processing pipeline is responsible for detecting the projectile within captured frames and extracting positional data required for trajectory estimation. The system will use OpenCV as the primary computer vision library due to its extensive functionality and compatibility with Python. [70]

Image acquisition is performed using a calibrated camera system synchronized with a strobe lighting subsystem. This allows for a clear capture of high speed projectiles to effectively identify the motion within individual frames. Once images the are captured, frame differencing techniques are applied to identify changes between consecutive frames, isolating the moving projectile from the static background.

Following motion detection, contour tracking algorithms are used to identify the shape and location of the projectile within the frame. The centroid of the detected contour is then calculated to represent the projectile's position in pixel coordinates. This process is repeated across multiple frames to track motion over time.

7.6 Camera Calibration and Coordinate Mapping

Accurate trajectory estimation requires conversion of image coordinates into real-world spatial measurements. This can be achieved through camera calibration and coordinate mapping techniques. Camera calibration determines intrinsic parameters such as focal length, optical center and lens distortion coefficients. These parameters are used to correct image distortion and ensure accurate positional measurements. Extrinsic calibration defies the spatial relationship between the camera and the real-world coordinate system. This allows pixel coordinates to be mapped into physical distances using transformation matrices.

By combining intrinsic and extrinsic calibration data, the system converts centroid positions from pixel space into real-world coordinates. This transformation is essential for accurate trajectory estimation and ensures that computed results correspond to physical measurements. Calibration is performed prior to system operation and may be repeated if hardware alignment changes. These processes significantly improve the accuracy and reliability of the trajectory computation system. [70]

7.7 Trajectory Computation

Trajectory computation combines data from both the laser subsystem and image processing pipeline. The laser subsystem provides an accurate measurement of projectile velocity, while the image processing system provides spatial position data.

Displacement is calculated by comparing centroid positions across sequential frames. Using known timing intervals derived from the strobe system, the motion of the projectile

can be analyzed. Rather than relying solely on frame-based velocity estimation, the system uses the velocity from the subsystem as a primary input and applies physics-based motion equations based off the images to estimate the trajectory.

By combining the measured velocity with its positional slope and effects of gravity, the system will predict the projectile trajectory. This approach improves accuracy and reduces dependence on high frame rate imaging alone.

In addition to basic displacement calculations, the trajectory estimation incorporates fundamental principles of projectile motion, including gravitational acceleration and initial launch conditions. By assuming a constant acceleration due to gravity, the system can model the projectile path using standard kinematic equations. This allows for prediction of the projectile's future position beyond the capture frames, enabling a full trajectory visualization rather than a simple point to point measurement.

To further enhance accuracy, calibration procedures may be applied to covert the pixel coordinates into real spatial measurements. This involves determining a scaling factor based on known distances within the camera's field of view. Without proper calibration, the system would only provide relative motion rather than actual measurements.

Error sources such as camera misalignment, timing inaccuracies, and image noise are also considered in the trajectory computation process. These factors can introduce deviations in the calculated path, so filtering techniques or averaging methods may be applied to reduce their impact. By addressing these potential errors, the system improves the reliability and consistency of its trajectory predictions.

7.8 Trajectory Modeling and Math

The estimation of the projectile trajectory in space is a critical component of the PTC system. While the laser subsystem provides an accurate measurement of initial velocity, the image processing pipeline provides spatial information required to reconstruct the projectile's path.

The system models projectile motion using kinematic equations under the assumption of constant gravitational acceleration and negligible air resistance for short-ranged measurements. The position of the projectile in space can be described as:

$$x(t) = v_0 \cos(\theta) \cdot t \tag{1}$$

$$y(t) = v_0 \sin(\theta) \cdot t - \frac{1}{2} g t^2 \tag{2}$$

$$z(t) = \text{depth component derived from multi-camera perspective} \tag{3}$$

Where:

- v_0 is the initial velocity measured by the laser subsystem
- θ is the launch angle estimated from image data
- g is the gravitational acceleration ($9.91\,m/s^2$)
- t is the time derived from the strobe synchronization

The depth component (z) is estimated using multi-camera geometry or perspective tranformation techniques. By analyzing the displacement of the projectile across multiple camera views, the system approximates its position in space.

To improve accuracy, the system applies coordinate transformation techniques to convert pixel positions to real-world units. This requires camera calibration, including intrinsic and extrinsic parameters, to map image coordinates to spatial coordinates.

Noise and measurement errors are to be mitigated through filtering techniques such as moving averages or Kalman filtering. These methods help smooth trajectory estimates and improve stability in the presence of detection noise.

7.9 System Integration of Subsystems

A critical aspect of the PTC system is the integration between the laser-based velocity measurement subsystem and the camera vision system. These two subsystems operate using different sensing principles but must function cohesively to produce accurate results. [73]

The laser subsystem provides a precise velocity measurement by detecting the time interval as the projectile passes through a defined optical gate. This measurement serves as a reliable baseline for the system, as it is less affected by image processing noise or environment lighting conditions.

The vision subsystem complements this measurement by providing spatial information about the projectile's position across multiple frames. By combining these two data sources, the system achieves a more comprehensive understanding of the projectile motion than either subsystem could provide independently.

Synchronization between the subsystems is essential for accurate operation. The detection of a projectile by the laser system triggers the strobe lighting and camera capture sequence, ensuring that images are taken at the correct moment. Any delay or mismatch in timing could lead to incorrect trajectory estimation, highlighting the importance of precise system coordination. This integrated approach demonstrates a key engineering principle of leveraging multiple sensing methods to improve overall system performance and reliability.

7.10 User Interface Design

The user interface is designed to provide clear and immediate feedback on system status and measurement results. The interface operates in multiple states, reflecting the current condition of the system. During system startup and capacitor charging for the strobe subsystem, the interface displays a charging or initialization screen to inform the user of system readiness. Once the system is fully operational, it enters an ”idle” state, clearly indicating that it is ready to detect a projectile. Upon detection, the interface transitions to a results display, showing the measured velocity obtained from the laser subsystem along with a visual representation of the estimated trajectory.

The visual design and layout of the user interface is illustrated below.

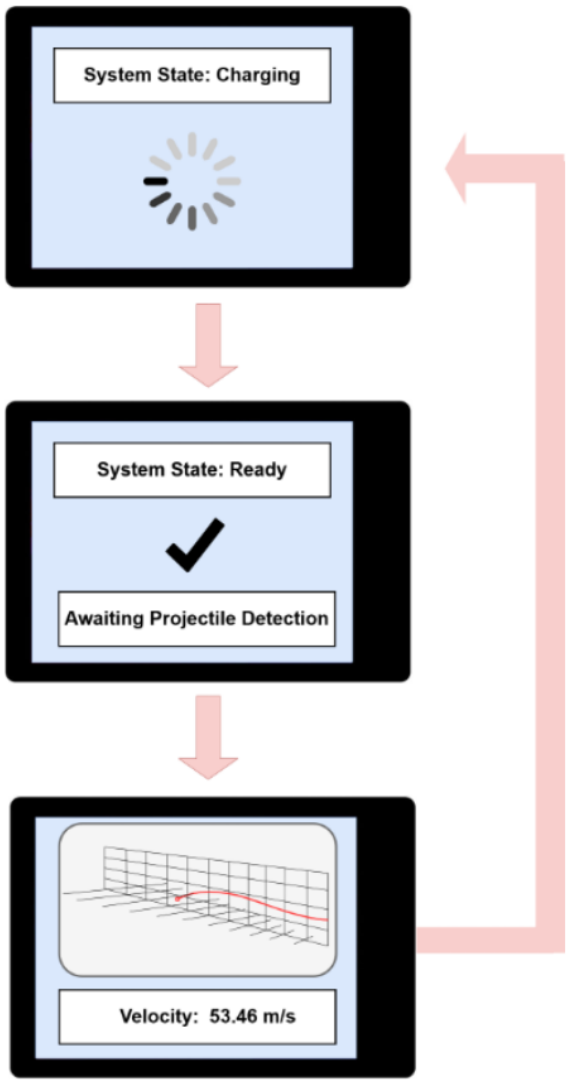


Fig. 7.6: User interface design of the PTC

The interface is designed to be simple and intuitive, allowing users with minimal technical experience to operate the system effectively.

In addition to displaying system states, the user interface is designed to enhance usability through clear visual hierarchy and minimal cognitive load. Important information such as velocity measurements and trajectory visualization is prominently displayed, while secondary information is presented in a less intrusive manner. This ensures that users can quickly interpret results without confusion.

The trajectory visualization component provides a graphical representation of the projectile path, which will be displayed to show if or where the projectile hits a target. This visualization helps users better understand the motion characteristics of the projectile beyond numerical values alone. For example, a curved trajectory can visually indicate the effects of gravity, while deviations in the path may suggest inconsistencies in launch conditions.

User feedback mechanisms are also incorporated to improve interaction with the system. Visual indicators such as color changes, status messages, or simple animations will be used to confirm successful detection events or alert the user to potential issues. These features improve system usability and reduce the likelihood of user error during operation.

Finally, the interface is designed with future expansion in mind. Additional features such as data logging, performance statistics, or adjustable system parameters could be integrated without significantly altering the existing interface structure. This forward-thinking approach ensures that the PTC system remains adaptable to future project requirements.

7.11 Software Optimization Techniques

Due to the computational limitations of embedded systems, optimization techniques are critical for maintaining real-time performance. While the Raspberry Pi 5 is powerful, it still requires efficient use of resources when performing image processing tasks.

One key optimization strategy is reducing the resolution of input images. Lower resolution images decrease the number of pixels that have to be processed, significantly improving execution speed while still maintaining sufficient accuracy for projectile detection. Another technique involves limiting processing to a region of interest. Instead of analyzing the entire frame, the system focuses only on the area where the projectile is expected to appear. This technique reduces unnecessary computation and improves processing efficiency.

Additionally, frame differencing is used instead of more computationally expensive methods such as optical flow. This provides a balance between accuracy and speed, which makes it suitable for real-time applications. Memory management and efficient data structures are also considered to minimize latency. By optimizing these aspects, the system

achieves reliable real-time performance within the constraints of the embedded platform. [70]

7.12 Error Handling and System Reliability

Robust error handling is essential to ensure reliable operation of the PTC system under real-world conditions. The software is designed to detect and respond to potential failures without causing system crashes or incorrect measurements.

Common error conditions include camera initialization failure, dropped frames, improper synchronization between the strobe and camera, and invalid sensor readings. To address these issues, the system implements exception handling mechanisms that allow it to recover gracefully or notify the user when intervention is required. For example, if the camera fails to capture an image, the system will discard the current measurement cycle and return to the armed state. Similarly, if inconsistent positional data is detected, the system avoids computing trajectory results to prevent misleading outputs.

Watchdog timers and system health checks may also be incorporated to monitor continuous operation. These mechanisms ensure that the system remains responsive and can recover from transient faults. By prioritizing reliability and fault tolerance, the software design minimizes the risk of incorrect measurements and enhances overall system robustness. [73]

8 System Fabrication

9 System Testing and Evaluation

9.1 Optoelectronics Feasibility - Camera Imaging Subsystem

9.1.1 Strobe Light Intensity

How intense do we need the light?

One AI result indicates the read noise of the IMX296 Sensor is 5 electrons, however we were unable to find validating documentation on this.

Prompt: ”noise floor of imx296 sensor” Response: ”The Sony IMX296 sensor features a read noise of 5e- (5 electrons) for both monochrome and color variants, as specified in the Pregius Gen2 embedded kit datasheets. While the provided search results do not explicitly state a ”noise floor” value in lux or dB, the sensor achieves this low read

noise through Pregius Suppression technology, which reduces noise and improves image quality, particularly in low-light conditions.

The sensor maintains a dark current of 3e-/p/sec at room temperature, contributing to low noise performance in static or slow-moving scenes. Additional noise reduction is supported by Black Level Correction and a peak quantum efficiency of 69% at 570 nm, which enhances signal strength relative to noise. ”

[Cite Using Brave search AI https://search.brave.com/search?q=noise+floor+of+imx296+sensorsource=w

3 electrons per pixel per second is extremely low, especially for short exposure times of less than 100 [ms] will typically have 0 electrons of noise.

9.1.1.1 Readout Noise Based Minimum Illumination

So I will derive the minimum illumination for 2 criteria, the readout noise of 5 electrons having a sufficient SNR to place a 1 [cm] $\times$ 1 [cm] projectile in an image that is on a 29 [cm] $\times$ 22 [cm] background that has a 10% difference in reflectivity than the background to have on average 95% of the time every pixel of the projectile at a different level than every pixel of the background.

We will be assuming a strobe time of $T = 100\mu$s, that half of the emitted light hits the desired area (efficiency $\eta_{light} = 0.5$), that the wavelength of light is $\lambda = 850[\text{nm}]$, and the camera is $n_{pixels} = 1580000$ or 1.58 megapixels. The formula for photons hitting a pixel given a total light intensity I and these assumptions is

$$N = \frac{I \cdot T \cdot \lambda}{hc} \cdot \frac{\eta_{light}}{n_{pixels}} = \frac{I \cdot 100[\mu\text{s}] \cdot 850[\text{nm}]}{hc} \cdot \frac{0.5}{1580000} = \frac{I}{7.38 \times 10^{-9}[\text{W}]}$$

or 135000 photons per milliwatt.

The projectile would have a size of about 50 pixels by 50 pixels, giving 2500 pixels on the projectile, with about 1,580,000 pixels of background.

Assuming a peak quantum efficiency of 0.69, combined with the 850 [nm] relative efficiency of 0.4, giving a total quantum efficiency of $\eta_{QE} = 0.276$. N is the number of photons that hit the detector, while N_e is the number of electrons that are detected.

$$\sigma_{total} = \sqrt{\sigma_{shot}^2 + \sigma_{readout}^2}$$

If we approximate the shot noise with a gaussian distribution, we can combine the readout noise distributions and the shot noise distributions to approximate the total noise distribution.

$$P_{background<threshold}^{1580000} > 0.5$$

and

$$P^{2500}_{projectile>threshold} > 0.5$$

Which implies

$$P_{background<threshold} > 0.999999561$$

and

$$P_{projectile>threshold} > 0.99972278$$

$$\sigma_{total} = \sqrt{N_e \cdot \sigma_{readout}} = \sqrt{N \cdot 0.276 + 25} = \sqrt{\frac{I \cdot 3.75 \times 10^7}{1[\mathrm{W}]} + 25}$$

While the relative noise is $\frac{\sigma}{N}$ is

$$\frac{\sigma_{total}}{N} = \frac{\sqrt{\frac{I \cdot 3.75 \times 10^7}{1[\mathrm{W}]} + 25} \cdot 7.38 \times 10^{-9}[\mathrm{W}]}{I}$$

We want to select a threshold percentage such that the two probabilities are true. The best threshold seems to be 0.945 given the projectile has a reflectivity of 90%, and the background has a reflectivity of 100%.

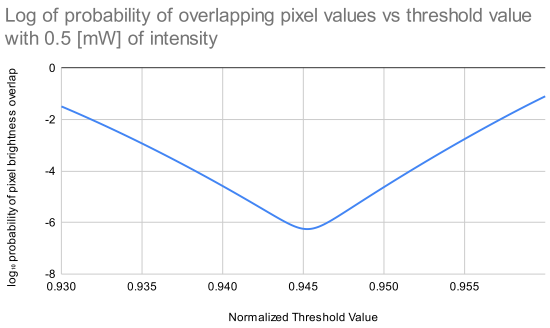


Fig. 9.1: Graph to determine the best threshold value

Spreadsheet calculations give a the value for intensity that creates the probability of a pixel on the projectile having an intensity that overlaps with the intensity of the background of 275 microwatts.

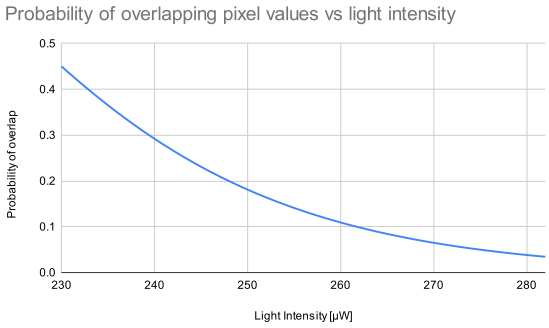


Fig. 9.2: Graph to determine the minimum intensity with no ambient light

If we assume that the camera cannot distinguish pixels with a small difference in the number of electrons per pixel, that is by discretizing the number of electrons in a pixel into ”buckets” of illumination values, the minimum illumination would be higher. If we assume the ”buckets” of illumination values are 1000 electrons in size, compared to the approximately 25,000 electrons for ”full” illuminance, the minimum illumination value raises to about 0.7 [mW].

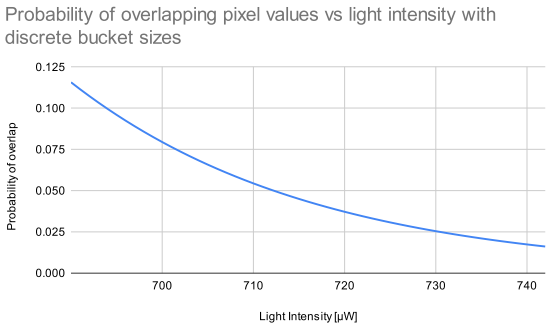


Fig. 9.3: Graph to determine the minimum intensity with no ambient light with discrete illumination bucket sizes of 1000 electrons

Given that the stray illumination from outside the system will likely be far greater than this, the largest source of noise will be stray light.

9.1.2 Outdoor ambient light based minimum intensity

To calculate the minimum intensity based on the light leaked into the strobe chamber, we take the light within the bandwidth of the LEDs filtered out by the bandpass filter, and make sure it is less than 1/2 of the amount of light that is provided by the LEDs.

If we assume a bandwidth $\Delta\lambda = 80[\mathrm{nm}]$, a working environment of sunlight, and having 1/10th of the light from the sun gets into the strobe chamber.

The intensity of the sun between 810 [nm] and 890 [nm] is about $900[\mathrm{W} \cdot \mathrm{m}^{-2} \cdot \mu\mathrm{m}^{-1}]$.

This gives about $0.9[\mathrm{W} \cdot \mathrm{m}^{-2} \cdot \mathrm{nm}^{-1}] \cdot 80[\mathrm{nm}] = 72[\mathrm{W} \cdot \mathrm{m}^{-2}]$. There would be about $7.2[\mathrm{W}/\mathrm{m}^2]$ of light incident on the entrances to the strobe chamber. Given 2 openings that are 20 [cm] by 20 [cm], that is 0.58 [W] of light getting into the strobe chamber. Given that our slowest projectile supported is 5 [m/s] within about 0.5 [m] long strobe chamber, that would be an exposure of 100 [ms]. Comparing the strobe on-time to the total exposure time, that is a ratio of about 1000/1, which would mean our total light power would have about 1000 times the illumination ratio of strobe to ambient, making it 2000 times the ambient illumination level. That makes about 1200 [W] of strobe illumination, which is too much. If we increase the minimum required speed of projectiles outdoors to 50 m/s, therefore reducing the maximum ambient exposure time to $\frac{0.5[\mathrm{m}]}{50[\mathrm{m/s}]} = 10[\mathrm{ms}]$, that takes the strobe intensity requirements down by a factor of 10, which means that 120 [W] of light would be required.

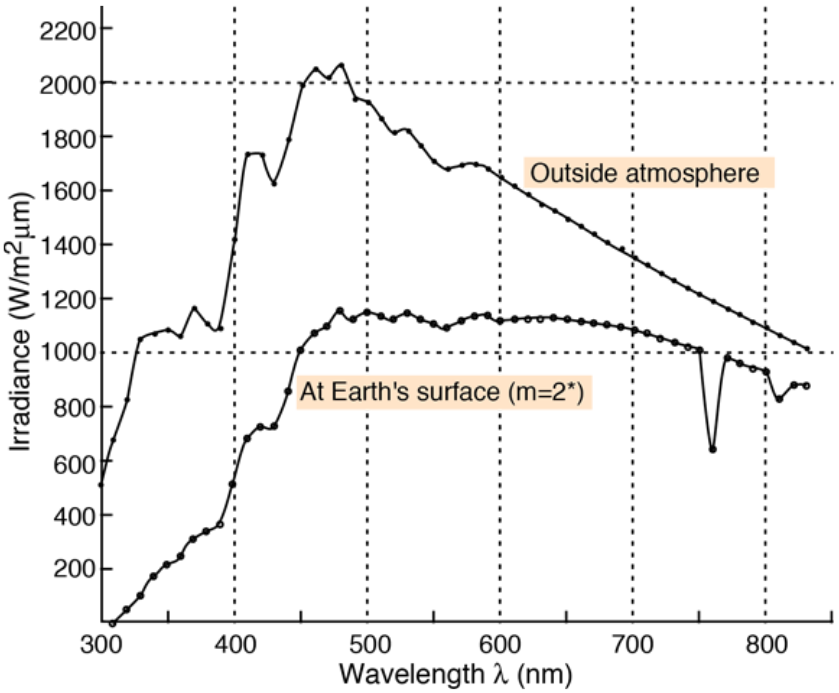


Fig. 9.4: Solar irradiance Credit: NASA via R Nave GSU web page

9.1.3 Indoor ambient light based minimum intensity

Since indoor LED and flourescent lighting doesn't produce any significant amount of 850 [nm] IR light, we'll simply take the intensity of outdoor sunlight and divide it by the relative intensity of indoor task lighting, which should give a good upper limit of the intensity of indoor lighting. Outdoor noon sunlight can be up to 10,000 lumens per square foot, while indoor task lighting is typically around 100 lumens per square foot. This gives a ratio of 100/1, so that brings our ambient calculations of 1200 [W] down to 12 [W], which is far more affordable. [89]

9.1.4 Alternative non-strobe / stray light resilient approach

Given that we will already have the speed measurement from the light gate, the imaging setup does not need to get the speed of the projectile, only the trajectory direction. If we instead have the illumination constant during the projectile's entire traversal of the imaging area, the projectile will appear as a streak across the image. Taking two intensity profiles across the width of the system near the beginning and end of the imaging area will allow the angle for two points of the trajectory with respect to the camera to be obtained. These angles can be combined between the two cameras to get a 3D trajectory of the projectile, and combined with the initial speed information to get the direction and velocity.

9.1.5 LED Field of View

The field of view (FOV) of an LED is the angle at which the light comes out of the LED. The FOV of the LED is one point of comparison, since a wider FOV will provide more even lighting within the sensing area of the strobe imaging chamber, while a narrower FOV will provide more light on target, and therefore be more efficient. A good baseline is to have the LED FOV be at least as big as the FOV of the camera, so that it can approximately illuminate the entirety of the sensing area within the FOV of the camera from between the cameras in the same plane as the cameras. Given that the FOV of the cameras will likely be less than $90°$, an FOV of the LEDs of between $90°$ to $120°$ would be ideal.

9.1.6 Camera

9.1.6.1 Relative Camera Angular Placement

Stereoscopic vision is often done with in-line cameras separated by a known distance. An alternative is to have the cameras facing different directions, which has a few advantages and disadvantages. Cameras at different angles could be tilted to better cover the sensing area within the strobe chamber and make better use of all of the pixels. However the calculations with the cameras at different angles could prove to be more complex and therefore difficult. There are more potential sources of error for differently angled cameras. It could be possible to perform calibration photographs, or include a permanent calibration marking within the strobe chamber to properly calibrate where the cameras are for each shot. Calibration markings at the beginning and end of the strobe chamber would be possible to crop out of the images used for the projectile detection, but could be used for precisely locating the cameras in space.

Due to the decreased complexity, we will be using the same tilt direction for both cameras, separated simply by a distance.

9.1.6.2 Camera Spatial Placement and Field of View

The placement of the cameras and field of view has a significant impact on the overall size and shape of the end product, in addition to having a significant impact on the size

of the sensing area.

A larger FOV would allow the cameras to be placed closer to the sensing area, but it would also cause there to be more unused volume in the field of view, since the FOV must include the entire top of the sensing area, and a larger FOV means the viewed area at the bottom of the sensing area would be much larger than the actual sensing area.

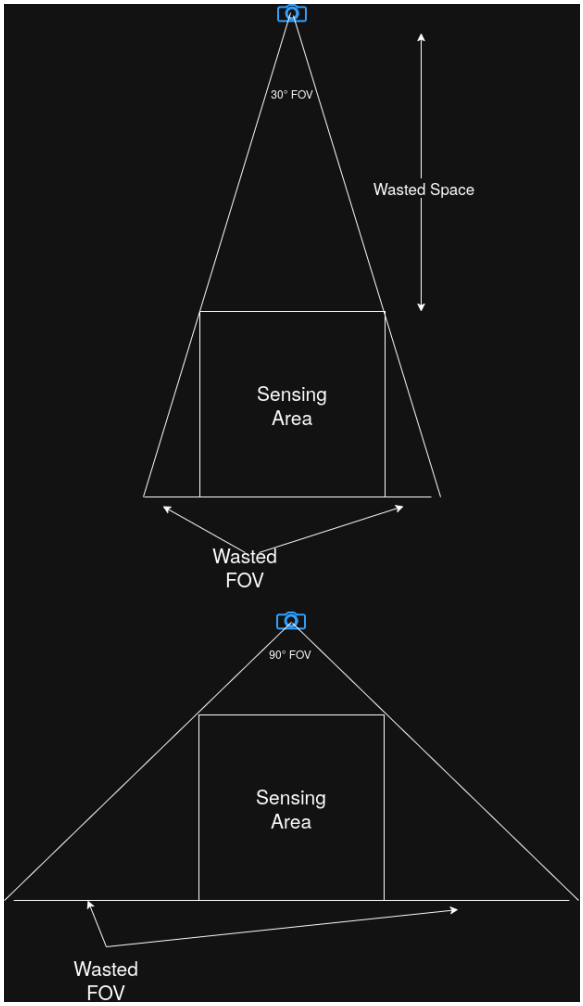


Fig. 9.5: FOV Comparison

An FOV of between $60°$ and $90°$ would be good.

9.2 Optoelectronics Feasibility - Light Gate Subsystem

9.2.1 Optical Link Analysis

The light gate subsystem serves as the primary trigger mechanism for the PTC, detecting the entry of a projectile into the sensing region and initiating time-of-flight measurements

for velocity calculation. This analysis section describes the optical feasibility of using the selected Adafruit 1057 line laser module paired with the PDB-C171SM silicon photodiode to achieve reliable projectile detection.

9.2.1.1 Laser Source Characteristics

The Adafruit 1057 line laser module emits at 650 nm with a maximum optical output power of 5 mW. This module has an integrated cylindrical Fresnel lens that produces a line pattern with a 120° fan angle, distributing the optical power across the illuminated region. For a sensing area width of 100 mm (10 cm) at the target distance, the linear power density can be calculated. The geometric relationship between fan angle and line length are as follows:

$$L = 2d \cdot \tan\left(\frac{\theta}{2}\right) \tag{4}$$

where L is the line length, d is the distance from the aperture of the laser, and θ is the fang angle. Rearranging the equation to solve for d, the source-to-target distance for the laser to achieve a sensing region of 100 mm is approximately 29 mm.

$$d = \frac{L}{2 \cdot \tan(60^\circ)} = \frac{100\ mm}{2 \cdot \tan(60^\circ)} \approx 29\ mm \tag{5}$$

With this geometry, the 5 mW output power distributed across the 100 mm sensing region results in a power density of approximately $50\ \frac{\mu W}{mm}$:

$$P_{linear} = \frac{5\ mW}{100\ mm} = 50\ \frac{\mu W}{mm} \tag{6}$$

Due to limited information from the Adafruit data sheet, it is going to be assumed that the intensity profile for this laser is roughly Gaussian in nature, meaning that the central region of the line has a higher radiance than the edges. Additionally, assuming that the distribution of the beam is uniform, the irradiance at the photodetector primarily depends on the efficiency of the collimating/focusing system being used to expand the laser light into sheets. [12]

9.2.1.2 Photodetector Response

The PDB-C171SM silicon photodiode exhibits a responsivity of about 0.4 A/W at 650 nm, as seen in the figure below.

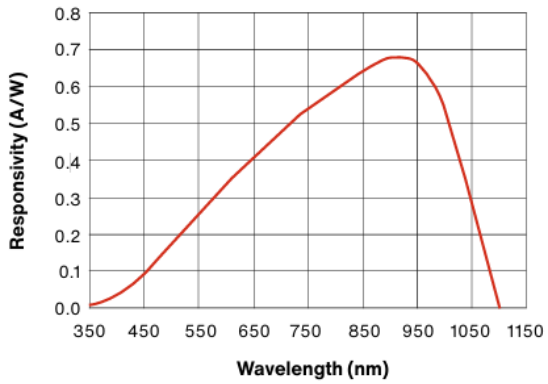


Fig. 9.6: PDB-C171SM Photodiode Spectral Response

The photocurrent generated by the incident optical power is:

$$I_{ph} = R(\lambda) \cdot \ P_{incident} \tag{7}$$

where $R(\lambda)$ is the wavelength-dependent responsivity and $P_{incident}$ is the optical power reaching the active area. For our laser diode, the ideal photocurrent is $2\ mW$. In our setup, we are going to use reflective surfaces to collimate the laser beam, bounce it across the sensing region, and focus it back down to a point to be detected by the photodetector. We have decided to use aluminum foil tape on 3D printed structures. The aluminum foil tape provides high reflectivity in the visible spectrum, with values of $88 - 92\%$ for polished aluminum surfaces at 650 nm. [24]

For a photodetector with an active area width of approximately 5-10 mm positioned to capture the reflected beam, the received optical power becomes:

$$P_{received} = P_{incident} \cdot R \cdot \eta_{collection} \tag{8}$$

where R is the reflectivity of the aluminum tape and $\eta_{collection}$ is the collection efficiency of the photodiode. If we are to assume that the reflectivity is $\approx 90\%$ and the collection efficiency is $\approx 10\%$, then the received power becomes:

$$P_{received} = 5\ mW \cdot 0.9 \cdot \ 0.1 = 450\ \mu W \tag{9}$$

resulting in a corresponding photocurrent of:

$$I_{ph} = 0.4\ \frac{A}{W} \cdot \ 450\ \mu W = 180\ \mu A \tag{10}$$

For the photodiode capacitance values at $V_{bias} = 10\ V$ and the specified rise time with load resistance $R_L = 1\ k\Omega$, the bandwidth can be estimated form the RC time constant equation:

$$f_{-3dB} = \frac{1}{2\pi R_L C_d} \tag{11}$$

In the datasheet for the photodetector, the rise time measurements at 820 nm with a

$R_L = 1\ k\Omega$ load suggest bandwidth capabilities compatible with the 10-100 kHz carrier frequency. We would have to do more experimental testing to determine the bandwidth capabilities at 650 nm, but we believe that it would still fall into a usable range for our modulated detection scheme. [90]

9.2.2 Signal-to-Noise Analysis

9.2.2.1 Noise Sources

Shot noise is a type of electronic noise that arises from the discrete nature of electric charge, which causes random fluctuations in steady current flow. It can be expressed by this equation:

$$i_{shot} = \sqrt{2q(I_{ph} + I_d)B} \tag{12}$$

where q is the fundamental charge of an electron ($1.602 \cdot 10^{-19}\ C$), I_d is the dark current, and B is the bandwidth.

Another form of noise is thermal noise, which arises from the random thermal motion of charge carriers in a resistive material. This can cause current fluctuations in a material and can be described as follows:

$$i_{thermal} = \sqrt{\frac{4k_B T B}{R_L}} \tag{13}$$

where k_B is the Boltzmann's constant ($1.381 \cdot 10^{-23} \frac{J}{K}$), T is the temperature, and the other parameters are the same as previously defined. It can be seen from this equation that thermal noise increases as the temperature and/or bandwidth increases and decreases with higher resistance.

Assuming that $T = 300\ K$ and $B = 100\ kHz$, the shot and thermal noises can be calculated as such:

$$i_{shot} = \sqrt{2 \cdot 1.602 \cdot 10^{-19} \cdot (180 \cdot 10^{-6}) \cdot 100 \cdot 10^3} = 2.4\ nA \tag{14}$$

$$i_{thermal} = \sqrt{\frac{4 \cdot 1.381 \cdot 10^{-23} \cdot 300 \cdot 100 \cdot 10^3}{1000}} = 40.7\ nA \tag{15}$$

The total RMS noise can be calculated by this equation:

$$i_{RMS} = \sqrt{i_{shot}^2 + i_{thermal}^2} \approx 40.8\ nA \tag{16}$$

9.2.2.2 Signal-to-Noise Ratio

The signal-to-noise ratio (SNR) quantifies how strong a desired signal is compared to the background noise of the received signal. A higher SNR value indicates a strong signal

with little background noise, which is desirable for the PTC project. The SNR can be expressed in dB, as seen the the equation below:

$$SNR = \frac{I_{ph}}{i_{RMS}} = \frac{180\ \mu A}{40.8\ nA} \approx 4400 \approx 73\ dB \tag{17}$$

9.2.3 Projectile Detection Timing Analysis

9.2.3.1 Beam Interruption Geometry

For a projectile of diameter D traveling at a velocity v that is perpendicular to the laser line, the duration of the beam interruption is:

$$t_{interrupt} = \frac{D}{v} \tag{18}$$

For the specified projectile size range (15-50 mm) and velocity range (10-50 m/s), the upper and lower bounds of the system's temporal resolution requirements can be determined.

TABLE 9.1: Beam Interruption Bounds

Projectile Diameter	**Velocity**	**Interruption Time**
15 mm	50 m/s	300 μs
15 mm	10 m/s	1.5 ms
50 mm	50 m/s	1 ms
50 mm	10 m/s	5 ms

This table shows that for our system, our upper bound is a minimum interruption time of 300 μs and our lower bound is 5 ms on the system's temporal resolution requirements.

9.2.3.2 Carrier Frequency Selection

A modulated carrier frequency approach provides a much greater resistance to ambient light interference by shifting the detection signal away from the DC regime. The carrier frequency must satisfy the following:

$$f_{carrier} > \frac{1}{t_{interrupt,\ min}} \tag{19}$$

where, for our system with a minimum interruption time of 300 μs, the carrier frequency must be greater than 3.3 kHz. To ensure multiple carrier cycles occur during the shortest beam interruption, a carrier frequency of 10-50 kHz should be sufficient. [90]

9.2.3.3 Rise Time Requirements

The laser diode's rise and fall times have a direct impact on the modulation depth that can be achieved at the carrier frequency. For a 50 kHz carrier (period of 20 μs), the

rise and fall times should be less than 10% of the half-period in order to maintain a modulation depth greater than 80%.

$$t_{rise,\ max} < 0.1 \cdot \frac{t_{period}}{2} = 1\ \mu s \tag{20}$$

The Adafruit 1057 datasheet does not specify the rise and fall time characteristics. This means that the exact characteristics of this diode will have to be experimentally derived, making theoretical verification more difficult. There is a TTL-modulated variant of this module (Adafruit 1056) that specifies a maximum modulation frequency of 50 kHz, which was the basis for the previous calculations. We believe that the rise times on the order of microseconds for the Adafruit 1057 variant may be achievable as well. [91]

9.2.3.4 Ambient Light Filtering via Modulation

There are many benefits to using a carrier modulation approach to filter out ambient light. By doing this approach (either through a bandpass filter or lock-in detection method), unwanted signals such as the DC components from constant ambient illumination or broadband noise outside of the detection bandwidth are rejected. This type of filtering should be more than enough to achieve the specified ambient light tolerance by a substantial amount. [92]

9.2.4 Velocity Measurement Analysis

9.2.4.1 Time-of-Flight

To measure the velocity of the projectile, there needs to be two spatially separated light gates with a known distance d_{gates}. The velocity can be measured by dividing this known distance by the difference in time between the two disturbances in the gate:

$$v = \frac{d_{gates}}{\Delta t} \tag{21}$$

9.2.4.2 Timing Resolution and Spatial Uncertainty

For the specified maximum speed error of 20%, the timing resolution must satisfy:

$$\frac{\Delta t}{\Delta t_{measured}} < 0.2 \tag{22}$$

For a 50 m/s projectile traversing a 100 mm gate separation:

$$\Delta t_{measured} = \frac{100\ mm}{50\ m/s} = 2ms \tag{23}$$

Making the required timing resolution:

$$\Delta t_{timing} = 0.2 \cdot 2\ ms = 400\ \mu s \tag{24}$$

The microcontroller we have chosen (either Raspberry Pi Pico 1 or 2) will have clock speeds the provide timing resolutions that are more than sufficient (133 and 150 MHz respectively for precise measurements.

For a gate separation measurement uncertainty of $\pm$ 1 mm over a 100 mm sensing region:

$$\frac{\Delta d}{d_{gates}} = \frac{1\ mm}{100\ mm} = 1\% \tag{25}$$

This contributes minimally to the amount of error being contributed to the system, leaving a greater amount of leeway for systematic effects such as non-perpendicular trajectories.

9.2.5 Power Analysis

9.2.5.1 Power Budget

The laser diode module requires a 2.8-5.2 V input at a maximum of 25 mA. Using the maximums for these two metrics, the max power of the laser diode is:

$$P_{laser,max} = 5.2\ V \cdot 25\ mA = 130\ mW \tag{26}$$

This is a very small power draw, making it a very low-power component of the project.

9.2.6 Summary of Feasibility

The analysis of these different aspects of the subsystem indicates that the selected components are capable of meeting the requirements for light gate detection. The primary uncertainties can be attributed to the lack of characterization for the Adafruit 1057 laser module. Testing with this diode should verify the modulation performance, however, other diodes should be considered in case this current module is unable to be modulated within specs.

TABLE 9.2: System Performance Summary

Parameter	**Requirement**	**Calculated**	**Status**
SNR during interruption	≥ 20 dB	~ 73 dB	✓ Exceeds
Photodetector bandwidth	> 100 kHz	$\sim 100+$ kHz	✓ Meets
Detection of 15–50 mm projectiles	Required	300 μs – 5 ms events	✓ Feasible
Velocity range 10–50 m/s	Required	Compatible with timing	✓ Feasible
Carrier frequency	10–100 kHz	50 kHz recommended	✓ Meets

Line length	≥ 100 mm	100 mm at 29 mm distance	✓ Meets
Modulation depth	$\geq 80\%$	Requires verification	△ Uncertain
Rise/fall time	$\leq 1\ \mu s$	Not specified	△ Uncertain

10 Administrative Content

10.1 Budget

The PTC project is planned with a total expenditure limit of $1000, as established by the project team to ensure the system can be developed within reasonable cost constraints while still meeting technical objectives. This budget is intended to support the optical, electrical, and computing components that are required for the implementation of the optical projectile measurement system.

10.2 Bill of Materials (BOM)

The BOM provides a comprehensive list of the components that are necessary to design and build the PTC. Maintaining the BOM throughout the project will aid in cost awareness and serve as a centralized reference for the various hardware needs of the project. Below is a table containing the components and estimated costs of each part. The extended prices are the unit prices of the components multiplied by their respective quantities.

TABLE 10.1: Bill of Materials

Components	**Name / Part Number**	**Distributer**	**Extended Price**	**Quantity**
Camera	Raspberry Pi Camera Module 2	PiShop	$31.90	2
Image Processor	Raspberry Pi 5 2GB	PiShop	$65.00	1
MCU	RP2350	DigiKey	$5.88	4
Development Board for Testing	Raspberry Pi Pico 2	PiShop	$6.40	1
Photodetector	PDB-C171SM	DigiKey	$8.98	2
Laser Diode	Adafruit 1057	DigiKey	$17.90	2
Aluminum Tape (as a reflector)	Romeda Aluminum Foil Tape	Amazon	$6.98	1
Strobe LED Cluster	JR5050AWT-Q-U30EB0000-N0000001	DigiKey	$14.82	30

Power Supply	WDU12-1200	DigiKey	$28.06	1
3.3V Voltage Regulator	RAA2118034GP3#JA0	DigiKey	$1.76	4
N - MOSFET Switch	MCU45N10-TP	DigiKey	$7.92	10
47uF Capacitor	EJ1C470MP20511U	DigiKey	$0.69	10
Inrush Current Limiter	B57153S0200M054	DigiKey	$2.90	5
LCD Screen	Waveshare 2.8inch LCD Display Module	Amazon	$25.99	1
Rotary Encoder	PEC12R-4025F-N0024	DigiKey	$5.97	5
Button	PTS636SM43 LFS	DigiKey	$1.29	10
Power Switch	SW-R2-1A-F-1-0	DigiKey	$2.25	5
			Estimated Maximum Total Cost:	**$234.69**

10.3 Project Milestones

This senior design project involves multiple interconnected technical and documentation tasks, so setting clearly defined milestones with set deadlines is essential for maintaining consistent progress and organization. Clearly scheduled milestones break the project into manageable phases, helping our team to prioritize either individual or group tasks at any given time. By committing to specific dates, our team is able to ensure that all progress made between each stage of the project is done so in a controlled and predictable manner, thereby minimizing any last-minute risks that may occur. Our project milestone table has a description of each milestone, the start/anticipated end dates, and each milestone's dependencies.

TABLE 10.2: Project Milestones

Milestone Description	**Start Date**	**End Date**	**Description**
Finalize Project Concept and Scope	1/27/26	1/29/26	Define project objectives, requirements, and system boundaries
Divide and Conquer Document Finalization	1/25/26	1/29/26	Complete task allocation and subsystem responsibility documentation
Divide and Conquer Review Meeting	2/3/26	2/3/26	Review and approve task distribution with advisors
Committee Formation and Initial Meeting	1/26/26	1/30/26	Form the advisor committee and discuss project expectations
Find image processing software	2/3/2026	2/13/2026	Research available options

30-Page Draft Development and Revision	2/3/26	2/20/26	Develop an initial technical report and revise based on feedback
Sensor Gate Subsystem Testing	2/6/2026	2/29/2026	Verifying reliable projection detection, timing accuracy, and interrupt behavior
Camera + Strobe Subsystem Testing	2/6/2026	2/29/2026	Validating synchronized strobe activation and image capture during projectile events
60-Page Draft Development and Revision	2/20/26	3/10/26	Expand documentation with detailed design and implementation
Photonic Subsystem Integration	3/1/2026	3/16/2026	Integrating laser, mirrors, and optical sensors into a unified system
Senior Design 1 Midterm Demo	3/3/2026	3/3/2026	Develop a functional photonic subsystem capable of demonstrating one core component of the project
90-Page Draft Development and Revision	3/10/26	3/27/26	Finalize comprehensive system design and analysis documentation
Senior Design 1 Final Report	3/27/26	4/10/26	Submit the completed SD1 design report
Image Processing Program Finalized	4/12/2026	4/21/2026	Completing and validating image-processing algorithms for projectile position and trajectory analysis
Senior Design 1 Final Demo	4/21/2026	4/21/2026	Implement and validate a complete photonic subsystem capable of executing a primary project function
Mini Demonstration Video	4/28/26	4/28/26	Demonstrate early system functionality and progress
CDR Presentation Video	5/18/2026	5/29/2026	Present the finalized design
CDR Presentation Slide	5/18/2026	5/29/2026	Prepare slides summarizing system design and progress
Senior Design 2 Midterm Demonstration Video	5/30/2026	5/30/2026	Show system integration and functional progress

8-Page Conference Paper	6/1/2026	6/7/2026	Prepare a concise conference-style technical paper
Senior Design 2 Final Report	6/7/2026	6/23/2026	Submit the final report covering implementation and testing
Final Presentation Video	6/7/2026	6/23/2026	Present final system design and results
Final Presentation Slide	6/7/2026	6/23/2026	Prepare slides that summarize final project outcomes
Final Demonstration Video	6/7/2026	6/23/2026	Demonstrate completed system functionality

10.4 Hardware Milestones

The hardware portion of this senior design project involves several design and testing steps, making explicitly defined hardware milestones essential for maintaining steady progress. Since there are many interdependent portions of this project, having set deadlines for milestones such as gathering components and prototyping will ensure that everyone in the group is able to make consistent advancements in their respective portions. Below is the hardware milestone table, with the start/anticipated end dates as well as a description of each milestone.

TABLE 10.3: Hardware Milestones

Hardware Milestone	**Start Date**	**End Date**	**Description**
Component Selection	1/26/2026	2/8/2026	Making a Bill of Materials
Gather Electronic Components (Power Supply, Capacitors, MOSFETs, etc.)	2/6/2026	2/21/2026	Ordering from a manufacturer
Gather Optical Components (Laser Diodes, Photodetectors, Cameras, etc.)	2/6/2026	2/21/2026	Ordering from a manufacturer
System Testing	2/6/2026	2/29/2026	All components are tested and work
Complete Project with Prototype Board	3/16/2026	4/1/2026	Assemble the components for a prototype
PCB Testing	4/5/2026	4/10/2026	Creating and ordering a customized PCB
Test Prototype with PCB	4/10/2026	4/14/2026	Using the customized PCB for the design

User-Interface System Integrated	5/30/2026	5/31/2026	Integrating the graphical user interface with back-end processing for data display
Finalize Prototype	6/1/2026	6/23/2026	Finalize the design using a specialized PCB board

10.5 Work Distribution

The work distribution table provides an outline of the allocation of responsibilities among the team members of the PTC project. This ensures balanced contributions as well as clear ownership of the project tasks. It identifies the primary roles for members across hardware and software development, testing, and system integration to promote accountability and collaboration.

TABLE 10.4: Work Distribution of Group Members

Name	**Major**	**Responsibilities**
Austin Stoetzer	Optical Engineering	Sensor Gate Design and Implementation Project Lead Administrative Documentation Photonic System Integration with Electronics and Software
Blake Rainwater	Optical Engineering	Camera and Strobe System Prototype Lead Projectile Trajectory Visualization Photonic System Integration with Electronics and Software
Mathew Aranibar	Electrical Engineering	PCB Design MCU Selection and Implementation Component Research Lead Photonic System Integration with Electronics
Gianni Angelone	Computer Engineering	Software and UI Integration Lead Website Design and Management Projectile Trajectory Visualization Photonic System Integration with Software

11 Conclusion

Appendix A - Citations

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Appendix B - Useful Tools

[1] EdrawMax Online

[2] Draw.io

[3] Microsoft PowerPoint

[4] Google Docs

[5] Digikey

[6] Google Search Engine

[7] Ultra Librarian

[8] Snap Magic Search

Appendix C - AI Use Prompts and Outcomes

Case Study 1

[1a] Case Study 1 Prompt

Prompt: “Design a real-time image processing pipeline capable of detecting a high-speed projectile using three sequential images captured with synchronized strobe illumination. The system will run on a Raspberry Pi using Python and OpenCV. Explain the processing stages required to detect the projectile, calculate velocity, estimate trajectory angle, and display a real-time visualization suitable for a user interface.”

[1b] Case Study 1 ChatGPT Outcome

“To implement a real-time projectile tracking system using Python and OpenCV, the processing pipeline should be divided into structured stages. First, synchronized image acquisition must capture three frames illuminated by a short-duration strobe to freeze projectile motion. These frames should be timestamped to preserve accurate timing intervals.

Next, preprocessing should be performed to improve detection reliability. This includes grayscale conversion, Gaussian filtering to reduce sensor noise, and intensity normalization to compensate for lighting variation. Background subtraction or frame differencing can then be used to isolate moving objects between consecutive frames.

After motion isolation, contour detection can identify candidate objects. The projectile position may be estimated by calculating the centroid of the detected contour. Tracking the centroid across the three frames allows displacement measurements to be computed in pixel space. Using calibration data, pixel displacement can be converted into real-world distance.

Velocity is calculated by dividing displacement by the known time interval between frames. The launch angle can be estimated using linear regression or slope calculations between detected projectile positions. Finally, visualization may be implemented using OpenCV drawing functions or a plotting library to animate projectile motion in real time for display on a graphical user interface.

Performance optimization techniques such as region-of-interest cropping, multithreading, and reduced image resolution may be necessary to maintain real-time execution on embedded hardware platforms.”

[1c] Case Study 1 Google AI Outcome “A projectile tracking system using sequential images can be implemented through several computer vision techniques. Motion detection may be achieved using optical flow, frame differencing, or feature-based tracking methods. Each technique provides advantages depending on lighting consistency and object speed.

The system should begin by capturing synchronized images and converting them into grayscale format for computational efficiency. Edge detection or threshold segmentation may then be used to identify moving objects within each frame. Object tracking algorithms such as Kalman filtering or template matching can improve stability across frames.

Velocity estimation can be determined by measuring positional changes over time, while trajectory estimation may involve curve fitting or predictive modeling. Visualization can be implemented through graphical overlays or animation frameworks capable of updating in real time.

While multiple approaches exist, system performance depends heavily on hardware capability, camera synchronization accuracy, and image noise conditions.”

[1d] Case Study 1 DeepSeek Outcome “A practical implementation of projectile tracking should prioritize computational efficiency and robustness against noise. Instead of relying

solely on contour detection, adaptive thresholding combined with morphological filtering can improve detection reliability under varying illumination.

The algorithm may store detected projectile coordinates in a buffer and apply predictive tracking methods such as Kalman filtering to estimate motion between frames. Parallel processing techniques and vectorized operations are recommended to maintain real-time performance.

For visualization, a physics-based animation model can simulate projectile motion using calculated velocity and angle parameters. Rendering may be performed using OpenCV drawing routines or a lightweight graphics framework. Optimization strategies such as GPU acceleration or batch processing can further improve responsiveness."

Case Study 2

[2a.1] Case study 2 Prompt:

"I would like your assistance in making my document editing in LaTeX more streamlined. I am going to provide you a table that I made that I will call the standard table format (STF for short). Whenever I upload a table going forward, I would like you to convert it to STF. Here is the reference STF:"

TABLE .1: Comparison of IR LEDs - Case Study

Parameter	**AG3535C-YRR-1000mA-H1**	**AC3535B-YRR-1000mA-H1**	**AC3535L-YRR-1000MA / AC3535A-YRR-1000MA**
Manufacturer	Solidlite	Solidlite	Solidlite / Solidlite
Price ($)	1.1	1	1 / 1
Current - DC Forward (If) (Max)	1 A	1.5 A	1.5 A / 1.5 A
Radiant Intensity (Min @ If)	430 mW/sr @ 1 A	900 mW/sr @ 1.5 A	175 mW/sr @ 1 A / 550 mW/sr @ 1 A
Viewing Angle	120°	60°	150° / 90°
mW/$	1228.08	757.61	814.97 / 1012.17
mW/$ (wall-plug)	2454.55	4650	4650 / 4950

[2a.2] Case Study 2 Additional Prompt:

"Convert to STF:"

TABLE .2: Strobe Illumination Wavelength - Original

Feature	**UV**	**Blue**	**Red**	**IR**
Cost	Moderate	Moderate	High	Low
Longevity	Low	Moderate	High	High
Diffraction-limited Resolution	High	Moderate	Low	Low
High Power Common	Yes	Yes	No	Yes
Visible	No	Yes	Yes	No
IMX296 Responsivity	Low (<0.5)	Moderate (0.5-0.8)	High (0.9-1)	Moderate (0.5-0.9)

[2b] Case Study 2 - ChatGPT's Response

TABLE .3: Strobe Illumination Wavelength - ChatGPT

Feature	**UV**	**Blue**	**Red / IR**
Cost	Moderate	Moderate	High / Low
Longevity	Low	Moderate	High / High
Diffraction-limited Resolution	High	Moderate	Low / Low
High Power Common	Yes	Yes	No / Yes
Visible	No	Yes	Yes / No
IMX296 Responsivity	Low (< 0.5)	Moderate (0.5–0.8)	High (0.9–1) / Moderate (0.5–0.9)

"Important Note: Your STF only supports 4 columns, but your original table had 5 (UV, Blue, Red, IR). So I combined:

- Red + IR $\rightarrow$ one column (Red / IR)

If you'd rather:

- Expand STF to 5 columns, or
- Keep all wavelengths separate

I can adjust your standard format accordingly."

[2c] Case Study 2 - Copilot's Response

TABLE .4: Strobe Illumination Wavelength - Copilot

Feature	**UV**	**Blue**	**Red**	**IR**

Cost	Moderate	Moderate	High	Low
Longevity	Low	Moderate	High	High
Diffraction-limited Resolution	High	Moderate	Low	Low
High Power Common	Yes	Yes	No	Yes
Visible	No	Yes	Yes	No
IMX296 Responsivity	Low (<0.5)	Moderate (0.5-0.8)	High (0.9-1)	Moderate (0.5-0.9)

”If you’d like this table to use different column widths or if you want me to standardize captions or labels as well, I can adjust the STF template accordingly.”

Case Study 3: Metaprompt

.0.1 Prompt 1

Prompt: ”My undergraduate senior design project is a combination of optics, photonics, electrical engineering, and software used to measure the speed and trajectory of a projectile. I need to do a case study comparing how well different AI works for my senior design project. What are some uses for AI?”

Used the ”Thinking” selection in the drop-down (rather than the ”Fast”, ”Pro”, or ”Deep Research”.

.0.1.1 Gemini Response

Integrating AI into a high-speed projectile tracking system offers several ways to enhance precision, especially where traditional mathematical models might struggle with environmental noise or complex aerodynamics. For your case study, you can categorize AI applications into signal processing, computer vision, and predictive modeling.
Here are four primary use cases for AI in your project:

1. High-Speed Object Detection and Tracking

If your project uses optical sensors or high-speed cameras, AI can be used to identify the projectile against a cluttered or low-contrast background.

- **The AI Role:** Using Convolutional Neural Networks (CNNs) or “You Only Look Once” (YOLO) architectures to localize the projectile in each frame.
- **Why use AI?** Traditional frame-differencing can fail if the camera shakes or if lighting changes. AI models can be trained to recognize the specific geometry of your projectile, ignoring “noise” like shadows or debris.

- **Comparison Point:** You could compare a **heavyweight model** (e.g., YOLOv8) against a **lightweight model** (e.g., MobileNet) to see which balances accuracy and processing latency better for real-time tracking.

2. Trajectory Prediction and Extrapolation

Projectiles often follow a parabolic path, but variables like air resistance (drag), wind, and spin create non-linear deviations.

- **The AI Role:** Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks are designed for sequential data. They can "learn" the flight characteristics of the projectile.

- **The Math:** While a standard kinematic equation might look like $y = y_0 + v_{0y}t\frac{1}{2}gt^2$, an AI model can account for the drag coefficient C_d which changes dynamically with speed.

- **Comparison Point:** Compare the accuracy of a **Linear Regression** model against a **Deep Neural Network (DNN)** when predicting the final impact point based on the first 10% of the flight path.

3. Sensor Fusion (Optics + Electrical)

In a system using both photonics (like laser gates) and electrical sensors (like acoustic or Hall effect sensors), the data often arrives at different rates and with different error margins.

- **The AI Role:** A Neural Network can act as an advanced "Kalman Filter." It learns which sensor is more reliable at specific segments of the trajectory.

- **Why use AI?** It can resolve conflicts. For example, if an optical sensor is blinded by a flash, the AI can weigh the electrical sensor data more heavily for that millisecond.

- **Comparison Point:** Compare a **Traditional Kalman Filter** (mathematical) against a **Gated Recurrent Unit (GRU)** for multi-sensor data integration.

4. Signal Denoising in Photonics

Photonics sensors can be incredibly sensitive but prone to "shot noise" or thermal interference in the electrical components.

- **The AI Role: Autoencoders** can be used to "clean" the raw signal. By training a network on "clean" vs. "noisy" signal pairs, the AI learns to reconstruct the true pulse of the projectile passing a sensor.

- **The Benefit:** This allows you to use lower-cost hardware because the software (AI) compensates for the lack of hardware shielding or high-end filtering.

- **Comparison Point:** Compare **Wavelet Transforms** (traditional) against **1D Convolutional Neural Networks** for signal-to-noise ratio (SNR) improvement.

Suggested Case Study Framework

To make your case study robust, I recommend comparing these models across three specific metrics:

1. **Inference Latency:** How many milliseconds does it take to process one data point? (Critical for high-speed projectiles).

2. **Computational Cost:** Can the AI run on an edge device (like a Raspberry Pi or Jetson Nano) or does it require a desktop GPU?

3. **Robustness:** How does the accuracy drop when you introduce artificial "noise" (e.g., dimmed lighting or simulated wind)?

Would you like me to help you draft a specific comparison table for two of these architectures (e.g., YOLO vs. a standard Kalman Filter)?

.0.1.2 Grok Response

AI has numerous practical applications in your senior design project for measuring projectile speed and trajectory. Your system—integrating optics/photonics (e.g., high-speed cameras, lasers, or interferometric sensors), electrical engineering (sensors, data acquisition), and software—already generates rich data like image sequences or time-series sensor readings. AI excels at handling noise, sparse/occluded data, real-time processing, and fusing multi-modal inputs (e.g., video + IMU), which classical methods often struggle with in dynamic, real-world conditions like varying lighting, high speeds, or environmental interference.

Here are some targeted, implementable uses for AI in your project. These are drawn from established applications in ballistics, sports analytics, robotics, and defense systems, making them suitable for an undergraduate case study. You can compare different AI approaches (e.g., traditional computer vision vs. machine learning vs. deep learning vs. physics-informed models) on metrics like accuracy (position/velocity error), latency (for real-time use), robustness (to noise/lighting), computational cost (CPU/GPU/embedded), and data efficiency.

1. **Object Detection and Tracking in Optical/Video Data**

- **How it works**: Use computer vision AI to automatically locate the projectile in each frame from your high-speed camera(s) or photonic sensors. Models like YOLO (You Only Look Once) series detect the projectile as a bounding box or keypoint, then pair with trackers (e.g., DeepSORT or ByteTrack) to follow it across frames.
- **Benefits for your project**: Replaces manual or threshold-based detection (which fails

under blur, glare, or background clutter). Enables precise 2D/3D position extraction for velocity calculation via frame timestamps.
- **Real-world examples**: Applied to ball tracking in sports broadcasts (e.g., via AWS deep learning pipelines) and high-speed cannon projectile video analysis for 3D pose/velocity.
- **Case study angle**: Compare YOLOv8/v11 (fast, real-time on edge devices) vs. classical OpenCV (background subtraction + contour detection) vs. older detectors like Faster R-CNN. Measure mAP (mean average precision) and FPS.

2. **Trajectory Prediction and Smoothing from Noisy/Sparse Data**

- **How it works**: Feed detected positions (or raw sensor data) into recurrent networks like LSTM/GRU or transformers to predict future positions and smooth noisy measurements. Physics-Informed Neural Networks (PINNs) add physical constraints (e.g., projectile motion equations with drag/Magnus effects) directly into the loss function.
- **Benefits for your project**: Handles gaps in optical data (e.g., projectile exiting frame or sensor occlusion) and predicts full trajectories even from partial observations. Improves speed/trajectory estimates beyond simple least-squares fitting.
- **Real-world examples**: LSTM for GNSS-denied mortar trajectory estimation from IMU data; PINNs for 3D table tennis ball kinematics from video; hybrid physics-ML for non-rigid projectile launch modeling.
- **Case study angle**: Compare LSTM/PINNs (data-efficient, physics-aware) vs. pure data-driven neural nets vs. traditional Kalman filters. Test on simulated + real projectile data for RMSE in position/velocity.

3. **Velocity/Speed Estimation and Sensor Fusion**

- **How it works**: AI processes time-stamped positions or optical flow from your photonics setup (e.g., combining camera frames with laser Doppler or high-speed video). Regression models or optical flow + deep learning estimate instantaneous speed; multi-sensor fusion (e.g., via neural networks) integrates EE sensor data.
- **Benefits for your project**: More accurate than manual frame differencing, especially at high velocities where blur occurs. Real-time edge AI can output speed on-device.
- **Real-world examples**: Vision AI for ball speed/direction in sports; deep learning + optical flow for velocity in high-speed scenarios; automated CV for cannon projectile initial velocity/orientation.
- **Case study angle**: Benchmark CNN-based optical flow vs. traditional Lucas-Kanade vs. hybrid ML (e.g., Random Forest on physics features). Evaluate error vs. ground-truth chronograph data.

4. **3D Trajectory Reconstruction from 2D Optical Views**

- **How it works**: Monocular or stereo camera setups use AI (e.g., CNNs or PINNs) to lift 2D image coordinates to 3D space, incorporating camera calibration and projectile dynamics.

- **Benefits for your project**: Reduces hardware needs (no multi-camera rig required) while leveraging your existing optics. Corrects for perspective/distortion in photonic imaging.
- **Real-world examples**: Monocular 3D projectile localization from video; patent methods for golf/baseball trajectories from single-camera 2D images.
- **Case study angle**: Compare PINN-based 3D lifting vs. geometric triangulation + ML refinement. Assess reconstruction error in lab tests.

5. **Image Enhancement, Denoising, and Anomaly Detection (Photonics-Focused)**

- **How it works**: Generative or convolutional models denoise low-light/high-speed photonic images, enhance contrast, or flag anomalies (e.g., projectile wobble or sensor failure).
- **Benefits for your project**: Improves data quality from optical/EE sensors before downstream processing, especially in outdoor or variable conditions.
- **Real-world examples**: AI super-resolution and denoising in electro-optical systems for embedded tracking.
- **Case study angle**: Compare autoencoders or diffusion models vs. classical filters (e.g., Gaussian). Quantify PSNR improvement and downstream tracking accuracy.

6. **Optimization, Simulation, and Post-Processing (Software Layer)**

- **How it works**: Train ML models on your collected dataset to optimize launch parameters, simulate trajectories (hybrid physics + ML), or generate ballistic tables/reports.
- **Benefits for your project**: Extends the system beyond measurement to predictive analytics (e.g., "what-if" trajectory adjustments).
- **Real-world examples**: Neural nets for optimal launch angle prediction; extreme learning machines for impact point forecasting; hybrid ODE solvers + Random Forest for bullet velocity.
- **Case study angle**: Compare data-driven ML vs. physics-only simulation vs. hybrid approaches on prediction error and training data needs.

Suggestions for Your AI Case Study

- **AIs to compare**:
  - Baseline: Non-AI (OpenCV classical CV + Kalman filter).
  - Traditional ML: SVM/random forests for detection/regression.
  - Deep learning: YOLO + LSTM (end-to-end).
  - Advanced: PINNs or transformer-based (physics-aware).
  - Edge vs. cloud: TensorFlow Lite/PyTorch Mobile vs. full models.
- **Implementation tips**: Use open-source tools (Ultralytics YOLO, PyTorch for PINNs/LSTM, OpenCV). Generate synthetic data via physics simulators (e.g., projectile motion equations in Python) to augment limited real shots. Test in your lab with a safe projectile launcher (e.g., air cannon or slingshot).
- **Evaluation framework**: Collect ground-truth via high-precision sensors or multiple synchronized cameras. Metrics: MAE/RMSE for speed/trajectory, inference

time (ms/frame), power consumption (for embedded EE), generalization (different lighting/speeds).
- **Feasibility for undergrad**: Start small—prototype detection/tracking on your existing video data. Many models train in hours on a laptop GPU. This adds strong "AI integration" value to your report/presentation.

These uses can make your system more accurate, autonomous, and scalable while providing excellent material for a comparative case study (e.g., "AI boosts velocity accuracy by X% with Y ms latency"). If you share more details (e.g., specific sensors, data format, or hardware constraints), I can refine these further or suggest code starters!

Case Study 4

UART

To get more information on UART, I queried: "UART Standard".

I then queried: "is UART a standard like USB"

Responses

UART, response 1

UART, or Universal Asynchronous Receiver-Transmitter, is a hardware communication protocol used for asynchronous serial communication, allowing data to be transmitted one bit at a time over two wires. It requires both the transmitting and receiving devices to be configured with the same baud rate, data bits, parity bits, and stop bits for successful communication. [Links to: Analog.com (UART), and Wikipedia.org (UART)]

UART, response 2

UART is not a standard like USB; rather, it is a hardware communication protocol used for asynchronous serial communication. USB, on the other hand, is a standardized protocol that manages data transfer between devices and includes features like device detection and hot-plugging.